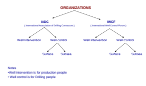
# WELL INTERVENTION COURSE




# HANDOUTS

# ***To Get The Max. Benefit:***

1. Be Alerted.
2. Do Not Hesitate To Ask Me at any time.
3. Your Question Is Valuable For Me and others.
4. We need to talk together a lot.
5. Respect different points of view and let us share information.
6. Your Suggestions Are Highly Appreciated Including Break Time Request.
7. Please Turn Your Mob. Phone Into Silent

# WELL INTERVENTION COURSE

**Course content:**

1. Well Control , risk assessment
2. Hydrates
3. Pressure & Volume
4. Well Completion
5. Well Interventions

    A- Coiled Tubing

    B- Wireline

    C- Snubbing

**Level 3** questions ( Operator level) focus on applying knowledge and skills

**Level 4** questions ( Supervisor level) focus both on applying knowledge and skills with an added focus on critical thinking and problem solving

* All exams ( Completion operations , completion Equipment, Wireline ,Coiled Tubing and Snubbing ) are 30 questions each in one hour

* One point per question

# WELL CONTROL

# Pressure control incident

## Impact

1- life loss

2- Environmental loss

3- Financial loss

4- Reputation

5- Extra governmental roles

**What is well integrity?**

Well Integrity is defined in Norsok D-010 as: “application of technical, operational and organizational solutions to reduce risk of uncontrolled release of formation fluids throughout the life cycle of a well”.

# Well Integrity Management System

Well Integrity Management System is a combination of technical, operational and organizational processes to assure a well's integrity over the wells' life cycle.

**Minimum Elements**

Well Integrity Policy and Implementation through Business Plan

Budget, Training, Audits, Immediate Response, Resources Roles

Responsibilities and Authority levels

**Risk Assessments, covering ALL aspects of the Well Integrity Management System**

- Minimum Barriers and Verification
- Equipment Standards and - Maintenance
- Operating Limits and Continuous Monitoring
- Well Handovers with Well Records
- Incident Reporting
- Compliance Audits

# Role of Risk Assessment - RA

Risk Assessment is a process that can be used to develop information to help with making such decisions involving well control.

# Risk Management- JSA

The main purpose of Risk Management is to formulate a decision- making process that may contribute to selecting Optimum Solutions in combination with Risk Reducing Measures

This decision-making process is called ‘Job-Safety-Analysis – JSA and should be implemented for:

- New or non-standard operations
- Operations involving use of new or modified equipment
- Any kind of operation that can be labelled hazardous
- Change in actual conditions which may increase the risk

# Risk Management

One of the Risk-reducing measures : Simulating Drills

Well Control Drills shall be initiated at unscheduled times, when conditions allow. To be performed under supervision of Well Supervisors. However, a drill should never be conducted during critical well activities.

Drills shall be recorded and reported as follows:

- Type of Drill
- Effectiveness and Timeline of Response
- Functioning of Equipment
- Lessons Learned for Follow-Up and Sharing

We must share 'any finding' so that we all get better at it.

Well control drills provide an opportunity to assess the effectiveness of the contingency plan and to identify and make good any inadequacies.

The purpose of Well Control Drills is to familiarize the crews with techniques that will be implemented in the event of a well control emergency

- It is important that returning crews have frequent Drills.

# HOW DOES AN *INFLUX* BEHAVE IN THE WELL?

Gas is compressible, and therefore its volume of gas is related to pressure and temperature. When we have a gas kick, the volume of gas must be allowed to expand in order to drop its pressure as it comes to surface. Gas, being lighter than fluids, will migrate upwards when the well is still shut in, creating higher shut-in surface pressures, if it is not expanded

## Oil Influx

Mostly oil influx always has some gas in solution, therefore an oil influx may behave somewhat similar to a gas influx , but to a lesser degree

### Water Influx

Water is virtually incompressible and does not expand significantly. When we have a water kick, surface pressure will not change

## H2S Gas

It is a toxic and very dangerous gas and it is relatively heavy. It may cause failure/brittleness of H2S non-resistant materials due to its sour gas effects

## Effect of not Bleeding off Pressure at Surface

When well is shut in, and when gas is allowed to migrate up the well bore without bleeding off, then there will be no room for expansion

Gas will carry original pressure as it moves up the wellbore

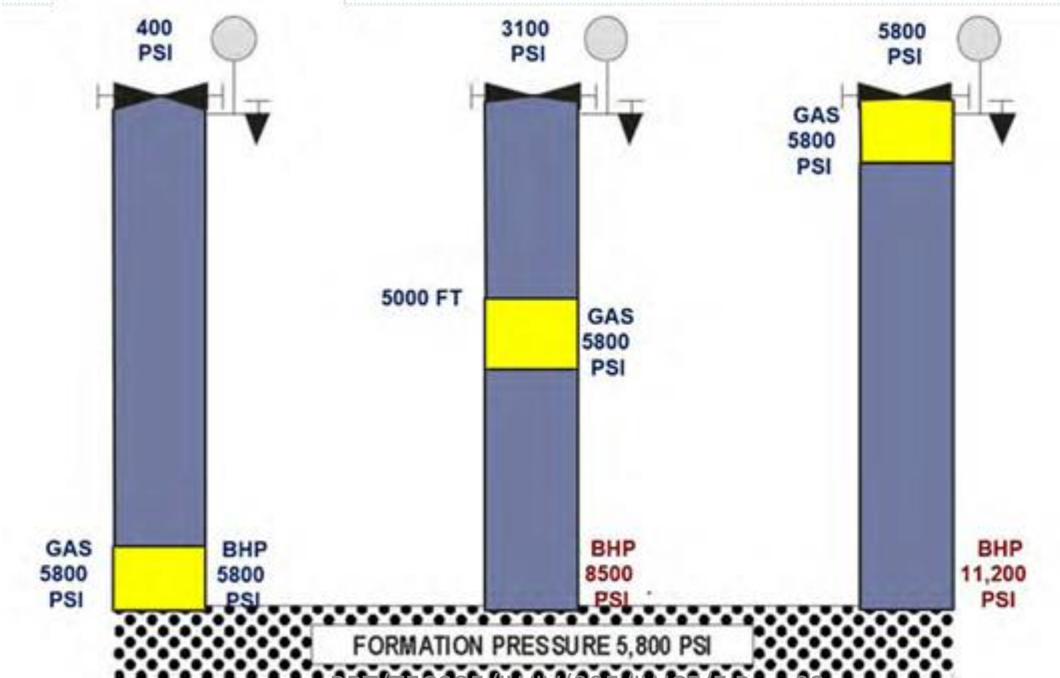
Therefore, the effects of gas migration in a closed well are:

- Gas Influx Pressure = Bubble Pressure] will remain the same
- Shut-in drill pipe and casing pressures will increase
- Bottom Hole Pressure  will increase,  possibly causing formation to fracture
- Pressure at any point below or above Gas Influx will increase

## Effect of not Bleeding off Pressure at Surface

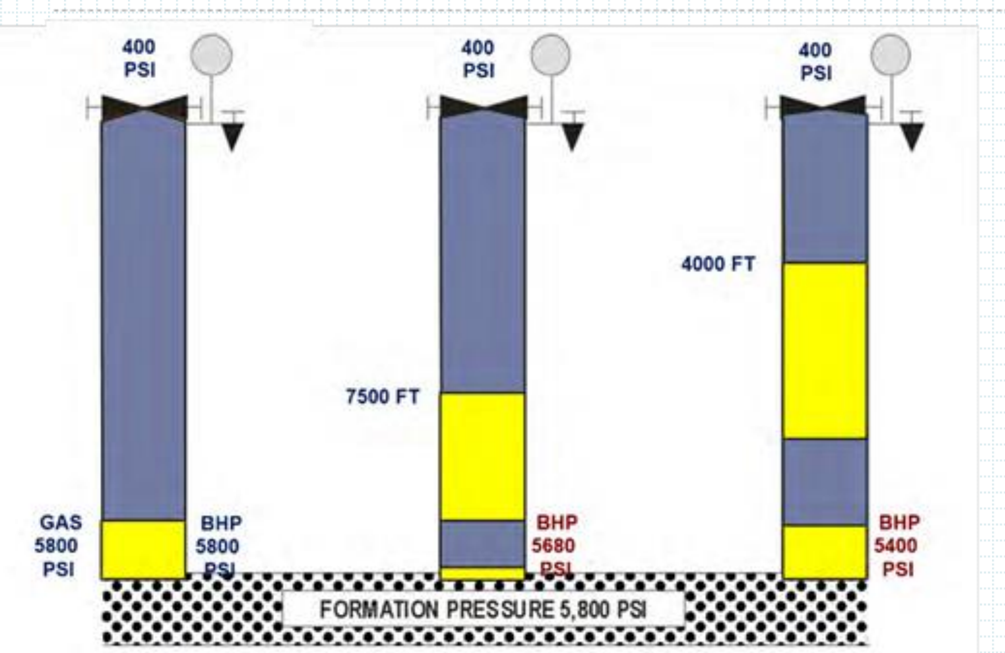
• The figure below shows the effect of not bleeding off the pressure at surface, while the gas bubble is slowly migrating, and not allowed to expand.

• The result is that BHP and pressure at surface increase to very high values, thus increasing risk to casing burst or formation fracture



**Effect of Bleeding off with Constant Pressure at Surface**

- The figure below shows the effect of bleeding off at surface, but keeping surface pressure constant.
- The gas bubble is migrating fast, and allowed to expand too much, because BHP is kept too low!
- The result is that we will invite more influx into the well and ultimately higher pressures if not controlled



## Surge and swab Pressure Effects

Movement of drill string the hole creates a “change in the BHP.

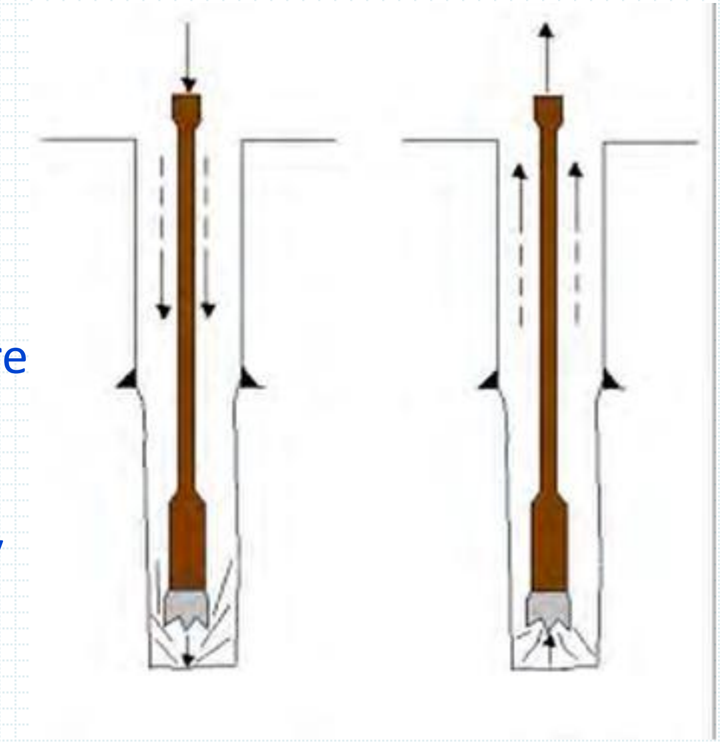
Tripping in the hole creates a surge pressure. It can cause formation to fracture which in turn can

lead to lost circulation

Tripping out  of the hole creates a swab pressure

It can reduce bottom hole pressure, Sufficiently

to allow influx to flow into the well



Surging and swabbing should always be minimized by controlling these four factors. some are controlled by the office, others by the personnel in the field

It is critical to use the “trip sheet” to detect any swab or surge.

By checking if the hole takes the proper amount of workover fluid [or if the right amount of workover fluid returns to the tank] due to metal displacement as any pipe is tripped in RIH or POOH

# BARRIER

What kind of well service disaster could happen caused by pressure?

Blow-Out

What causes Blow-Out?

Barrier failure

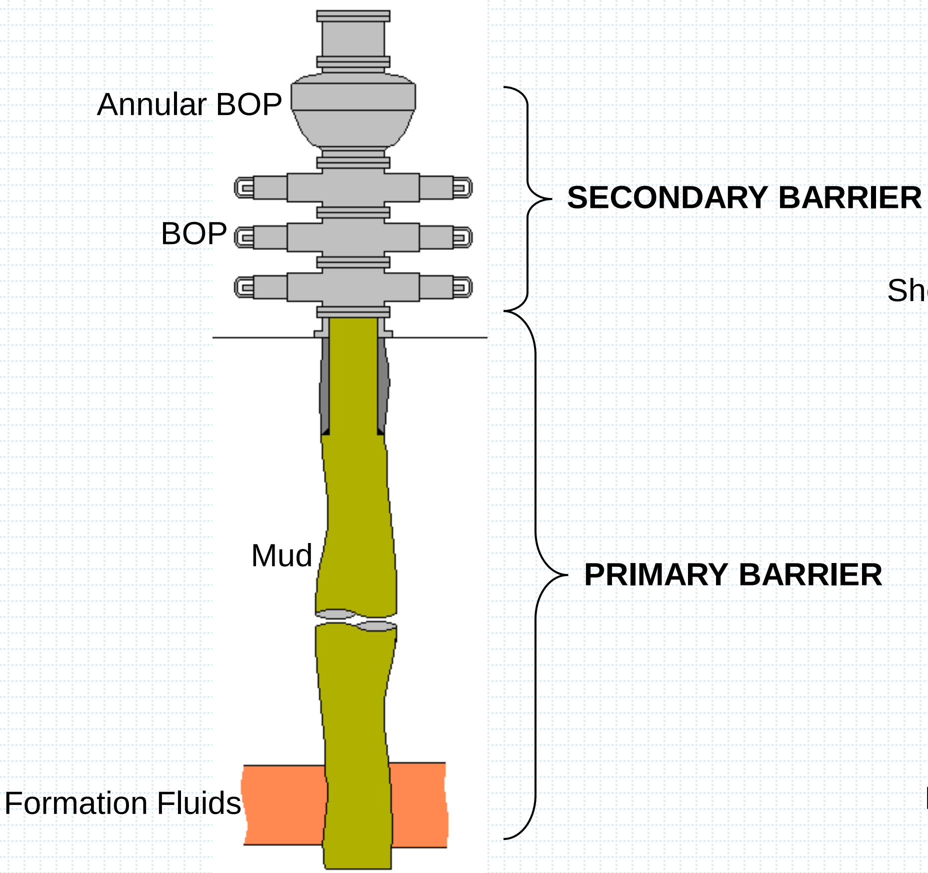
What is the equipment used to control a well from blow-out?

Barrier

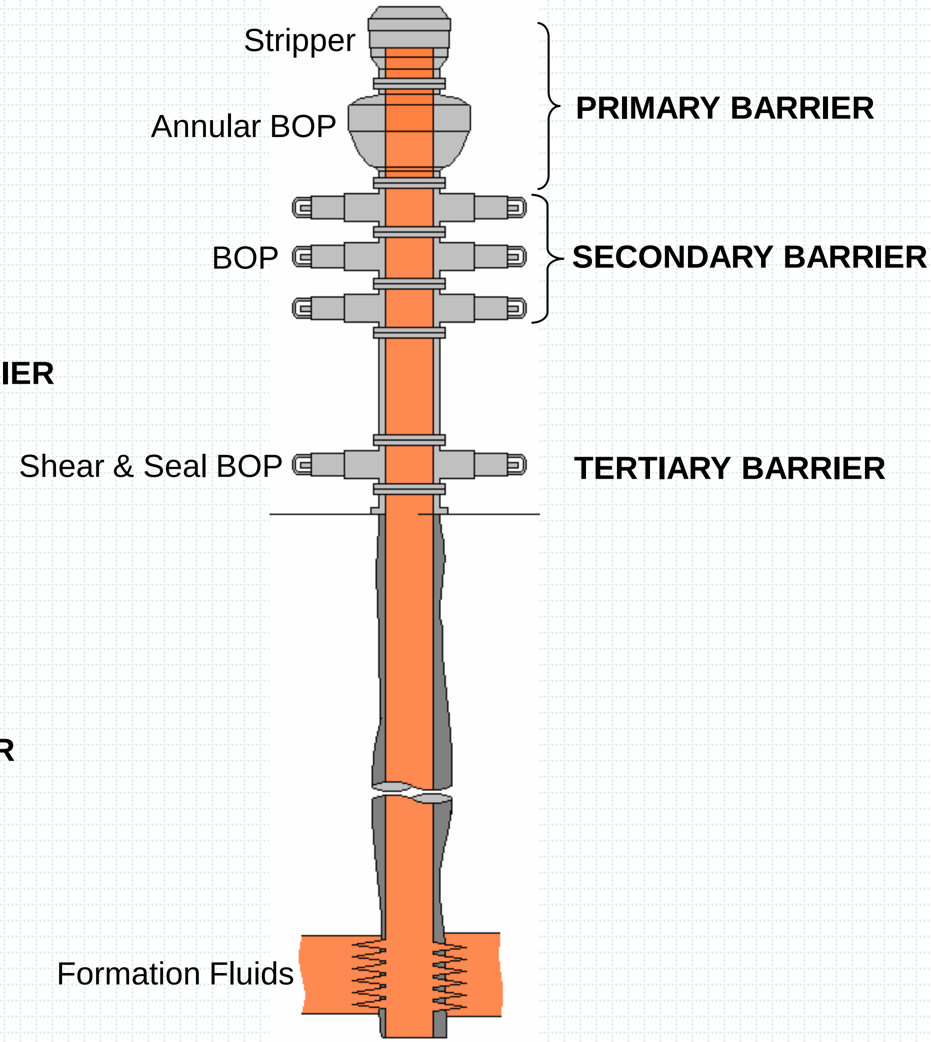
What is barrier?

Any device, fluid or substance that prevent flow of wellbore fluids.

## DRILLING
(Double Barrier Protection)




## WELL INTERVENTION
(Triple Barrier Protection)



## Barrier Classification:

1. **Primary Barrier** – 1st line of defense system
2. **Secondary Barrier** – 2nd line of defense system
3. **Tertiary Barrier** – 3rd line of defense system when 1st & 2nd system failed or been compromised. Must have the ability to shear pipe/line.

**Any barrier can be classified as a primary barrier as long as it prevents flow of the well**

## API recommendation of Barrier

**A minimum of two tested and available barrier regardless of type of operations**

What can happen when you unset a retrievable bridge plug and don't have enough weight above it?

## Barrier Type:

1. Mechanical Barrier

   1.1 Closed Type — Close through out operation.

   1.2 Closeable Type — Close when required.

2. Hydrostatic Barrier

   - *Liquids that creates HP greater that formation pressure ( within 200-300 psi ) and monitored all time*

## TEMPORARY BARRIER

A barrier that moves from a place to another like BOP. To be tested before we use and whenever we suspect a leak

(BOP , STRIPPER , PLUG,…….)

## PERMANENT BARRIER

A fixed barrier in the well like cemented casing ( before perforation ) .To be tested after installation and whenever we suspect a leak

( CASING AFTER CEMENTING AND BEFORE PERFORATING )

## ACTIVE BARRIER

A barrier which is in working condition during operation like stuffing box/stripper while in or out of hole

# WELL CONTROL THEORY

**Example of Barriers:**

| MECHANICAL BARRIER | |
|---|---|
| **CLOSED TYPE** | **CLOSABLE TYPE** |
| Stuffing Box | BOP |
| Grease Injection Head | Annular Preventer |
| Stripper | Xmas Tree |
| Wire line Plug | Subsurface Safety Valve |
| | |
| **HYDROSTATIC BARRIER** | |
| **Drilling Fluids** | **Fresh Water** |
| **Completion Fluid** | **Salt Water** |
| **DIESEL ?????** | **NITROGEN ???????** |

# Barrier test

- Temporary barrier

  Before we use

  When we suspect a leak

  Permanent barrier

  After installation

  When you suspect a leak

# BARRIER INTEGRITY

1. **MECHANICAL BARRIER**

   *Must be "Tested From Direction of Flow"*

2. **HYDROSTATIC BARRIER**

   *Must be diligently monitored for a period of time to Ensure Thermal Expansion & Contraction Effects Have Ceased & must be OBSERVABLE at all time.*

# Barrier Integrity test

## Why integrity test?

To confirm that PCE is holding the maximum expected surface pressure where we measures the following parameters

1- Pressure applied

2- Leaks

3- Test duration

4- Fluid used

# PRESSURE CONTROL EQUIPMENT CHECKLIST

Below checklist has to be filled by well intervention operator to be sure of

1- Number of barrier are adequate to the job

2- Barrier type are adequate to the job ( HP ,HT ,Sour gas , oil ,gas ……)

3- API test procedures are implemented

4- Double barrier theory are implemented for High pressure categories

5- PCE are visually inspected

6- API test procedures are implemented (WI-SF-C0M.02.03-03)

| ITEM | FUNCTION TEST | PRESSURE TEST | VALID CERTIFICATE | VISUAL INSPECTION | API TEST PROCEDURES | ADEQUTE FOR THE JOB |
|---|---|---|---|---|---|---|
| PRIMARY BARRIER | | | | | | |
| SECONDARY BARRIER | | | | | | |
| TERTIARY BARRIER | | | | | | |
| DOUBLE BARRIER | | | | | | |

# BARRIER PHILOSOPHY

## BARRIER ENVELOPES

Barrier system is normally an envelopes of a group of individual barrier elements.

Examples:

- Stuffing Box is part of Lubricator Envelope.
- Xmas Tree is part of Packer Tubing Envelope.
- Casing is part of Production Casing Envelope.
- Etc

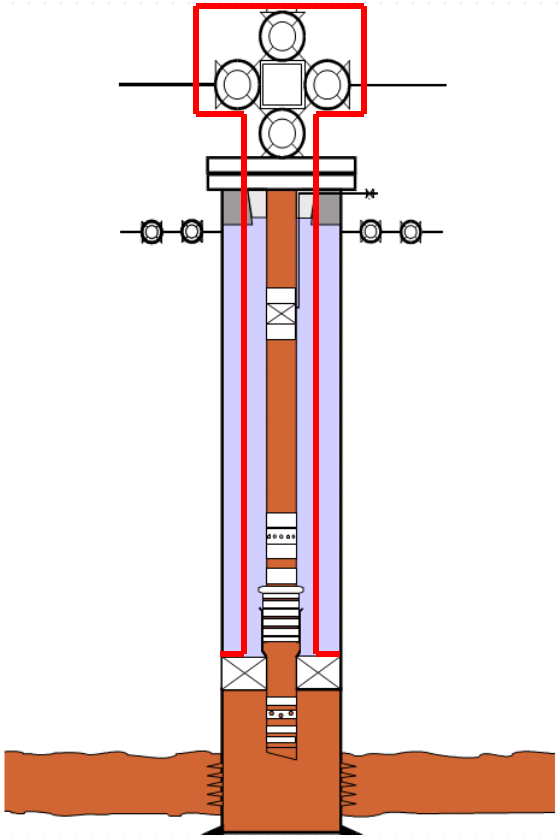
# BARRIER PHILOSOPHY

**Example of Barrier Envelope :**

**PACKER/TUBING ENVELOPE**

Barrier elements include:

- Packer
- Tubing
- Tubing Accessories
- Tubing Hanger
- Xmas Tree

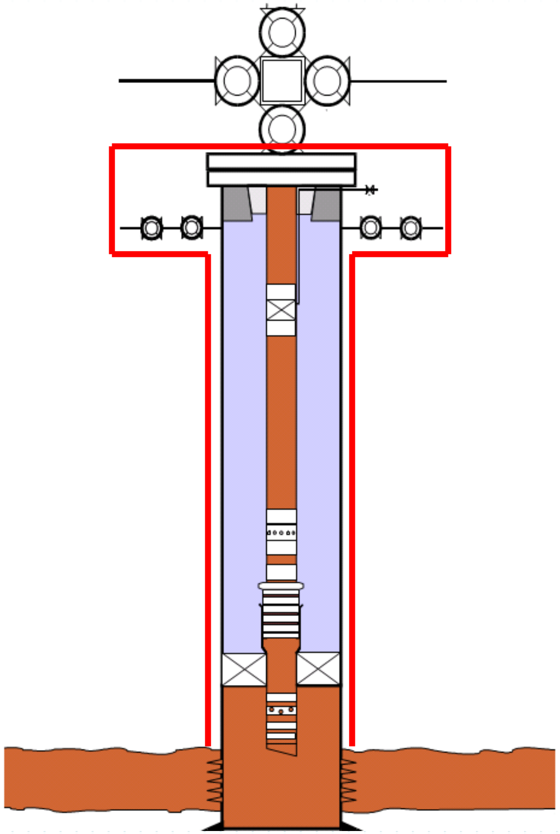


# BARRIER PHILOSOPHY

## PRODUCTION CASING ENVELOPE

Barrier elements include:

- Production Casing
- Side Outlets Valves
- Tubing Hanger
- Tubing Head
- Casing Head Housing

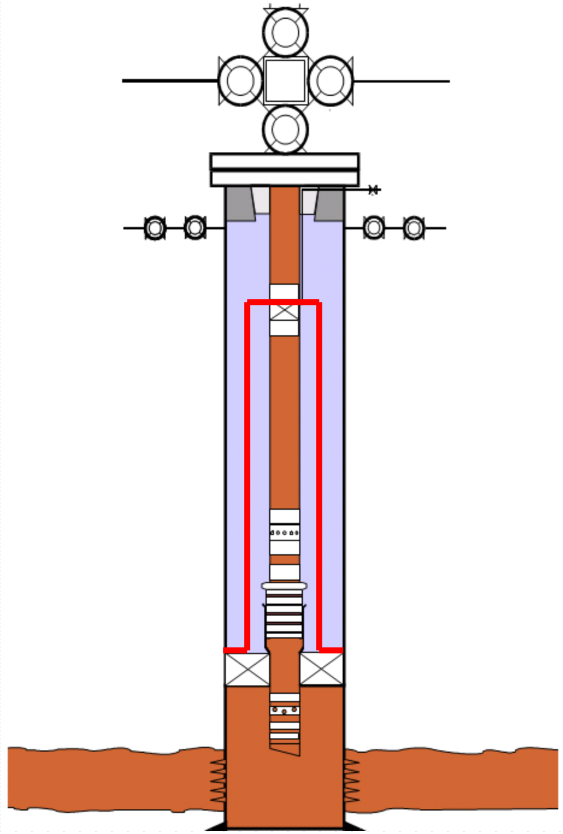


# BARRIER PHILOSOPHY

## PACKER/TUBING/DHSV ENVELOPE

Barrier elements include:

- Packer
- Tubing
- Tubing Accessories
- DHSV



**Barrier envelope contains well pressure**

1- X-Tree

**2- Tubing Hanger**

**3- Production tubing**

**4- Packer**

**5- Casing below Packer**

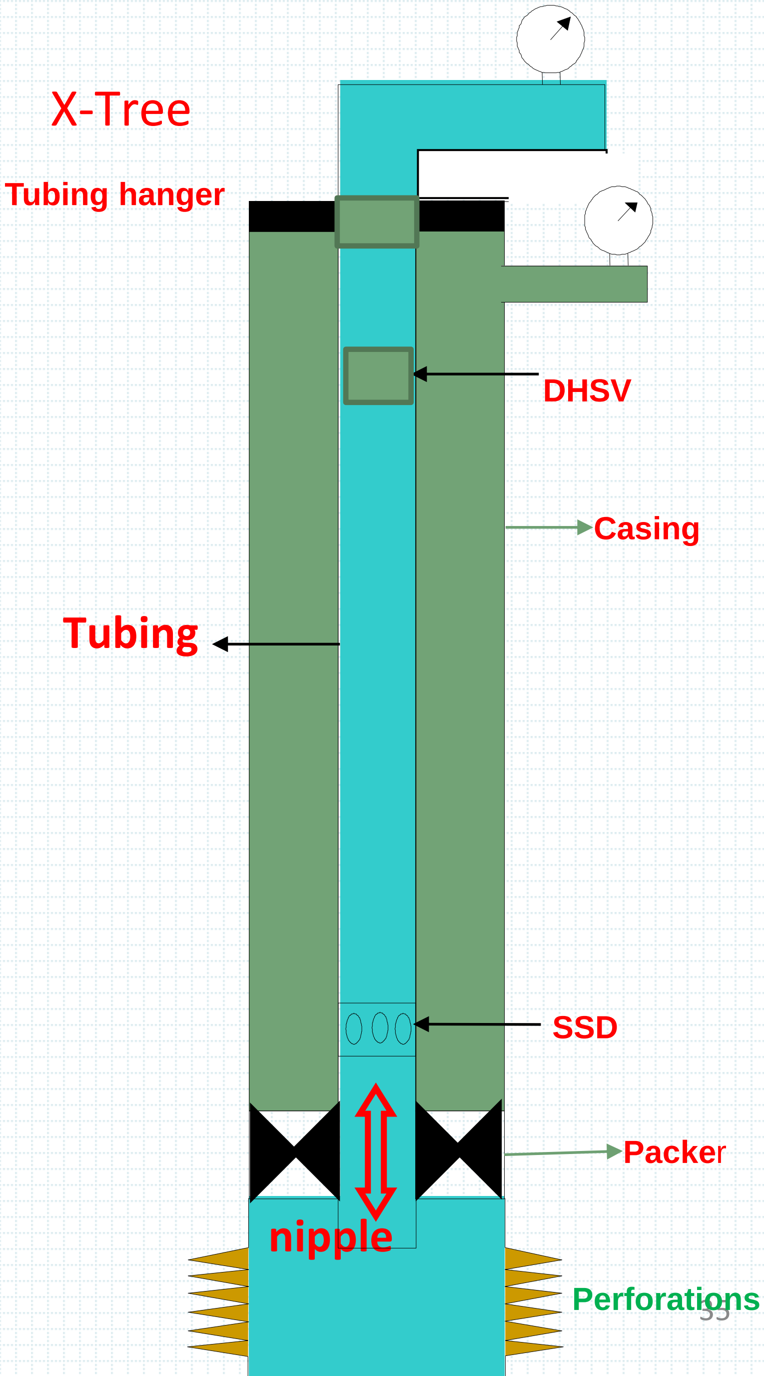
**Barrier envelope contains Annulus Pressure**

**1- Tubing Hanger**

**2- Side outlet valves**

**3- Packer**

**4- Casing above packer**



# Shear/seal BOP

Shear/seal BOP is the tertiary barrier. we call it ( shearing Barrier )

as it has the shearing capability

We use it when Primary barrier & Secondary barrier failed

We put directly on top of the X-Tree

In offshore platform where the deck is too high we use two

shear/seal devices One on top of X-tree and one at the platform

It should be operated from separate control Panel

# BOP pressure & function test

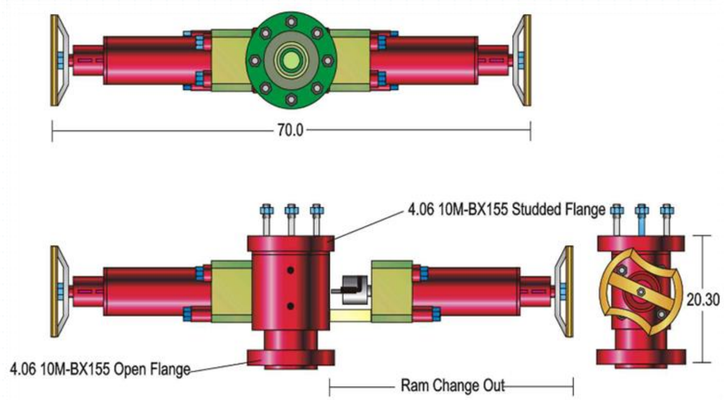
All BOP has to be function tested once hydraulic

Hoses are connected

Function test has to be recorded for further use

Function test has to be applied for each ram

separately



# BOP pressure & function test

When pressure test the BOP ,following is to be highlighted

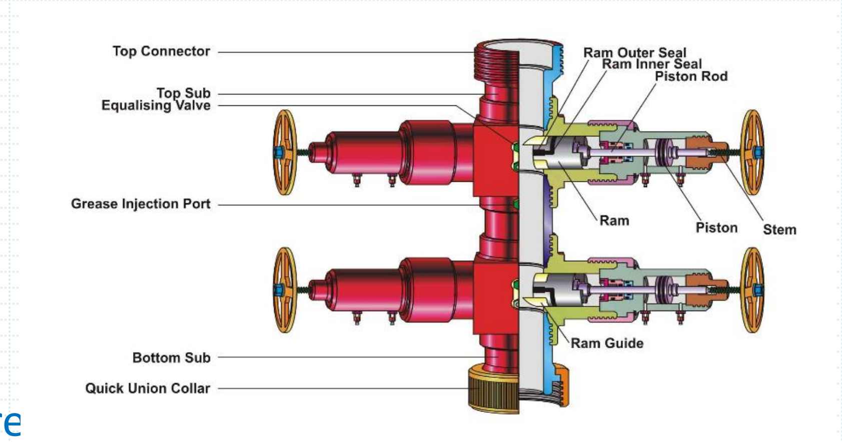
1- Test area should be evacuated from personnel

2- Never test with gas

3- BOP should be flushed to get rid of air inside

4- Never test with flammable liquids

5- Never exceed maximum BOP certified working pre




6- Test has to be made according standards and written procedures

7- Test has to be recorded on a chart and software application

8- We applied pressure test on BOP body and each individual pressure control actuator ( Blind , Pipe )

# Integrity test of Barrier elements

**1- PLUGS**

From above ( positive plug) or from below (inflow test)

**2- SLIDING SLEEVES**

Using deferential pressure between tubing & annulus

**3- DHSV**

Bleeding off pressure above and monitor build up

Apply pressure on control line and monitor leak

**4- X-TREE & GATE VALVES**

Apply pressure through kill wing or swab valve after bleeding off pressure on FWV

**5- PACKERS**

Apply pressure from the tubing and monitor build up in the annulus



**6- ANNULUS**

Apply pressure from the annulus valves and monitor for leaks

# Integrity test of Barrier elements

## General rules

1- Test from direction of flow

2- Never test with gas or flammable liquids

3- Be sure there is no Air in the system

4- Do not exceed maximum working pressure

5- All test has to be recorded

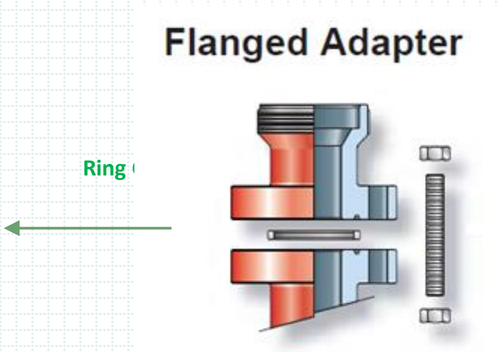
6- Use test standards & procedures

# Joints

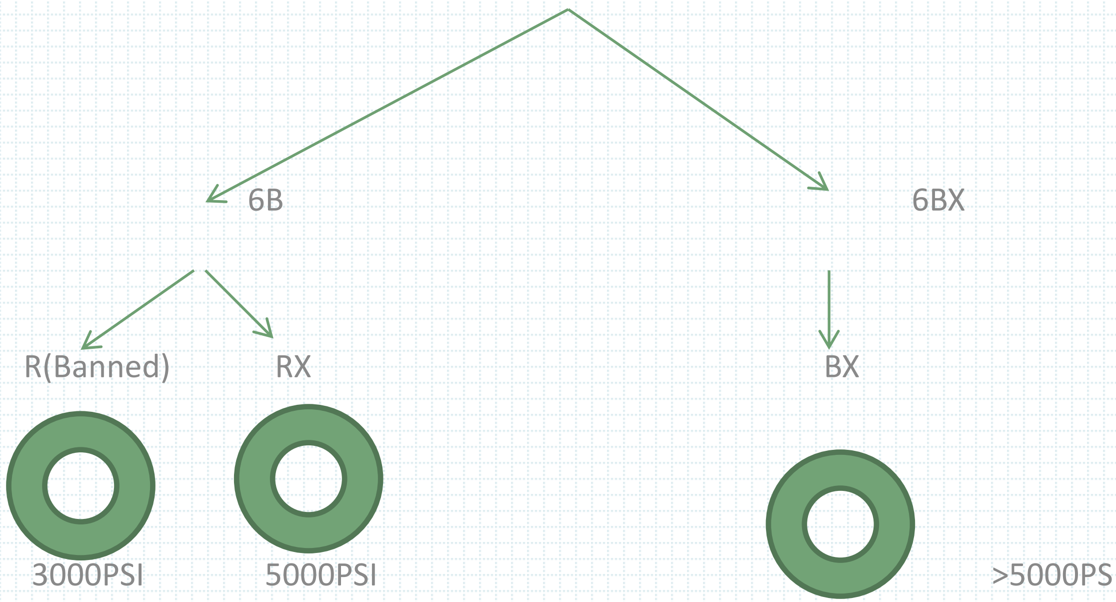
## 1- Ring Gaskets & Flanges

Ring Gaskets used one time only

Should be brand new when used



# API FLANGES



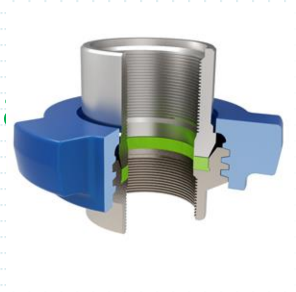

R & RX are interchangeable while 6B & 6BX are NOT interchangeable

(WI-SF-COM-04.02.05)

# Joints

## 2- Hummer union seal

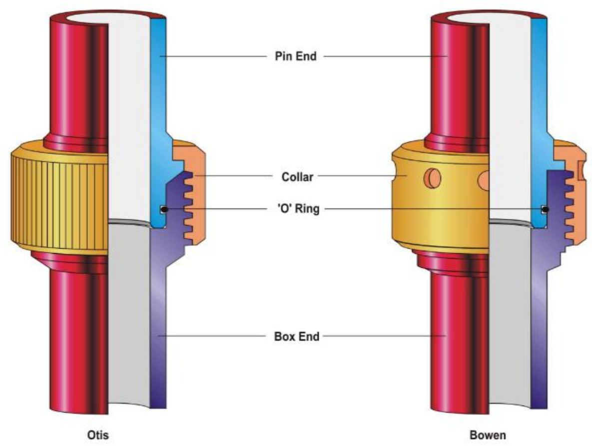
Hummer union seal should be visually inspected a



changed periodically and matches pressure

category and Fluid pumped

# Joints




3- Quick union seal

Use one time only

Should be brand new when used

Stored in Air condition store

Has a shelf life

CTU rig up should not include any quick connection , Flange type should be used

Flanges withstand torque

# Insert Materials ( Elastomer)

- **Characterized by their wear, temperature and chemical resistance.**

  - **Urethane** **(-40F<T<200F / Long wear)**
  - **Nitrile.** **(10F<T<350F / NO H2S) good for oil and wear but not for H2S**
  - **Viton.** **(0F<T<400F) excellent for H2S & chemicals**
  - **EPDM.** **(20F<T<500F / excellent for steam)**

**Packing life depends on:**

Hydraulic pressure.

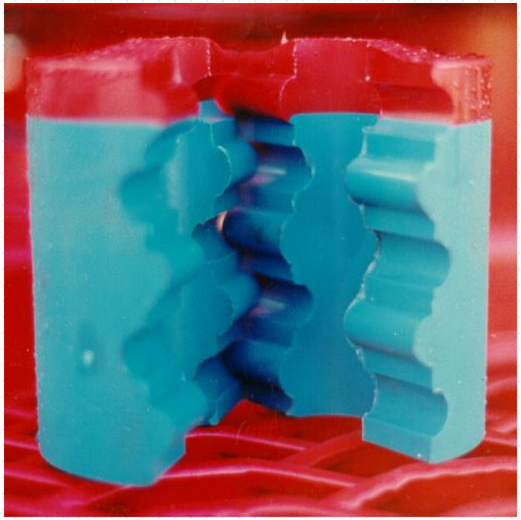
Lubrication.

Bushings condition.

# Insert Materials ( Elastomer)

Good elastomer

damaged one



Due to bad selection of elastomer

# EXPLOSIVE DECOMPRESSION OF ELASTOMERS/ SYNTHETIC RUBBER SEALS

- Rapid gas decompression, also known as *Explosive Decompression* often occurs when high-pressure gas molecules carbon dioxide in natural gas, $CO^2$ , in particular] migrates into an elastomer at a compressed state

- When the pressure surrounding the elastomer is suddenly released, the compressed gas inside the elastomer try to expand and exit the elastomer.

- Explosive decompression is major risk for any elastomeric seal operating in a high-pressure gas environment, over 1,000 psi.

# BOP Operating system

BOP is a hydraulic operated device with circuit pressure ranging from 1500 to 3000 psi

Engine on the powerpack feed the pump with mechanical power , then pump turns mechanical power into hydraulic power that pushes hydraulic oil through high pressure rubber hoses all the way to the BOP

BOP actuators (Pistons & cylinders) turns hydraulic power back into mechanical power that opens & close BOP rams

Open & close BOP rams is controlled from the control cabin of the unit by knobs or arms , these knobs or arms should be protected to prevent operating the BOP by mistake

# BOP Operating system

The picture shows a CTU control cabin where

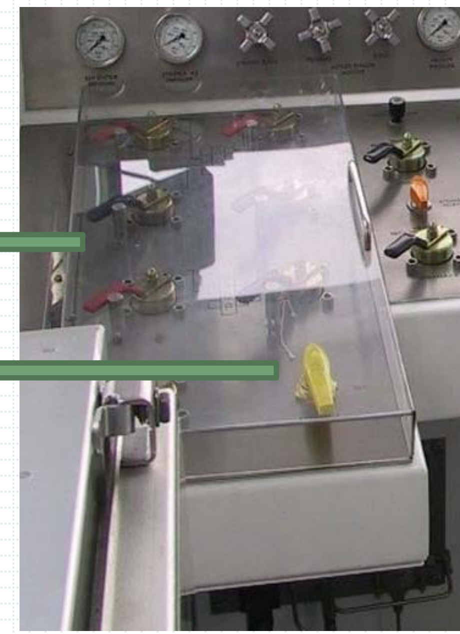
The arrow shows the place of the control arms

Of the BOP , to activate the bop you have to

1- Open the hard plastic cover

2- Open the yellow valve to allow pressure to inter

3- Activate the arm



Hard plastic cover

yellow valve

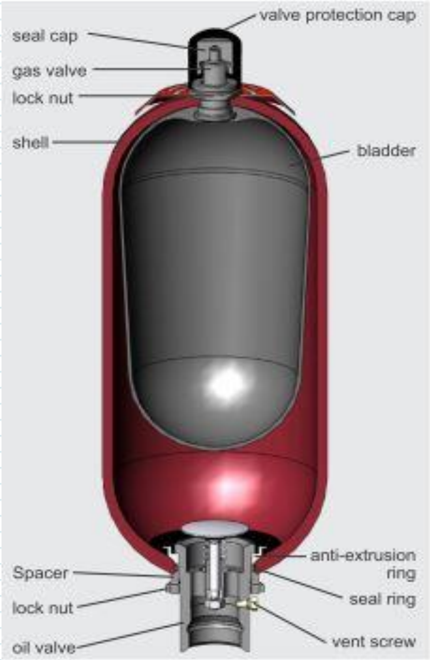
# Accumulators

Why we use accumulators → Back up if powerpack failed

What is inside it → Hydraulic oil and pressurized Nitrogen inside a bladder

Size of Accumulators should be calculated carefully (API RP-16ST)

If not correctly calculated the BOP will not activate



# Verification

Barrier verification criteria should be found in the following documents

1- Well Program

2- Operations Manual

3- Industry standards ( API , NORSC ,……etc)

4- Equipment manufacture Manual

# Well barrier test documentation

This chart must be signed by

[a] the person performing the test and

[b] a company representative to ascertain the completion of a satisfactory pressure test and must contain the following information

- Type of Test with details on depth and type fluid above, if relevant
- Test Pressure or Pressure Differential applied
- Equipment/Component/Barrier being tested
- Test Medium or Fluid
- Client name, Field/Well No, Date and Signatures
- Estimated volume of system pressurized,- Volume pumped and returned as per API 53 standards

# Checking PCE

All PCE equipment must be checked for

1- Pressure rating of the job

2- Damage

3- Wear & tear

4- The intended use (Temperature , H2S ,…….)

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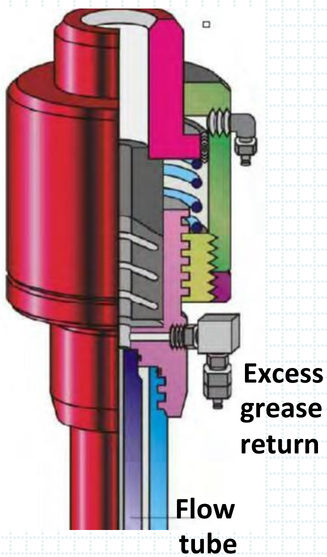
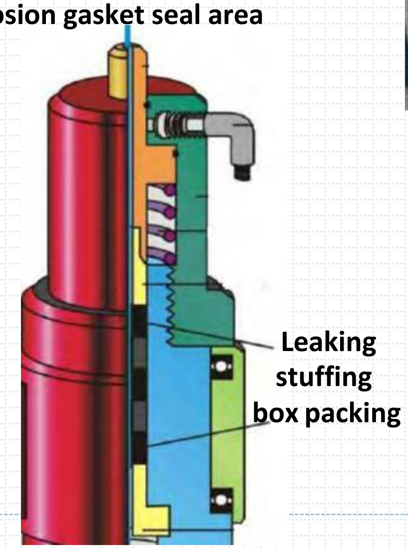
# Checking PCE

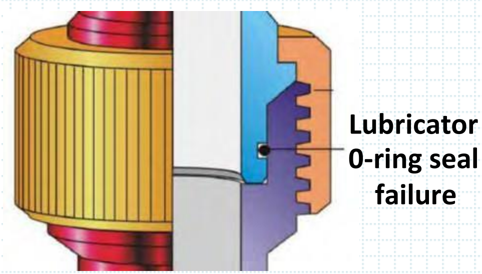




Erosion gasket seal area




Leaking piston seals on wireline bBOP

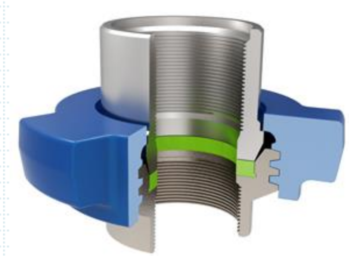


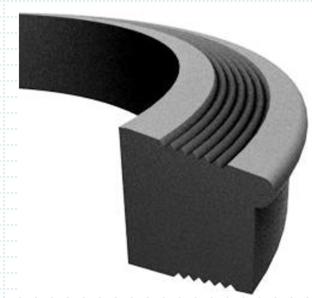
# Checking PCE

5- Elastomer type ( based on Pressure ,Temperature , fluid type)

All sealing surfaces must be cleaned and checked for damage

All adapters/connections must be checked fc





1- Pressure rating

2- Dimensions ( length , OD , ID ,……. )

3- Torque

4- Thread type

# PCE Hydraulic hoses

Hydraulic hoses connection is a major part of well control

Connections must be of same type & model

Connections must be clean of water , grease , sand ,…..

Quick coupling seal should be checked and replaced regularly

Threaded type should be checked using digital gauges

Against wear & tear

Hose external cover should be checked for wear & tear


All connections must be checked for leaks

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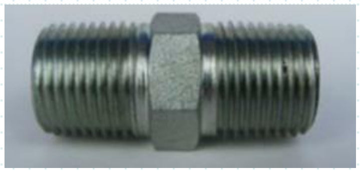
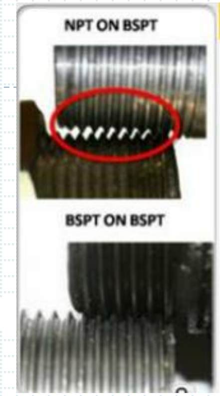
# Smaller Fittings



NPT

BSPT




Do we have a match?

PTFE TAPE

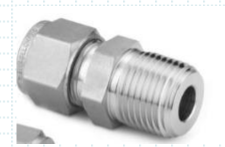
SEALANT


BSP x NPT

Note: Do nut use PTFE, Teflon, Tape in SSV and SCSSV installations

Note: Only use new fittings for permanent installations. Discard damaged fittings

Swagelok x NPT



NPT = National Pipe Thread
BSPT = British Standard Parallel Thread Taper

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# Pressure gauge limits

- During positive or negative pressure tests the pressures are monitored, recorded an documented.
- Only calibrated gauges or recorders should be used. If the accuracy of the pressure gauge or recorder is in doubt, it can be verified by using a dead weight tester. Faulty gauges or recorders should be replaced and recalibrated
- During a test the pressures may initially increase or drop, which may not be due to a leak, but due to thermal pressure effects of the test fluids
- Do not always trust the pressure gauges and recorders.
- A small leak may not be noticed on the pressure gauge , when testing with large volumes. Therefore , whenever feasible, the equipment tested should be monitored for physical leaks.

# Pressure gauge reading

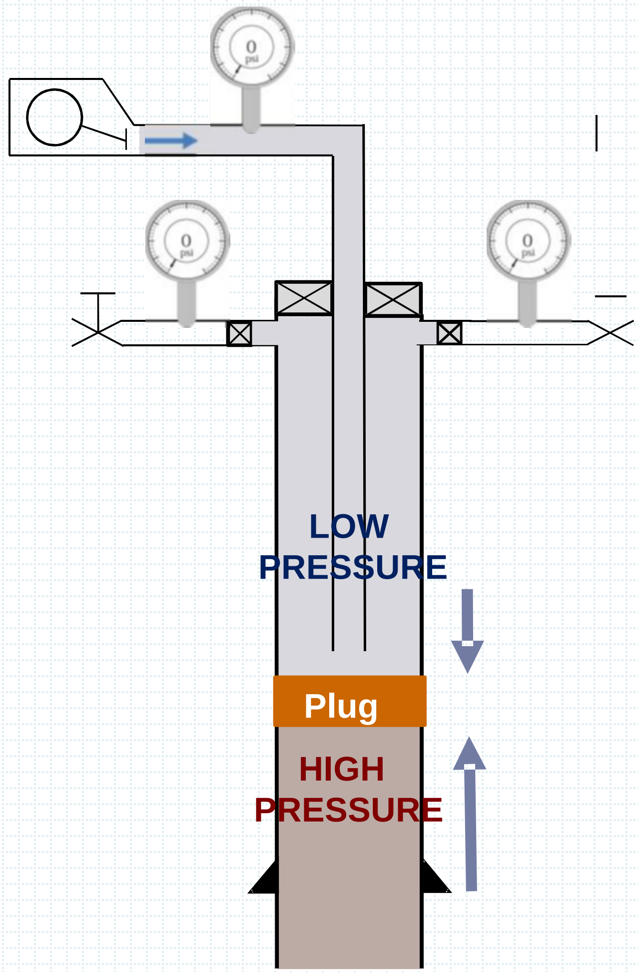
Bleeding down.

- If you are bleeding down a line, lubricator or other pressure vessel and the reading on the gauge shows zero, be aware of hydrates, that may have formed across a needle valve plugging it and there may still be pressure inside the lubricator.
- If two gauges on the same line or pressure vessel show different values, always question both gauges, until the cause has been established.

# INFLOW TEST

- Bleed of the pressure above (first stage )and monitor build up (second stage)
- It has nothing to do with fluid level in the well
- When bleeding , We should avoid creating deferential pressure that collapse the pipe
- If we set a barrier and we can not test it from direction of flow ,we should test it from above ( bonded material )

# Example of Inflow Test- Negative Test



# JOINT OPERATIONS MANUAL

Combines Operators' and Contractors' Procedures. Captures the procedural and emergency aspects. The manual is generally called the ***Joint Operations Manual*** also called ***Well Operation Manual*.**

To be available on site at all times for immediate reference and in case of an emergency

**Personnel in Charge**

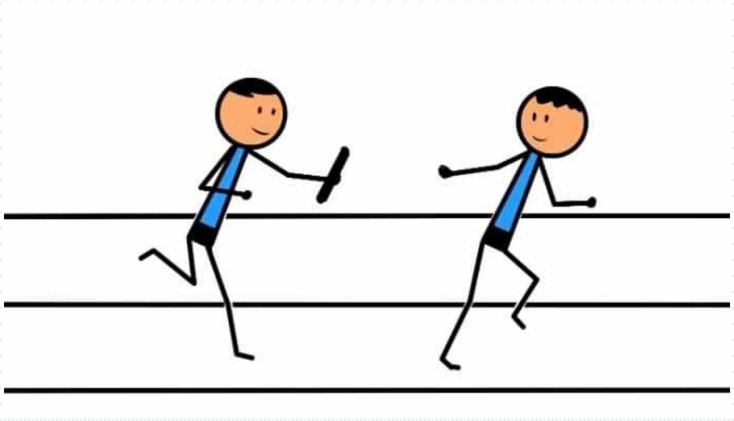
- The Equipment Operator is in charge to shut in the well, when pressure control problem arise during well intervention operations.


- Another person, specifically designated in the *Joint Operations Manual* can also shut in the well in case there is an Emergency.

# HANDOVER

The ***well handover*** is a process that formalizes the transfer of a well/or well operating **responsibility** and is endorsed by the use of related well handover documentation and templates

***Handover*** can be between:



Drilling and Production

Well Intervention and Production

Production and Well Intervention

Production and Maintenance

# Handover

Successful Handover should include

1- Complete & correct information

Date , time , location , job type , client , current situation ,………

2- Roles & responsibilities

Names of crew . Responsibility of each , …………

3- Clarify well barriers & well conditions

Well barrier types & classification , condition of each ,……..

4- Identify well parameters changes

Change in pressure , temperature , flow rate , composition of fluid

,……………………..

# MANAGEMENT-OF-CHANGE – MOC

**Definition:** System to Manage Changes and Deviations within a Well Project, such that HSE risks resulting from unforeseen ***consequences of change or deviation are identified and reduced,*** so that risks remain as low as practically possible.

**Change**

An action, alteration or intervention , either planned or unplanned, that requires adding, removing or modifying a system, program or plan

**Deviation**

Any non-compliance, planed or o unplanned requiring the appropriate level of approval


**What does it include?**

- Changes to Hardware/Equipment, HSE critical control systems [e.g. alarms and trip settings] and Conditions relevant to safe delivery of a well pore pressure and fracture gradient predictions.
- Changes to operational, maintenance, inspection procedures

# WELL CONTROL THEORY

## Definition

- Overbalance — Greater pressure than formation pressure
- Underbalance — Less pressure than formation pressure
- Inflow Test — Bleed pressure from ▲ & monitor build-up
- Positive Plug — Hold pressure from ▼ & ▲ ( we use bonded material in this case instead f chevron seal
- Barrier — Device, Fluid or Substance that prevent flow of hydrocarbon
- Formation Fracture pressure — Maximum pressure applied to the formation before it breaks
- Double barrier principle — There must always be two barrier elements between the well and the external environment. One barrier element is a back-up for the other.

# FORMATION DAMAGE

# FORMATION DAMAGE

Well Intervention most notably formation damages are;

* Exceeding formation fracture **Pressure**

**Fluid**

**Pressure** **Compatibility** **Cleanliness**

**Surge and swab Pressure are also good factors**

# WELL KILLING

Well Intervention Well Kill Method:

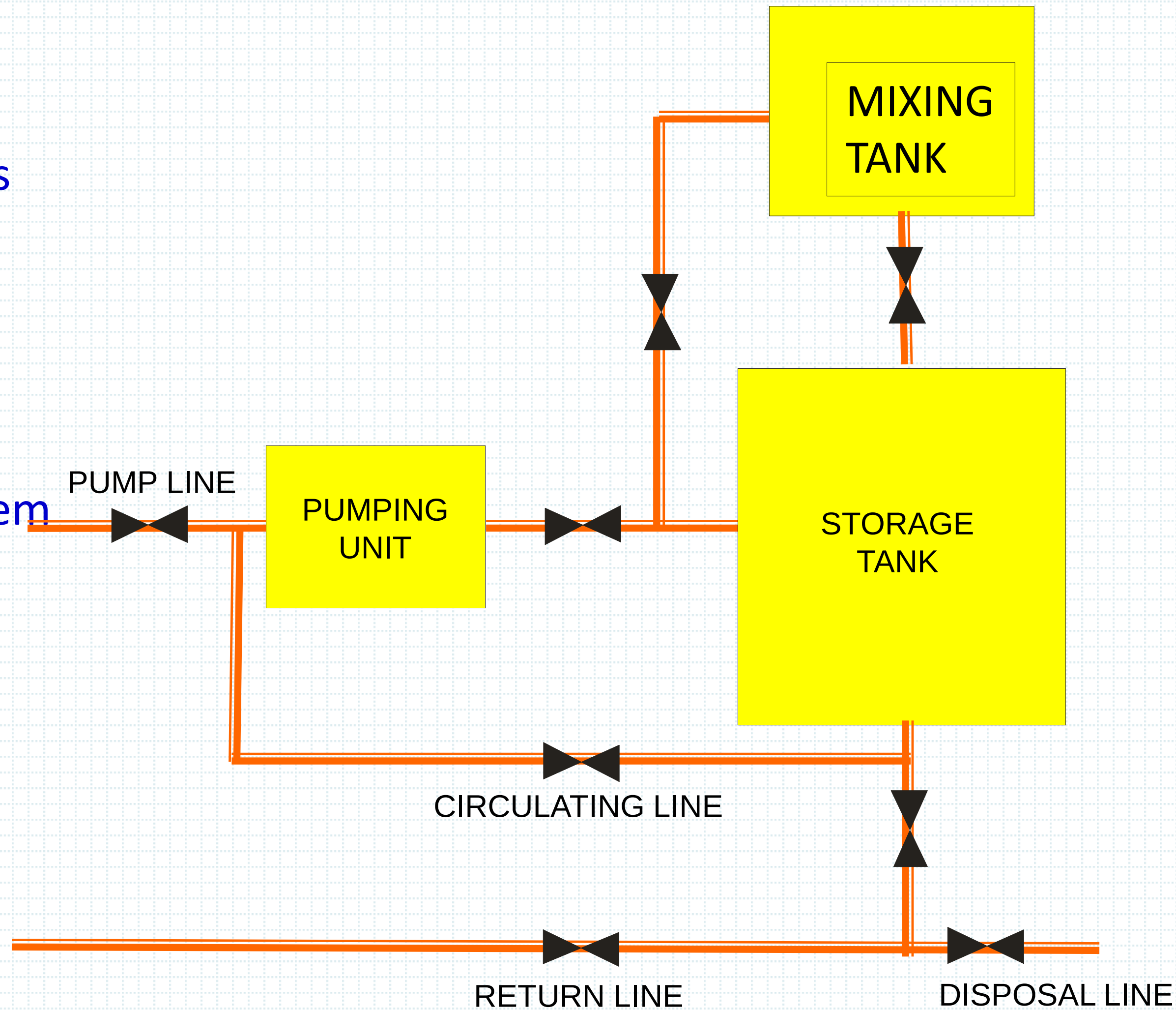
| | | |
|---|---|---|
| **1.** | **Bullhead** | *The most preferable method in case of emergency.* |
| **2.** | **Forward Circulation** | *Minimize formation damage.* |
| **3.** | **Reverse Circulation** | *Best planned kill method.* |
| **4.** | **Lubricate & Bleed** | *Last resort .* |

# INFORMATION REQUIRED

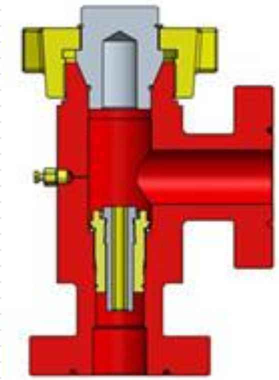
1. Fluid Levels In Tubing
2. Wellhead Pressure Rating
3. Formation Pressure
4. Formation Fracture Pressure
4. Casing & Tubing Sizes
5. Casing & Tubing Strengths
6. Injectivity Pressure
7. Maximum Allowable Surface Pressure (MASP)

# PUMP EQUIPMENT

1. Pump
2. Surface Pump Lines
3. Choke Manifold
4. Isolation Valves
5. Pressure Gauges
6. Fluid Disposal System
7. Mixing Tanks
8. Reserve Tanks
9. Fluid
10. Chemicals



# Chokes



Fixed chokes

For accuracy

Adjustable chokes

For simplicity

Both on the return line

# BULL HEADING

Pump kill fluid thru tubing and squeeze influx into formation , We use this method in EMERGENCY

Must apply injectivity test

**Advantages:**

- Quick & easy methods especially in smaller tubing
- No hydrocarbons brought to surface.

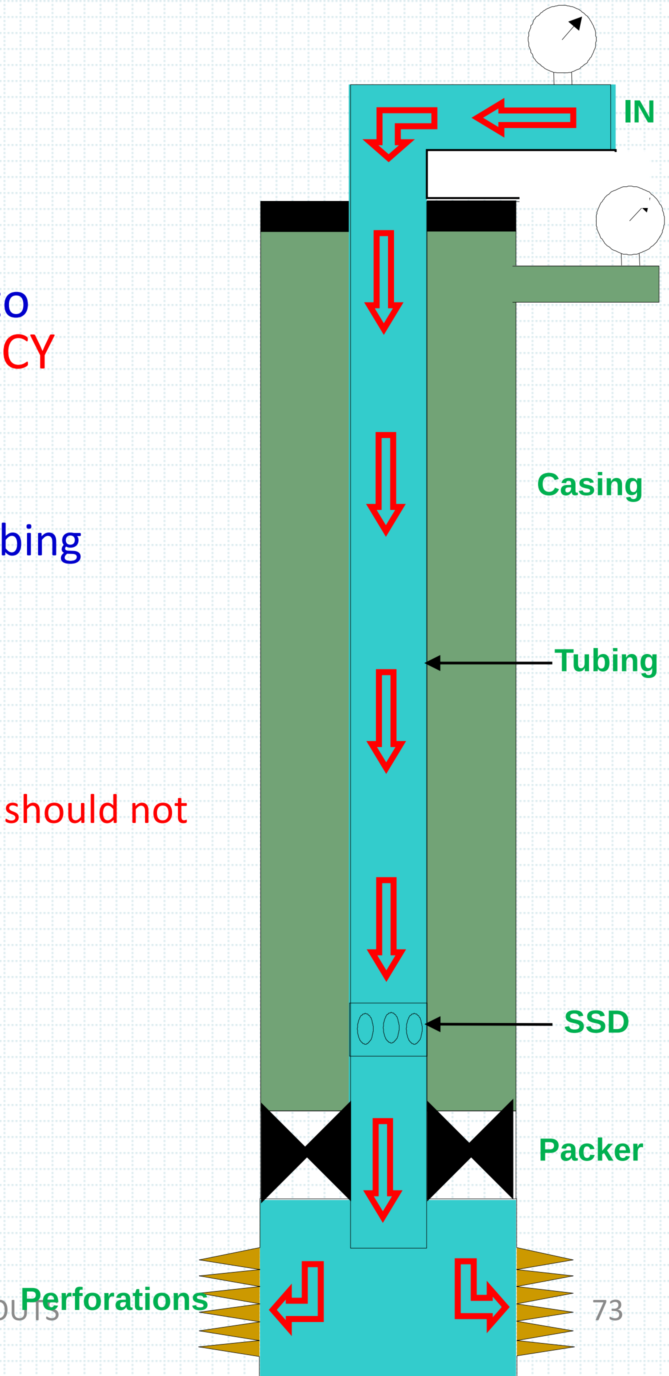
**Limitations**

1- Formation fracture pressure (While squeezing we should not exceed it to avoid formation fracture)

2- Tubing burst pressure

3- Max. pressure of surface equipment

4- Formation damage

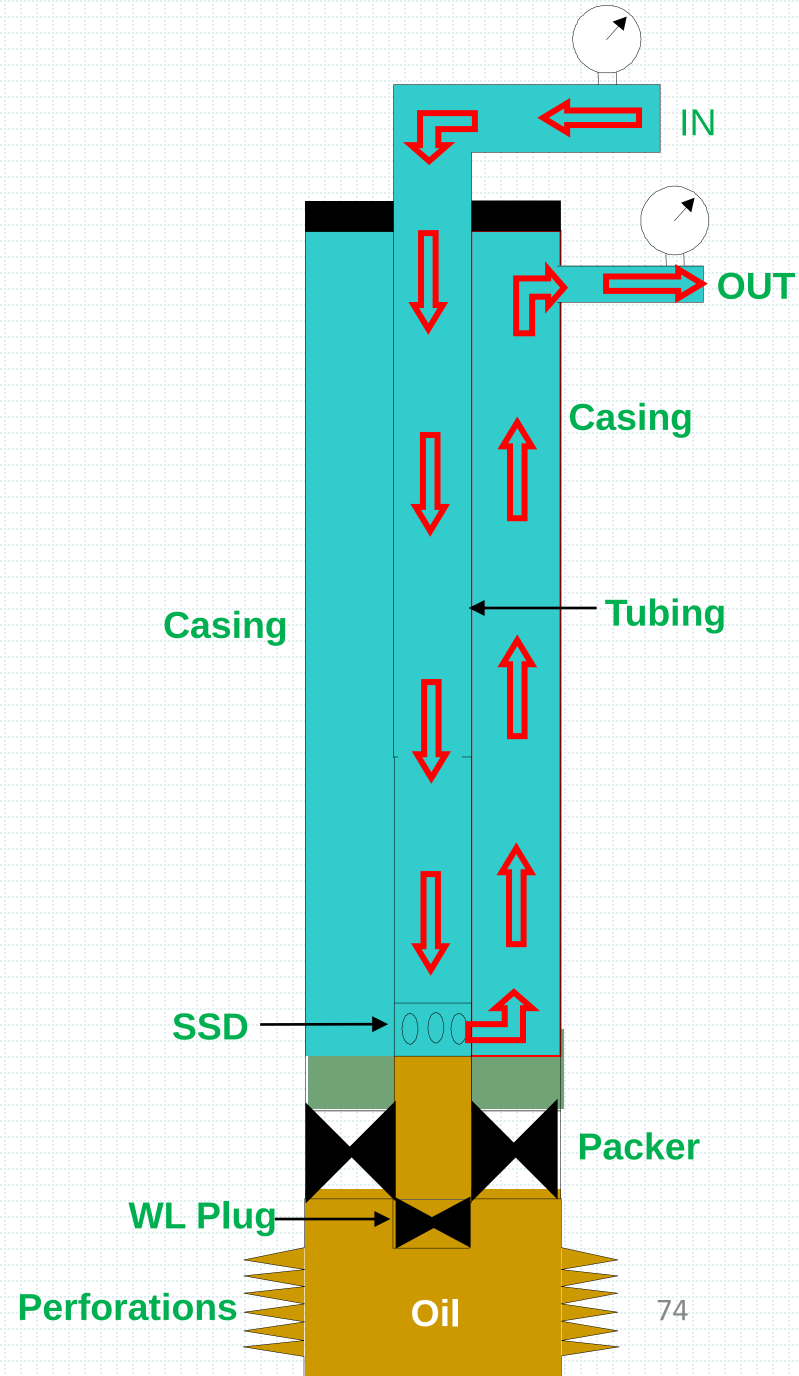


# FORWARD CERCULATION

Pump fluid thru tubing & return thru Annulus .

Disadvantages:

- Large volume pumped ( Tubing + Annulus )
- High surface pressure
- Influx will be beneath completion fluid

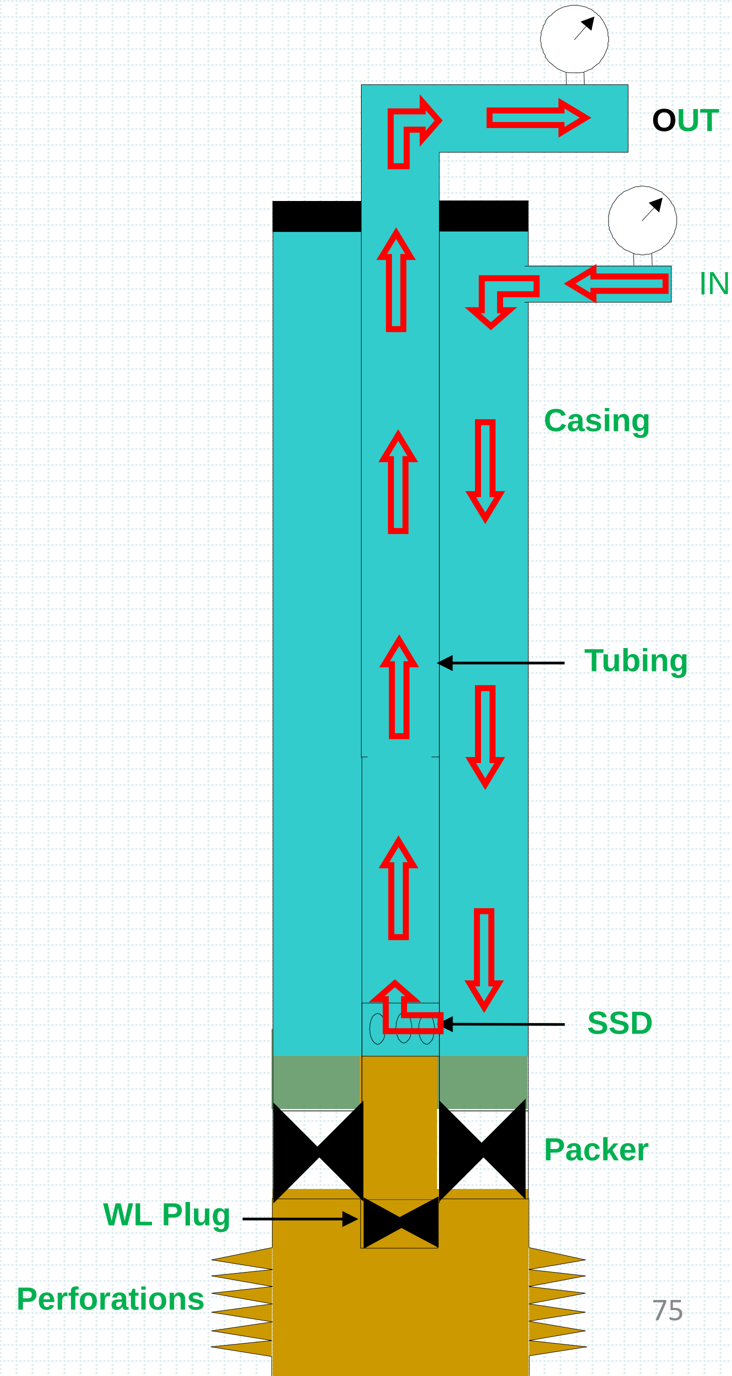


# REVERSE CIRCULATION

Pump fluid thru casing & return thru tubing.

Advantages:

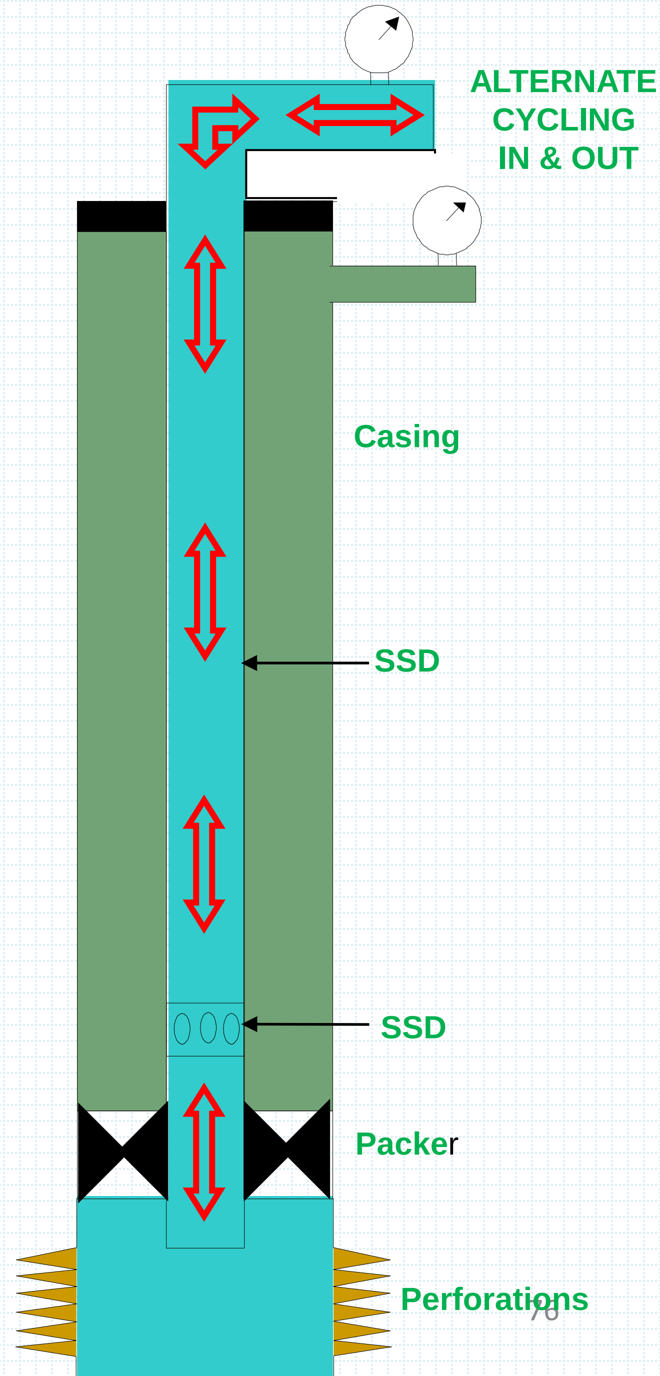
- Small volume pumped ( Tubing)
- Low surface pressure
- Influx will not enter Annulus
- Fast and PLANNED



# LUBRICATE & BLEED

Pump a small amount of kill fluid & bleed a small amount of gas while keeping constant BHP until well is dead.

- It is the alternative method of Bull Heading
- It is a slow process
- It is a well kill method that replaces the light fluid with a heavier fluid within the same volume



# OPERATIONAL CHALLENGES

Hydrates will only form if there is free water present in a system.

## Hydrates

Crystalline water structures filled with small molecules.

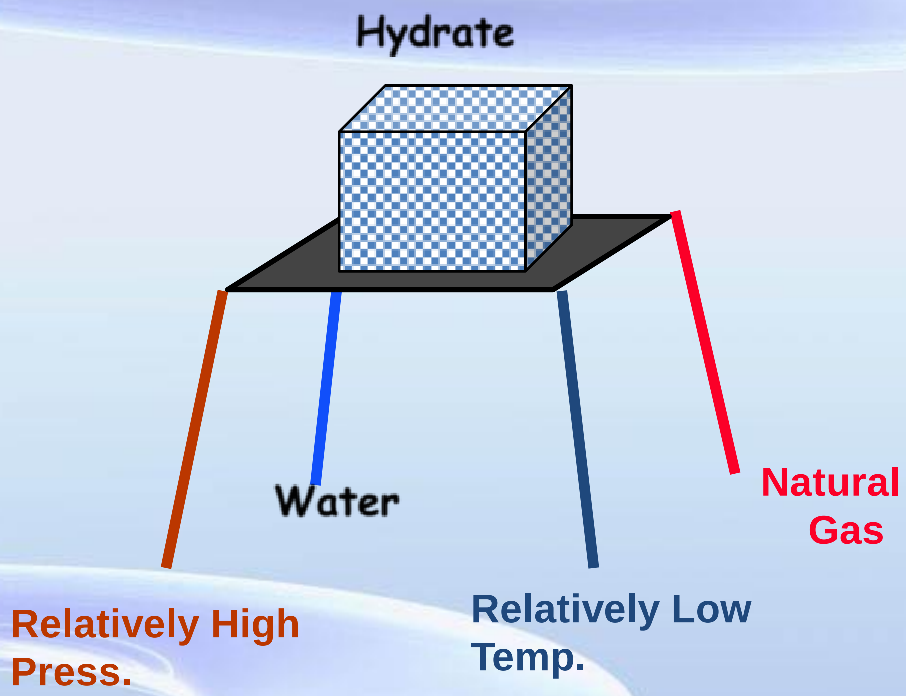
- Presence of free water
- Presence of light gas molecules
- Relatively high pressure
- Relatively low temperature

### Removing Hydrates

- Reduce pressure

  (*Risk: hydrate plug slips & gas above it, the plug may blow up).*
- Raise temperature
- Use inhibitor (*glycol, methanol or salt*).

# Elements Necessary for Hydrate Formation



# Hydrates

## Problems associated with Hydrates

1- Loose of production

2- Unable to RIH or POOH

3- Unable to circulate

4- Misinterpreting barriers

5- Blocking the whole system

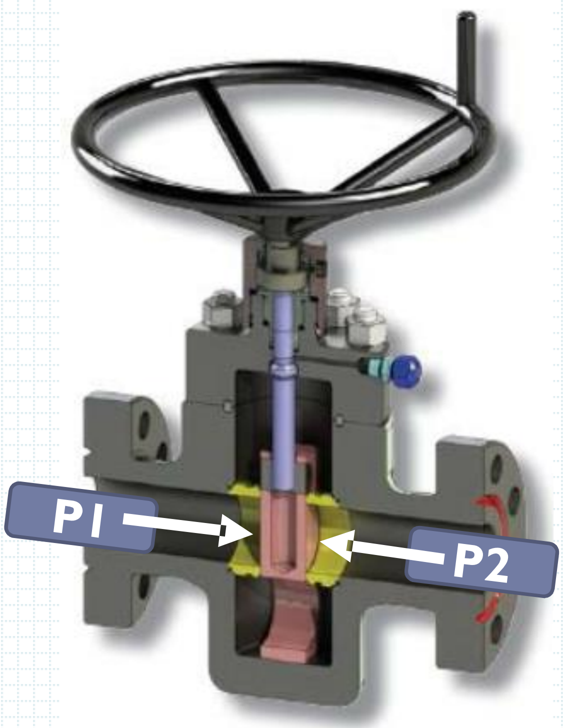
6- Damage down stream equipment if the removal is not controlled

Gulf of Mexico is a clear example of loosing Millions of US dollars due to the formation of Hydrates during the curing process of this accident

# Mechanical Barriers – Freeze Plug Method

Placing temporary barrier ,cryogenic ice plug, using liquid $N^2$ , methanol, liquid $CO^2$ or $CO^2$ dry pellets, within a flexible wrap, canister or using coil lines. Temperatures are then lowered to -80°F and a viscous gel pill is used as the 'freeze plug' or the wells hydrocarbons and water creates a ***hydrate plug.***

# Equalizing Pressure Across  Gate Valves




**P1 >>P2**

During Integrity Testing and during Well Intervention operations, Xmas Tree valves will be closed and opened repeatedly.

As a result of the ‘surge effect’, *opening* the valve under pressure can:

- Damage other equipment or tools downstream of this valve
- Damage the valve itself

Therefore we ***must*** equalize any pressure differential across valves, before we open any valve by at first “cracking” it slightly open to allow equalization.

If the valve cannot be cracked open, ***do not use*** a wrench, since this may shear the valve safety pin and make the valve completely inoperable

Use a pump to equalize across the valve before attempting to open again.

# PRESSURE & VOLUME

# Formation pore Pressure

Formation pore pressure is defined as the pressure exerted by the formation fluids on the walls of the rock pores.

Formation pore pressure changes by

* Well depletion

* Injection

# Formation Fracture pressure

Maximum pressure applied to the formation before it breaks down

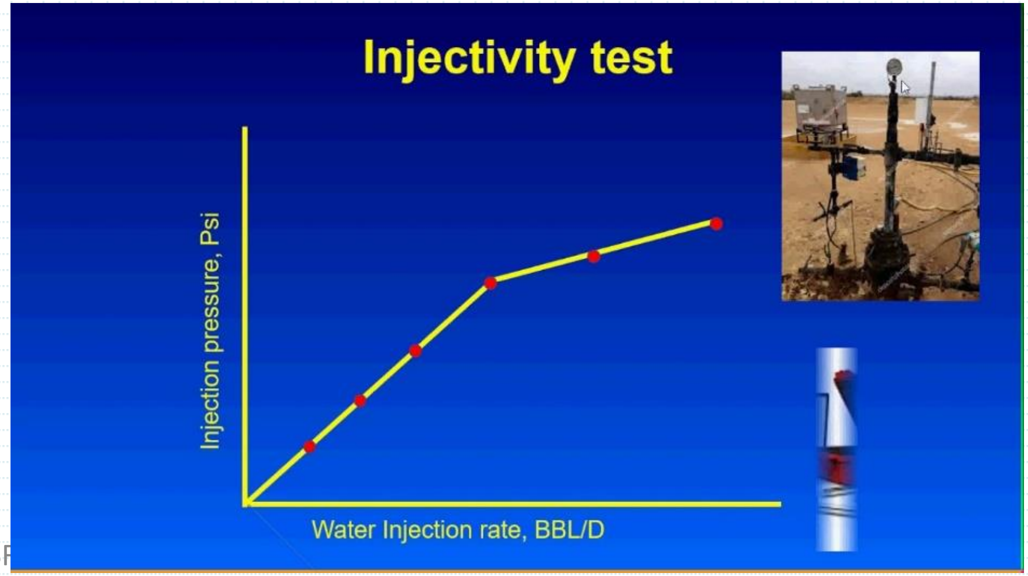
If pumping pressure exceeds formation fracture pressure , formation will be fractured and we loose production

# Formation Injectivity test

A procedure conducted to establish the rate and pressure at which fluids can be pumped into the treatment target without fracturing the formation.

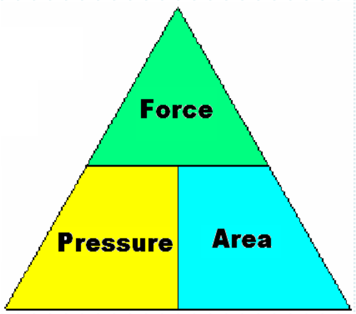
Test is a must to avoid formation fracture pressure Increasing the pumping pressure while well is closed in a small intervals and monitor flow rate

Do not exceed FFP as this will lead to loose production and forced to do side tracking to gain production from that well



# FUNDAMENTALS OF FLUIDS & PRESSURE

Pressure is defined as Force per unit area exerted by fluid i.e.

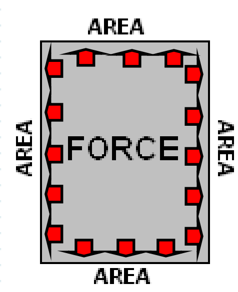



**Pressure = Force ÷ Area**

# FUNDAMENTAL OF FLUIDS & PRESSURE

**Basic laws of physics concerning fluids & gas**

- Liquids are not compressible
- Gases are compressible
- Liquids & gases are both **FLUIDS**
- Pressure in fluid is transmitted equally in all directions.



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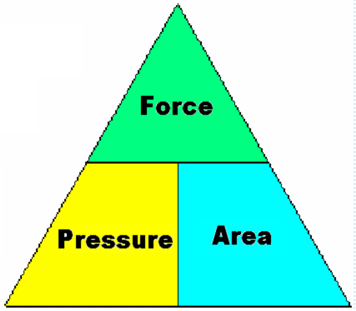



## Pressure is classified into 3:

1 **Normal Pressure**

- *Only due to hydrostatic pressure of fluids.*

2 **Abnormal Pressure**

- *Due to formation compaction, water squeezing from surrounding clays/shale, thermal expansion, earth movements etc.*

3 **Subnormal Pressure**

- *Less than normal pressure due to incompact formation (e.g. Loss circulation zones).*

# PROPERTIES OF GAS

- Gases differ from solids and liquids in the following ways:
- A sample of gas assumes both the shape and volume of the container.
- Gases are compressible.
- The densities of gases are much smaller than those of liquids and solids and are highly variable depending on temperature and pressure.
- Gases form homogeneous mixtures (solutions) with one another in any proportion.

# THE GAS LAWS

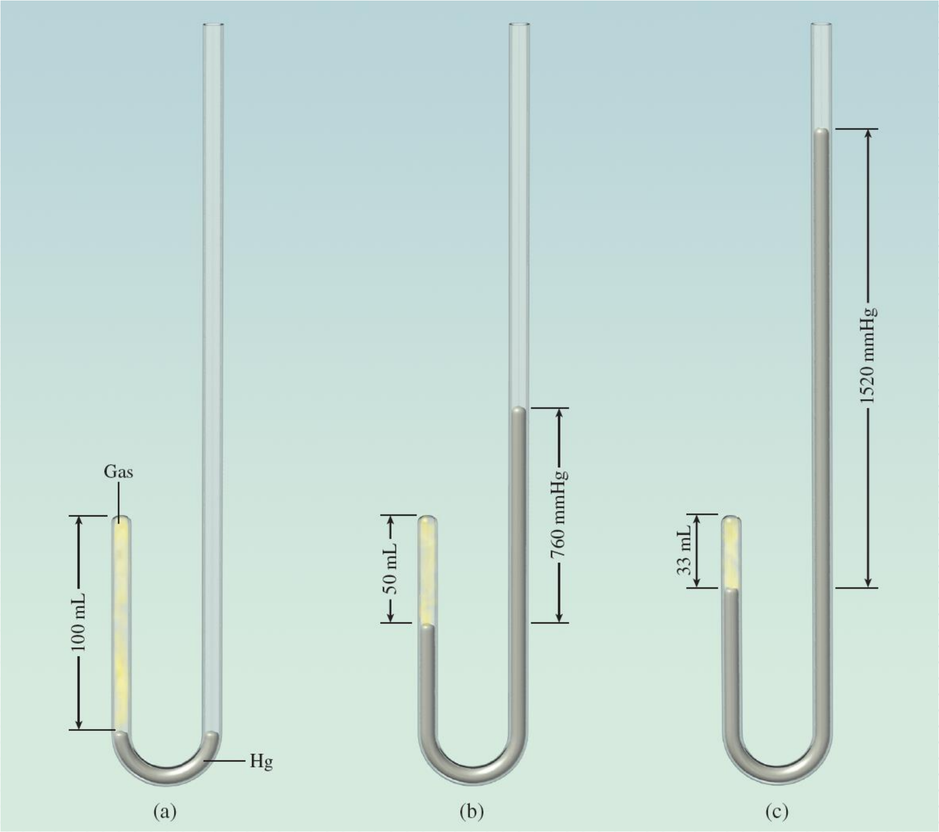
***Boyle's law*** states that the pressure of a fixed amount of gas at a constant temperature is inversely proportional to the volume of the gas.

|  | (a) | (b) | (c) |
|---|---|---|---|
| $P$ (mmHg) | 760 | 1520 | 2280 |
| $V$ (mL) | 100 | 50 | 33 |

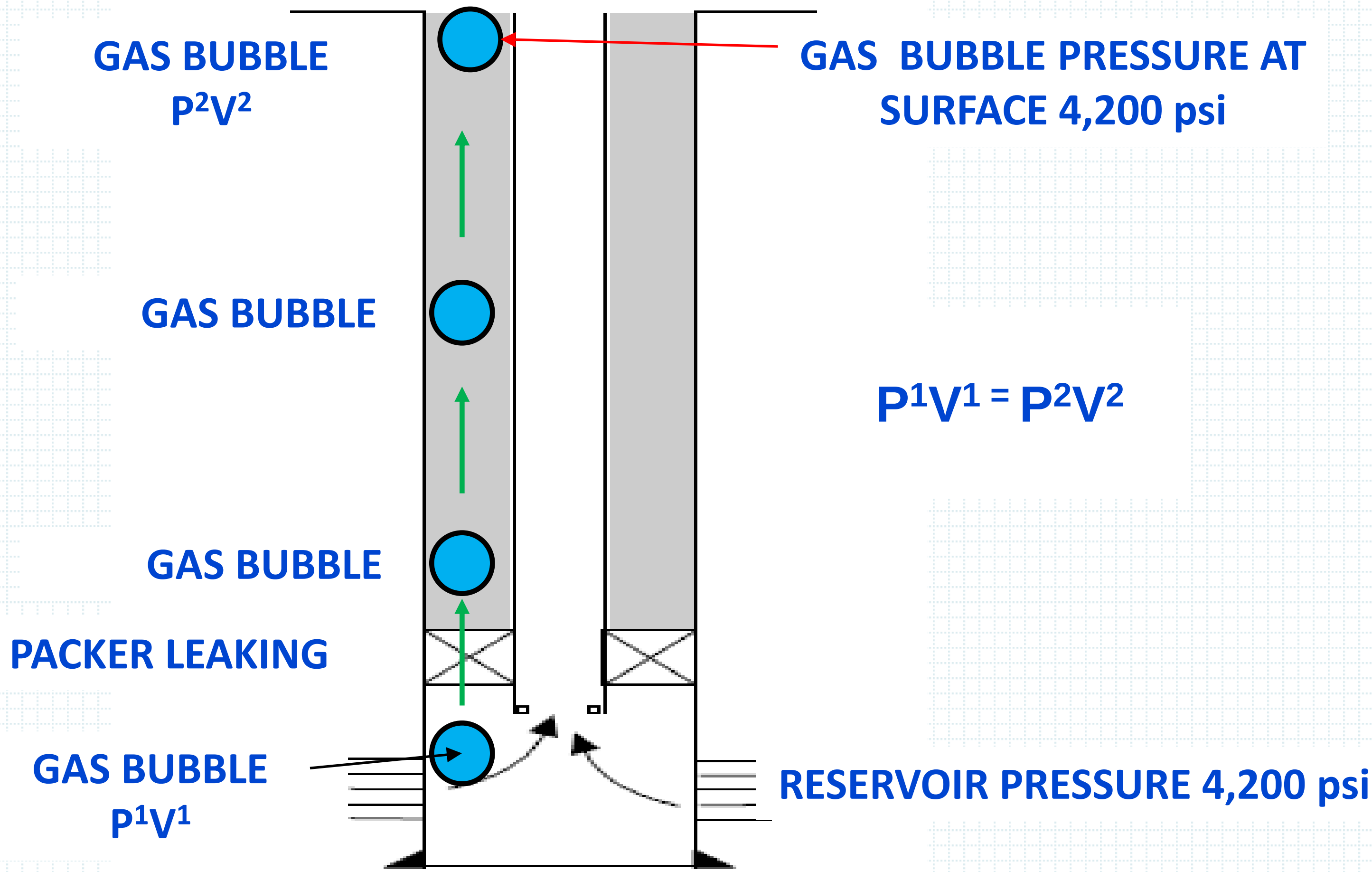
$$V \propto \frac{1}{P}$$

$$P_1V_1 = P_2V_2$$

**at constant temperature**



# GAS MIGRATING TO SURFACE WITHOUT BEING ALLOWED TO EXPAND

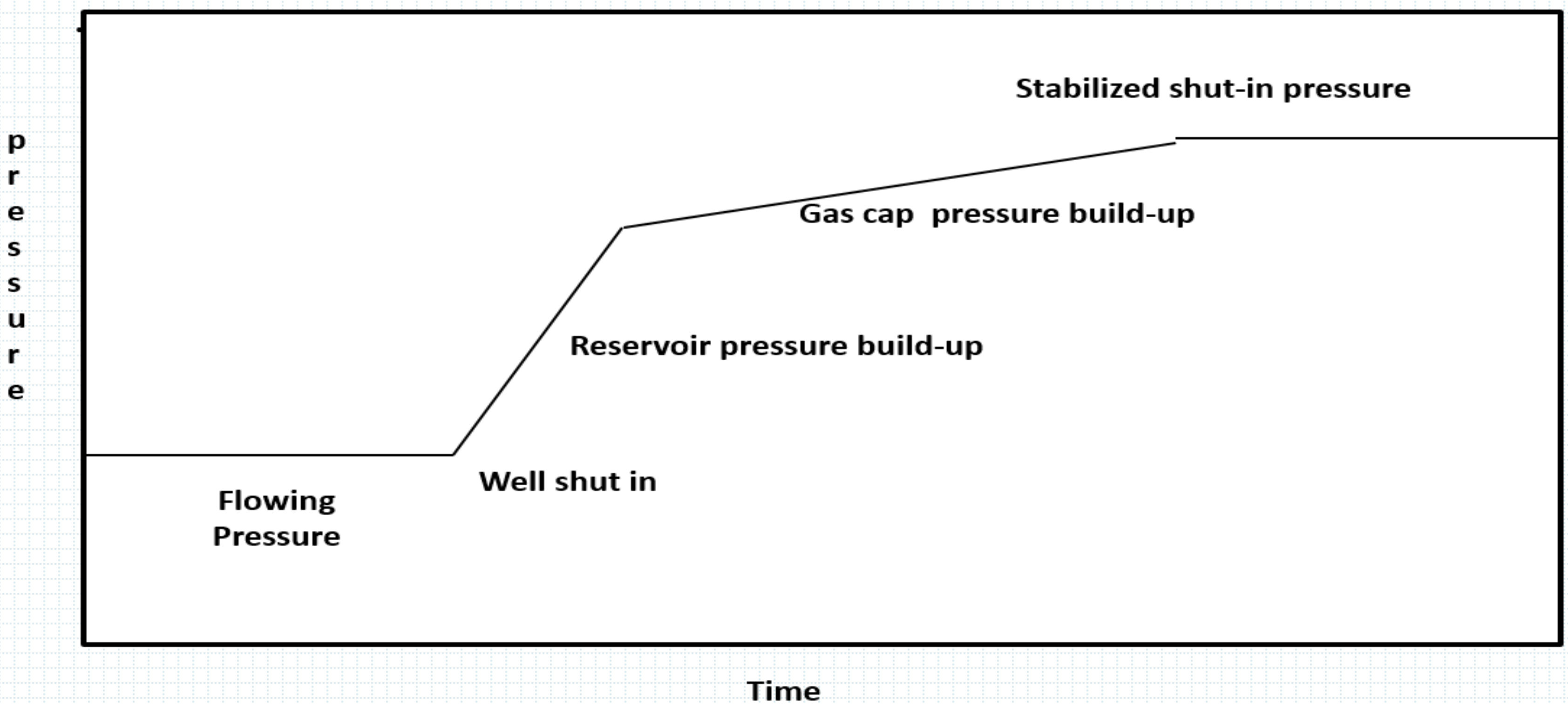


## Oil well shut in pressures

The below diagram shows the surface pressures, when an oil well is shut in. When the well is shut in the reservoir pressure initially causes a rapid pressure increase .

Most produced oil has gas in solution. Over time this gas migrates to surface forming a gas cap causing the shut-in pressure to increase until it stabilizes.

The shut- in build-up pressures is monitored and recorded over the life of the well. It can indicate dropping reservoir pressure, formation damage, “skin” and increasing “water cut”.



# HYDROSTATIC PRESSURE (HP)

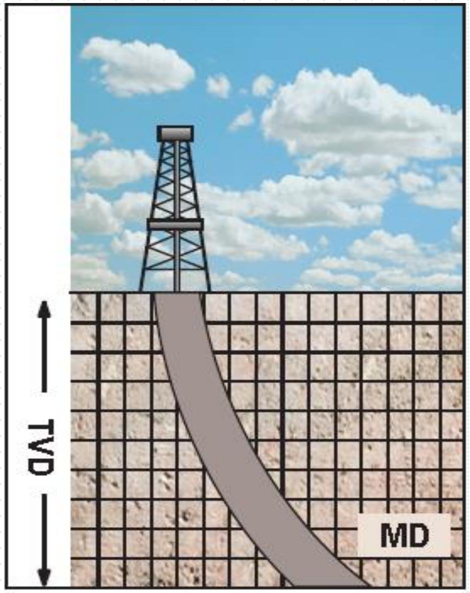
HP is the pressure developed by column of fluid at given TVD. “Hydro” means fluid & “static” means stationary. ∴ HP is a pressure created by vertical stationary column of fluid.

**HP = TVD X D X 0.052**

E.g. Find HP in psi of fresh water at 500 ft TVD.

HP = 500 ft × 8.33 ppg x 0.052

= 216.5 psi




**PRESSURE GRADIANT**

It is HP/FT

**HP = TVD X PG**

**PG= DX0.052**

= Force *(gal/ft³)* ÷ Area *(ins²)*

=7.48 / 144 = 0.052

To find BHA of 2 fluid column or more simply add the HP of the fluid column of the same unit of measurement together.

# HP calculations

- There is a margin of 1000 psi before reaching the annulus limitation. How much can the density of the tubing fluid be decreased before reaching the limit if the packer is set a 15000 feet MD, 13500 feet TVD?

HP = TVD X D X 0.052

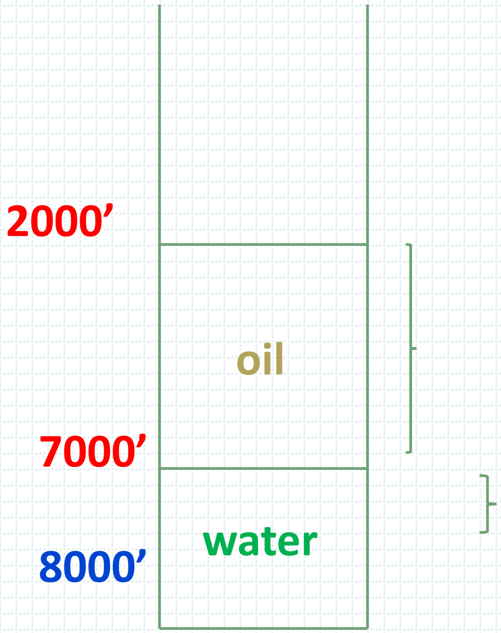
D = HP/TVD X 0.052

= 1000/13500 X 0,052 = 1.425 PPG

# example

Water Density 8.33 ppg
Oil density 8.00 ppg
Oil level 2600 MD & 2000 TVD
Water level 7890 MD & 7000 TVD
NEED HP @ 8000’
HP= Water HP+ Oil HP

HP= (1000x8.33x0.052) + ( 5000x8.00x0.052)



# Well Pressure

| | | |
|---|---|---|
| Well Data: | Well Depth: | 9150 ft MD (7900 ft TVD) |
| | Formation PG: | 0.570 psi/ft |
| | Gas PG: | 0.08 psi/ft |

**1 Find Well BHP:** **Formula: Formation PG x TVD**

$\therefore 0.570 \times 7900 = \mathbf{4503\ psi}$

**2 Find Maximum Surface Pressure:** **Formula: BHP – HP**

Gas HP $= 0.08 \times 7900 = 632$ psi

$\therefore$Max Surface Pressure $= 4503 - 632 = \mathbf{3871\ psi}$


**3 Working Pressure for Wellhead Equipment:**

2000 psi or 3000 psi or 5000 psi

**4 Find the minimum Kill Fluid Density:**

**Formula: Formation PG ÷ Constant = ppg**

$\therefore 0.570 \div 0.052 = \mathbf{10.95\ ppg}$

# Pumping

| | | |
|---|---|---|
| Given Data: | Casing Capacity: | 0.04049 bbl/ft |
| | Tubing Displ. Cap: | 0.00829 bbl/ft |
| | Tubing Capacity : | 0.01190 bbl/ft |
| | Pump Displacement: | 0.0899 bb/stroke |
| | Tubing Shoe @: | 9000 ft MD (7800 ft TVD) |

**1 How many strokes to displace Tubing String?**

**Formula: Volume (bbl) ÷ Pump Displacement**

∴ Volume **=** Tubing Capacity × MD ∴ 0.01190 × 9000 **=** 107.1 bbl

∴ Convert bbl into strokes **=** 107.1 ÷ 0.0899 **= 1191.3 strokes**

**2 How many strokes to displace the Entire Wellbore?**

( Casing volume – tubing displacement volume) / pump displacement

( 0.04049 - 0.00829 ) x 9000 / 0.0899 = 3223 strokes

# TOTAL VOLUME OF A WELL

1- Casing V – Tubing Displacement V

2- Tubing V + Annular V

3- Casing V-Tubing Displacement V( closed end)+Tubing V

( IN CASE OF CLOSED END )

## Pumping Time

| | | |
|---|---|---|
| Given Data: | Tubing Depth: | 8750 ft MD (8125 ft TVD) |
| | Tubing Capacity: | 0.00387 bbl/ft |
| | Annular Capacity: | 0.00970 bbl/ft |
| | Pump Rate: | 1.25 bpm |

**1 Calculate the time to pump bottoms up.**

**Formula: Annular Volume (bbl) ÷ Pump Rate**

Annular Volume **=** Annular Capacity × MD 0.00970 × 8750 **=** 84.88 bbl

Convert bbl into min **=** 84.88 ÷ 1.25 **=** **67.9 minutes**

**2 Calculate the time for complete circulation.**

**Tubing volume + Annular volume/ pump rate**

( 0.00387 + 0.00970 ) x 8750 / 1.25 = 95 minutes

- PUMP BOTTOMS UP

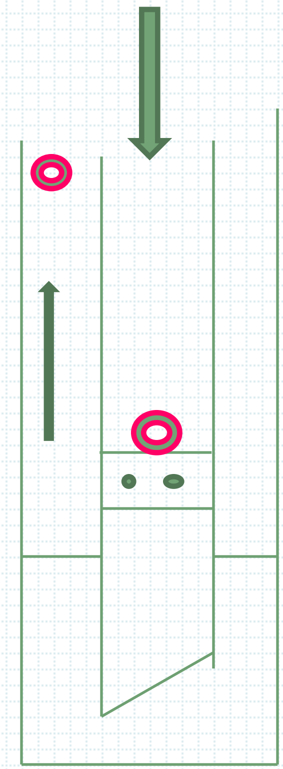
OR

- PUMP UP THE WELL

- ANNULAR VOLUME

AV 350 BBL

TV 150 BBL



# Volume

Find annular volume above packer in bbl of the following well.Table represent annular volume

Tubing: 3½", 9.2 ppf Casing: 9⅝", 47 ppf

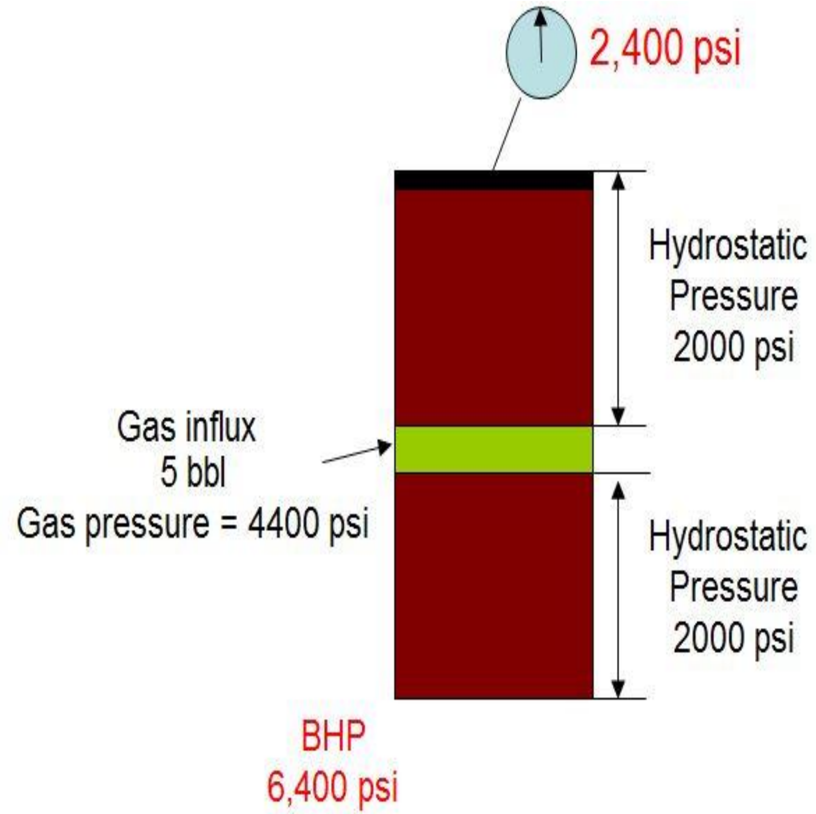
Packer: 9120 ft TVD 9750 ft MD

| Casing | ppf | $ft^3$/Lin.ft | Lin.ft/$ft^3$ | Bbl/Lin.ft | Lin.ft/Bbl |
|---|---|---|---|---|---|
| 9⅝" | 29.30 | 0.3607 | 2.7723 | 0.0642 | 15.5763 |
| 9⅝" | 36.00 | 0.3468 | 2.8835 | 0.0618 | 16.1812 |
| 9⅝" | 43.50 | 0.3308 | 3.0229 | 0.0589 | 16.9779 |
| 9⅝" | 47.00 | 0.3237 | 3.0892 | 0.0577 | 17.3310 |
| 9⅝" | 53.50 | 0.3100 | 3.2258 | 0.0552 | 18.1159 |

**Formula:**

**Bbl/ft x MD** = 0.0577 x 9750

= **562.5 bbls**




**Conclusion:**

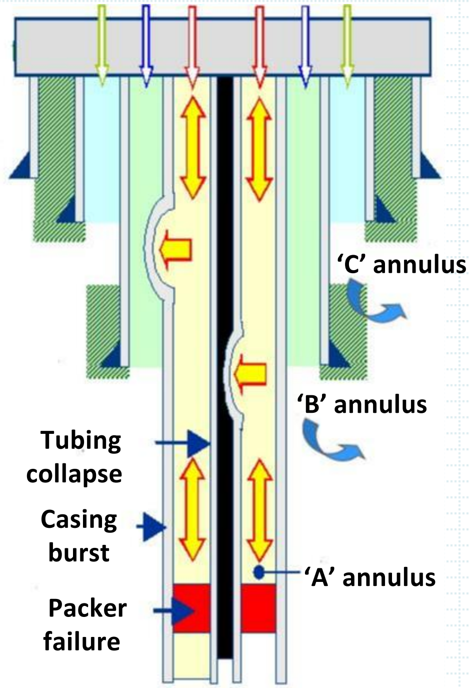
If the well is shut in and the gas influx is allowed to migrate, gas pressure will remain constant; however, **bottom hole pressure and tubing pressure will increase**.

# MAASP

The Maximum Allowable Annular Surface Pressure (MAASP) equals the formation breakdown pressure at the point under consideration minus the hydrostatic head of the mud/or influx in the casing. **During well control operations the critical point to consider is the casing shoe**

MAASP has to be reduced in value during the well life due to wear & corrosion of the pipes

# Annulus Pressure Monitoring and Build-up

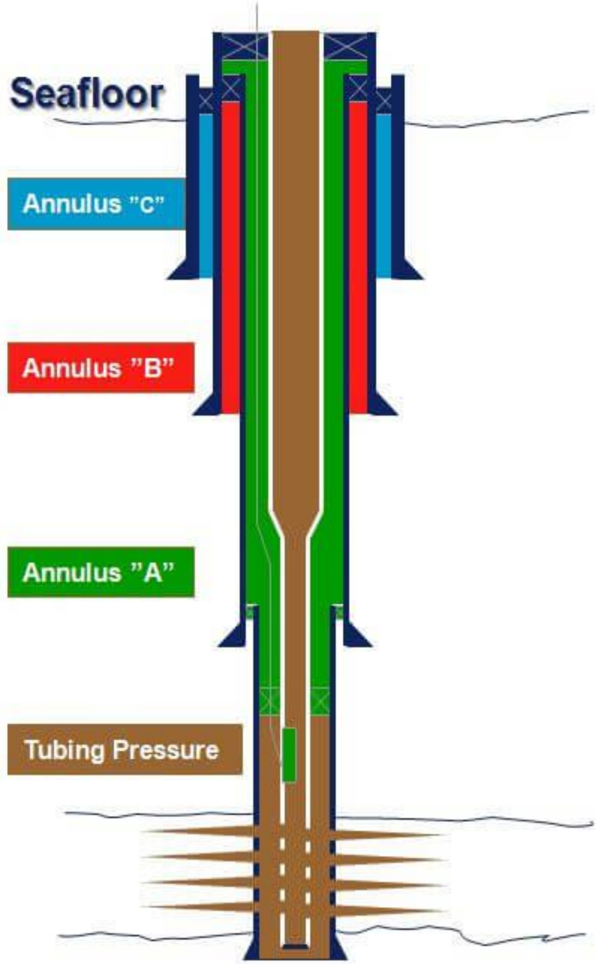


# MONITORING ANNULUS PRESSURES

The pressure in all accessible annulus (A, B and/or C) shall be closely monitored and maintained on a regular basis in all operations within minimum and maximum pressure range limits (MAASP) as defined in the completion design.

There are three main types of annular pressures encountered in wells:

A. Intentionally Applied Pressure
B. Thermally induced pressure
C. Sustained Casing Pressure



# A. Intentionally Applied Pressure (AP)

Pressure that intentionally applied by the operator for various purposes.

***<u>Examples:</u>***

Maintain a small positive pressure on the 'A' Annulus in operations & closely monitor.

**WHY???**

*<u>So that any gain or loss in annulus pressure can easily be seen</u>*.

Applied pressure in annulus and continuously monitored during a high pressure fracture stimulation.

**WHY???**

*<u>To balance the extra tubing pressure caused by the fracture stimulation.</u>*

The MAASP is the absolute maximum pressure for a given annulus that is not to be exceeded while applying pressure in annulus, since it represents the integrity limit for that annulus.

**If exceeded MAASP, the tubing may collapse or casing may burst.**

# B. Thermally Induced Pressure (TP)

Pressure that results from thermal expansion of trapped fluid in the annulus.

Monitoring annulus pressures is important for the managing TP.

***Examples:***

The annuls pressure should be closely monitored when starting up a high temperature new well to production.

**WHY???**

*As Thermal expansion of brine in annulus may increase the annulus pressure to above MAASP.*

The annulus pressure should be closely monitored after changing out the ‘A’ annulus contents for a lighter fluid.

**WHY???**

*As The new fluid will heat up and expand increasing the annulus pressure.*

***Action should be taken for SCP to Bleed pressure and record volume.***

# C- Sustained Casing Pressure (SCP)

SCP may be the result of degradation or failure of well barrier.
e.g. leak through casing or tubing, leak through cement or through

Monitoring and routines for annulus pressures will aid early detection of SCP and important for the managing SCP.

***<u>Examples:</u>***

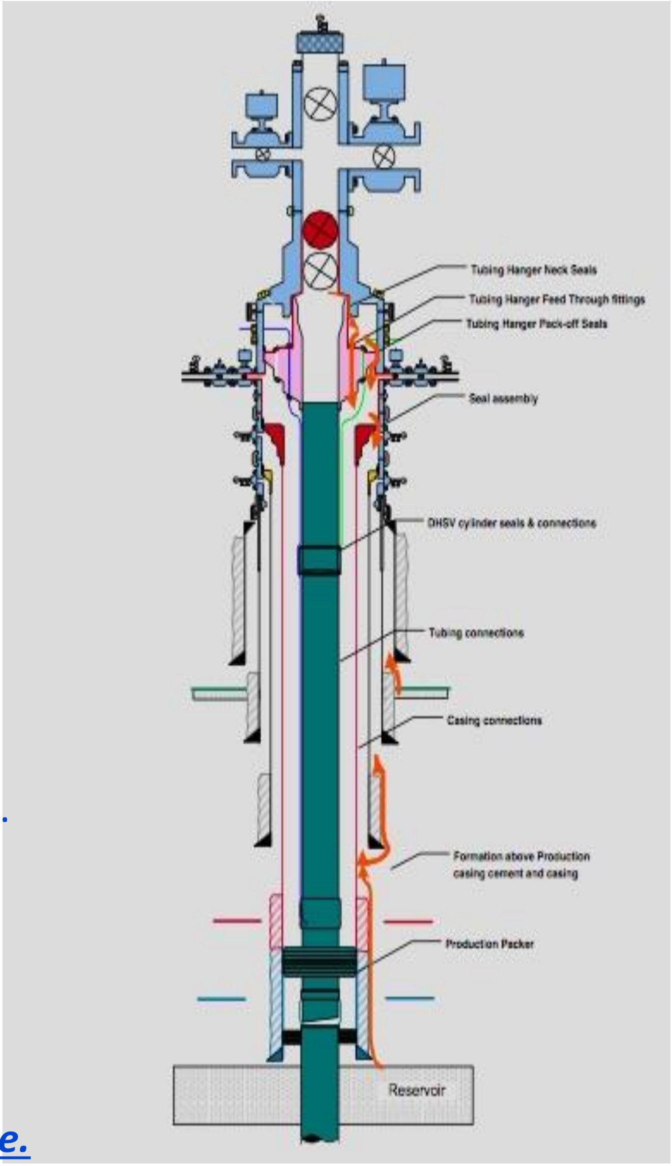
When running with full bore intervention tools through a SSD, the annulus 'A' pressure should be monitored. **WHY???**

*<u>To indicate if the tools have opened the SSD and communication achieved between ann. And tubing thru SSD.</u>*

It is important to monitor and maintain normal annuls 'B' pressure.

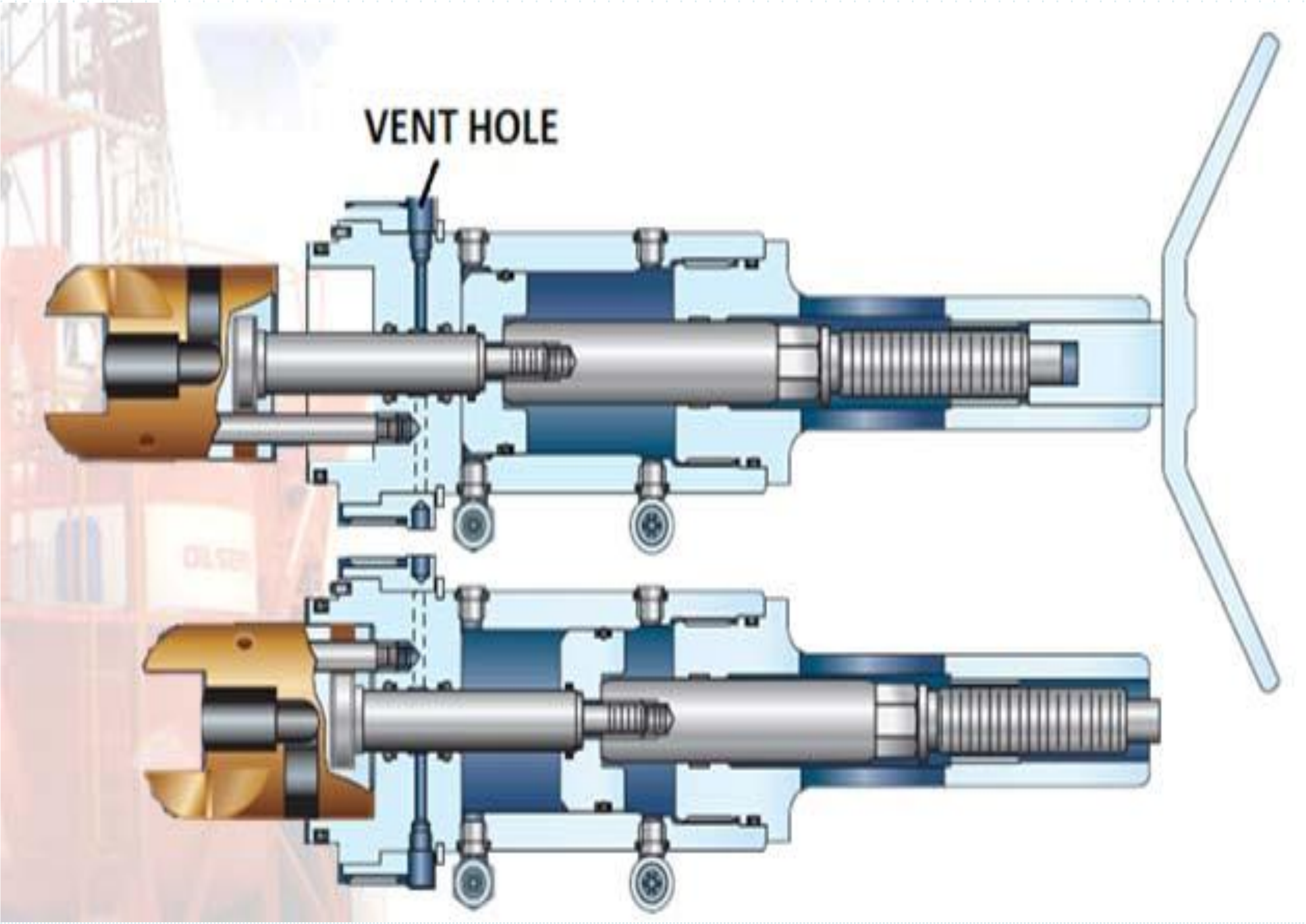
**WHY???**

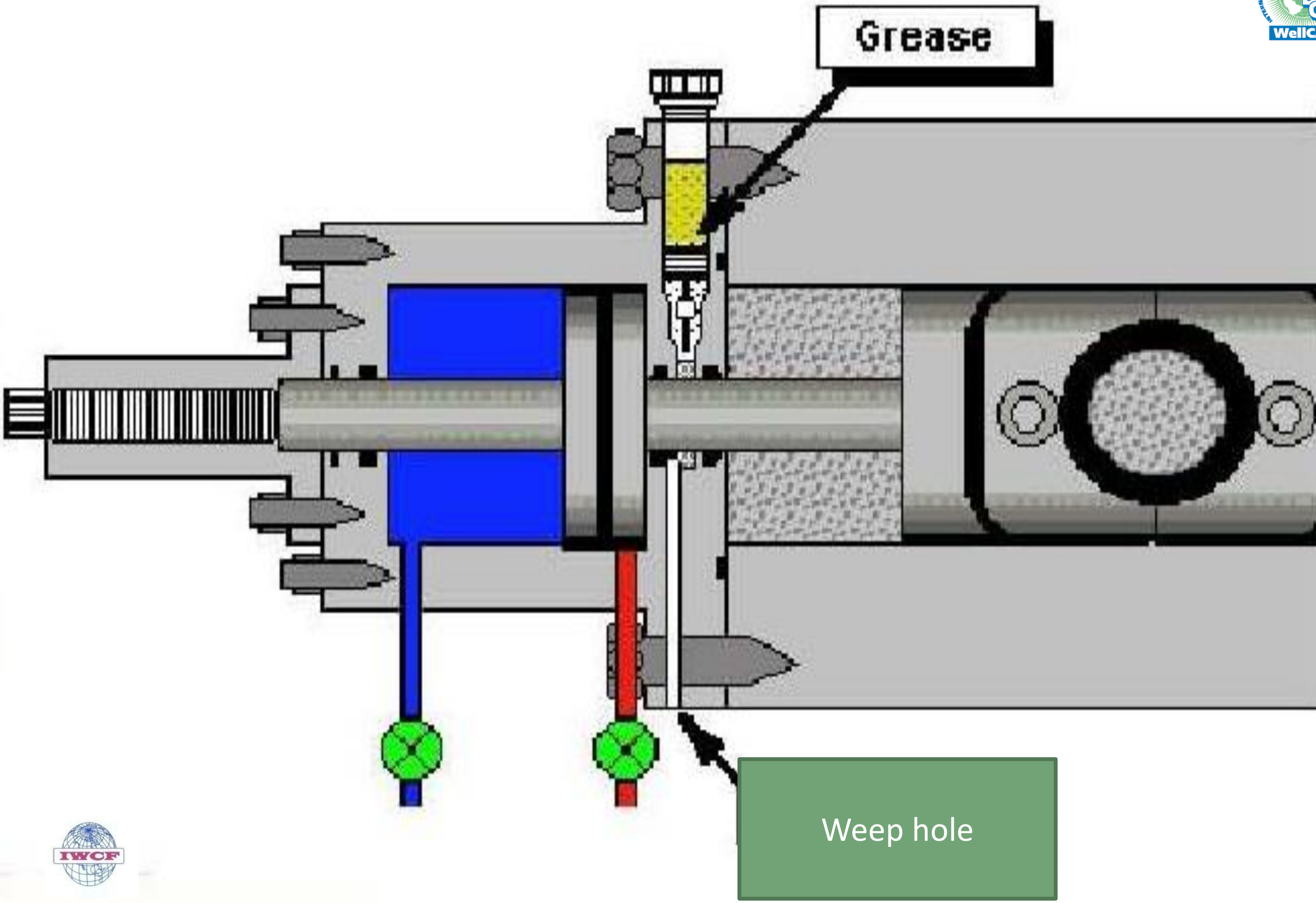
*<u>As over pressure of the annulus that build up between 2 casing due to communication with formation through cement failure may breakdown the casing shoe.</u>*



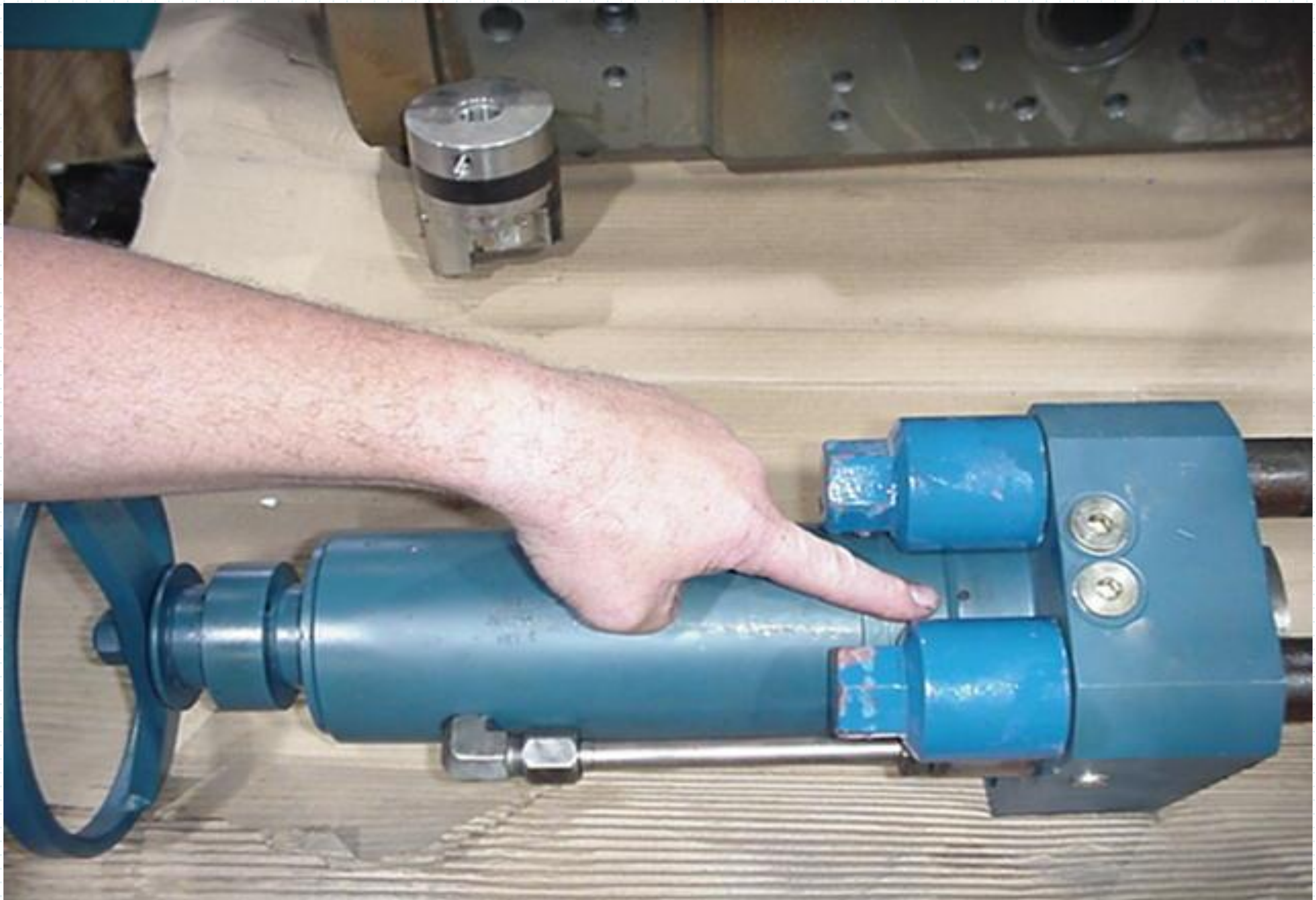

***<u>Action should be taken for SCP to Bleed pressure and record volume.</u>***

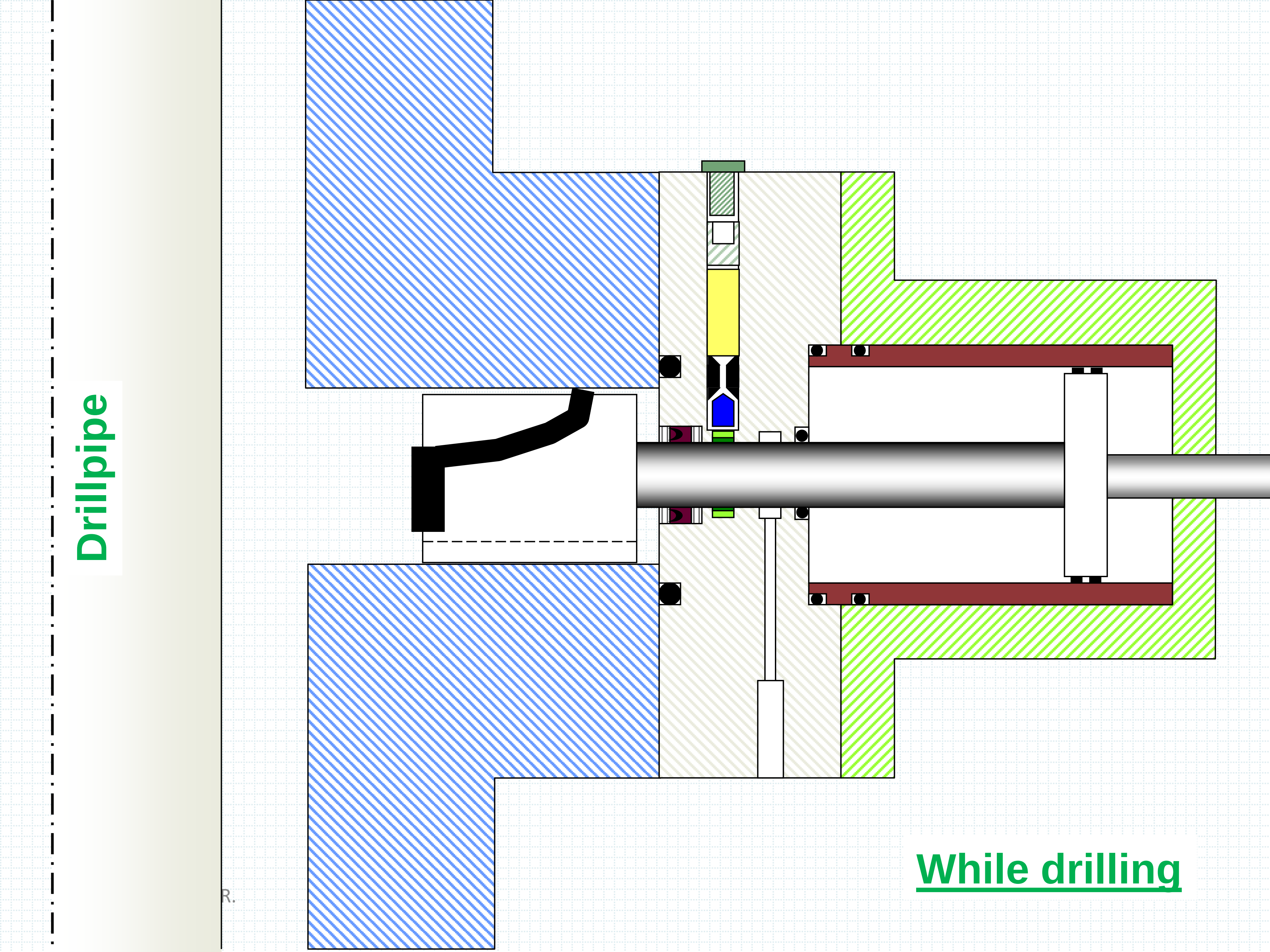
# WEEP HOLE

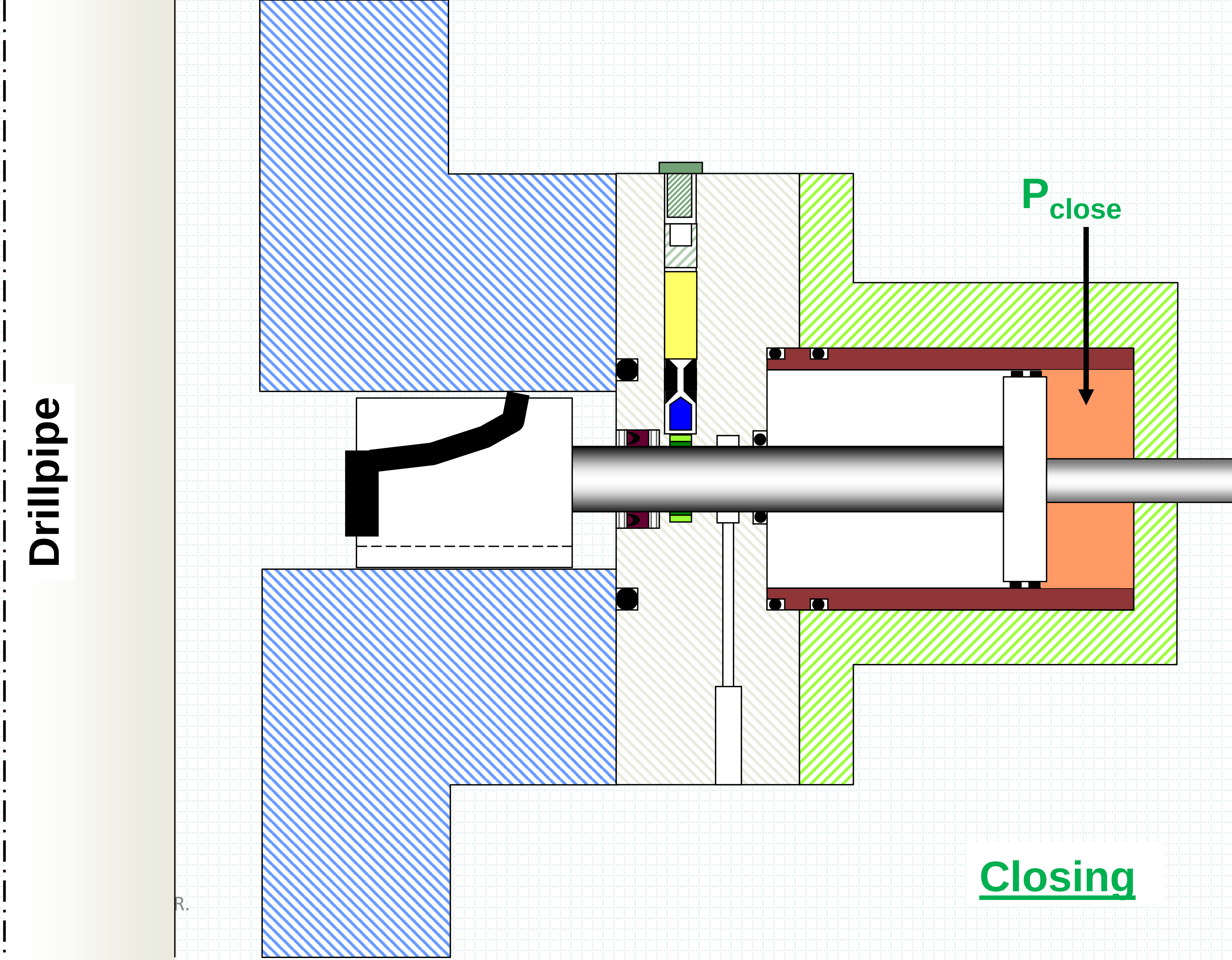



Grease
Weep hole

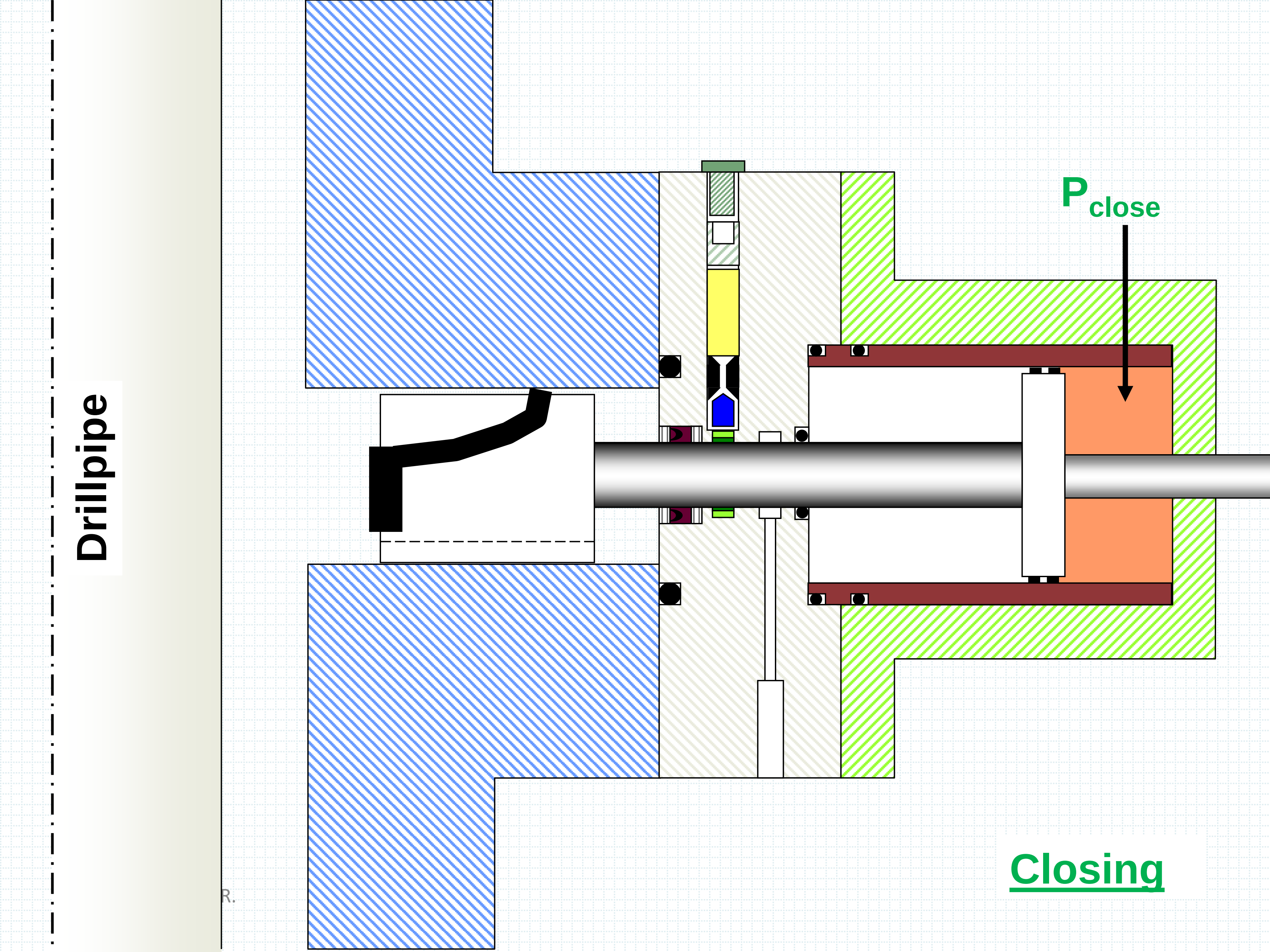
# Weep hole



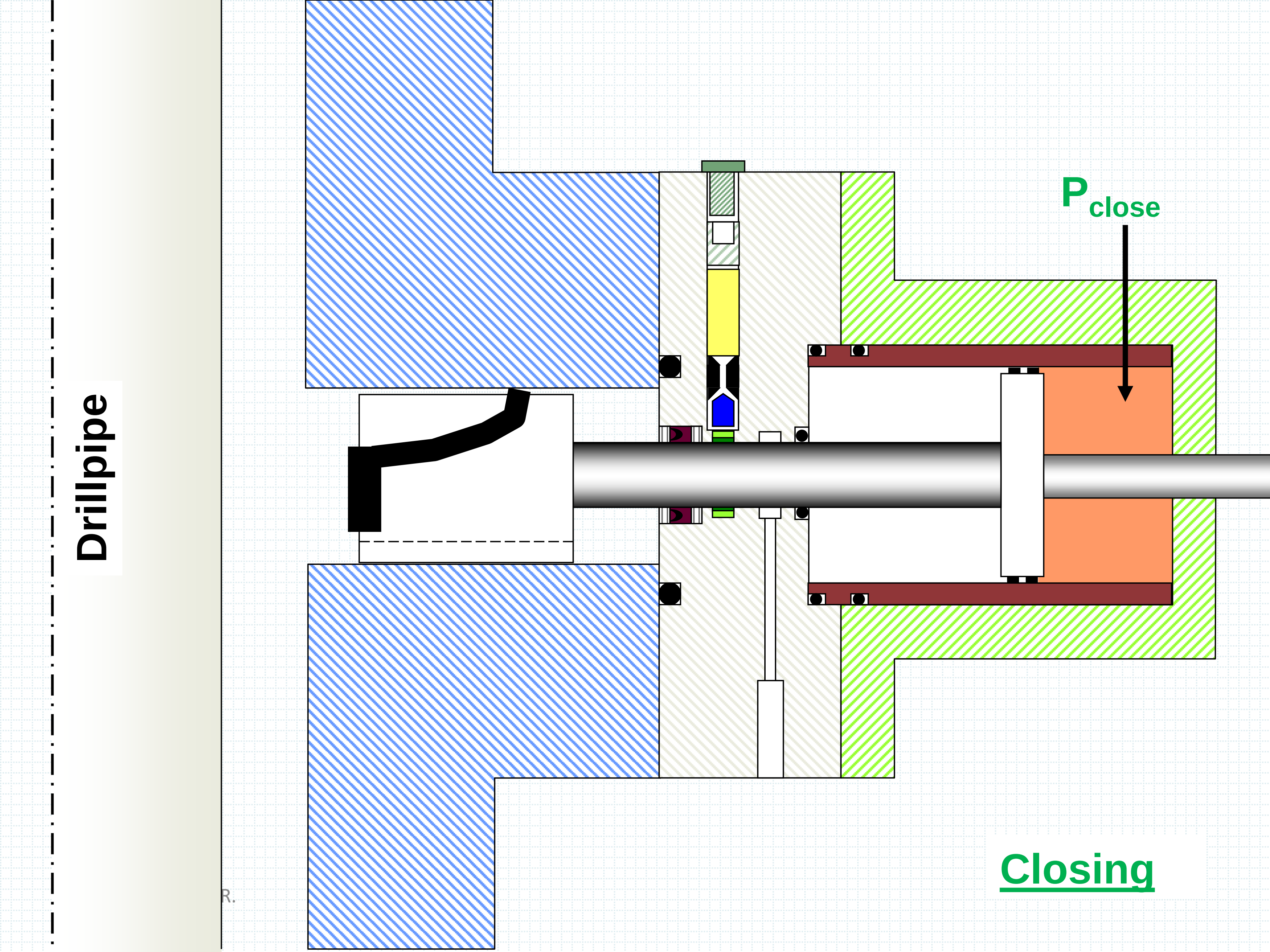

Drillpipe
While drilling



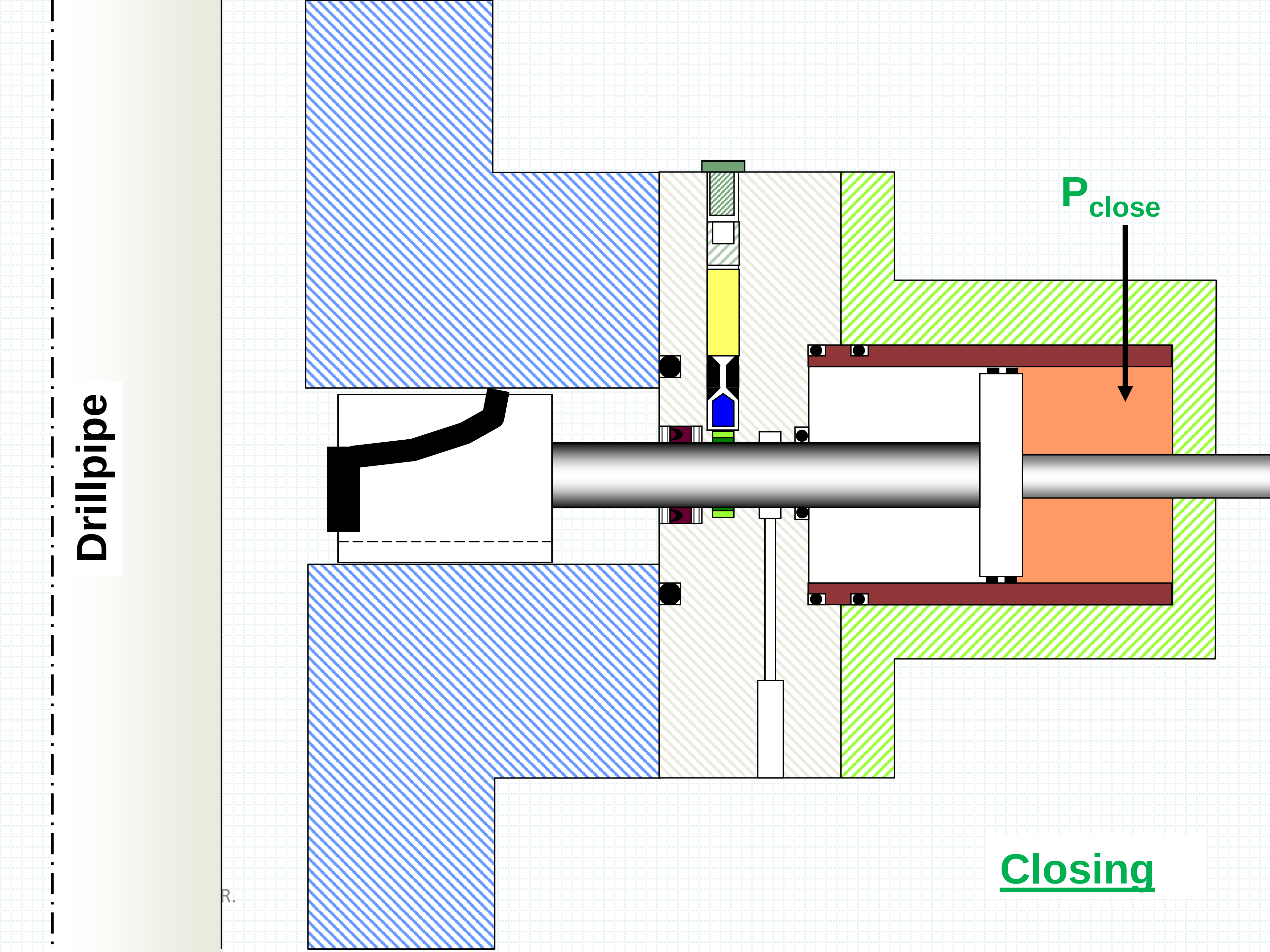

**Closing**

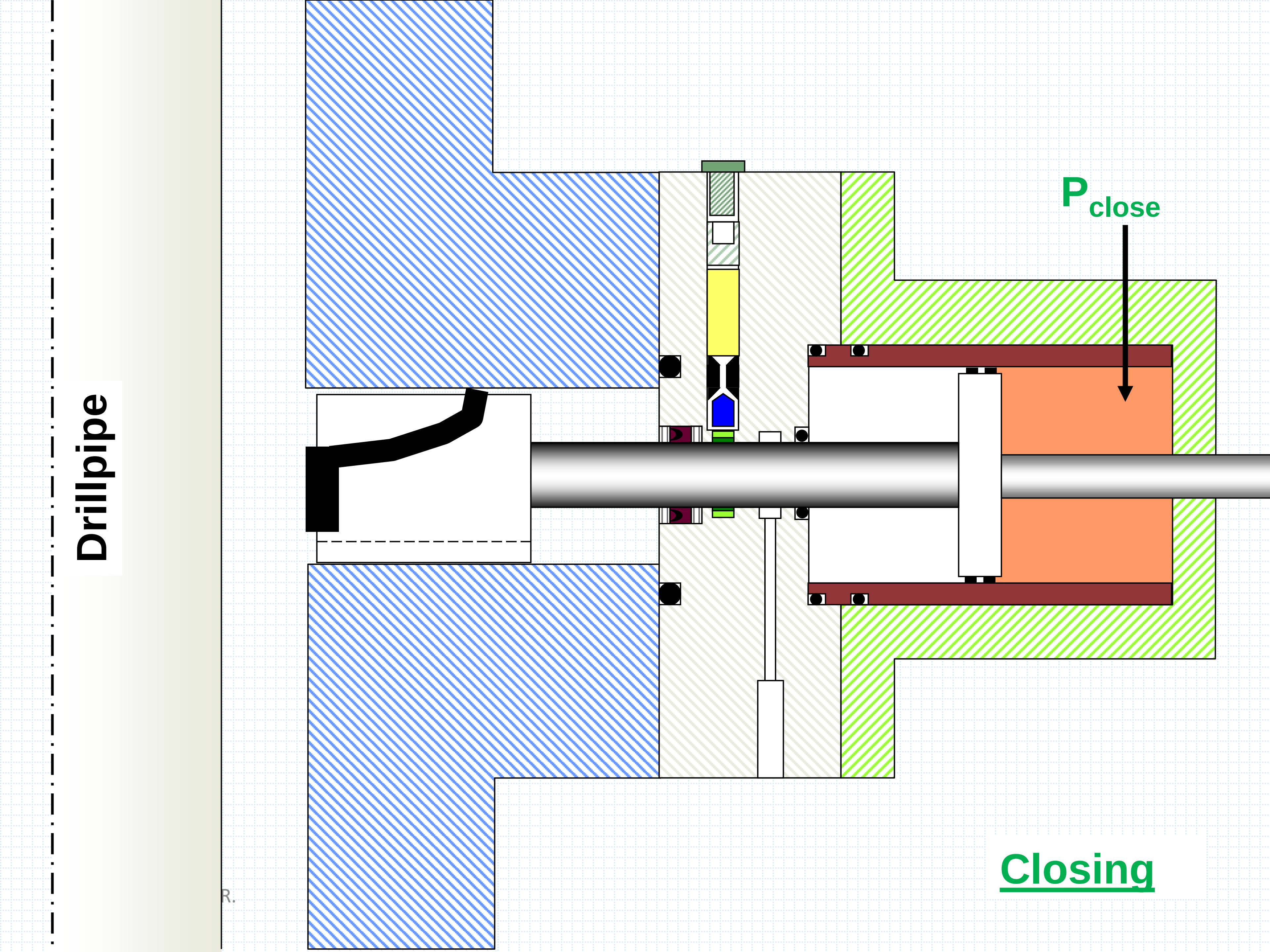


Drillpipe
P close
Closing
R.

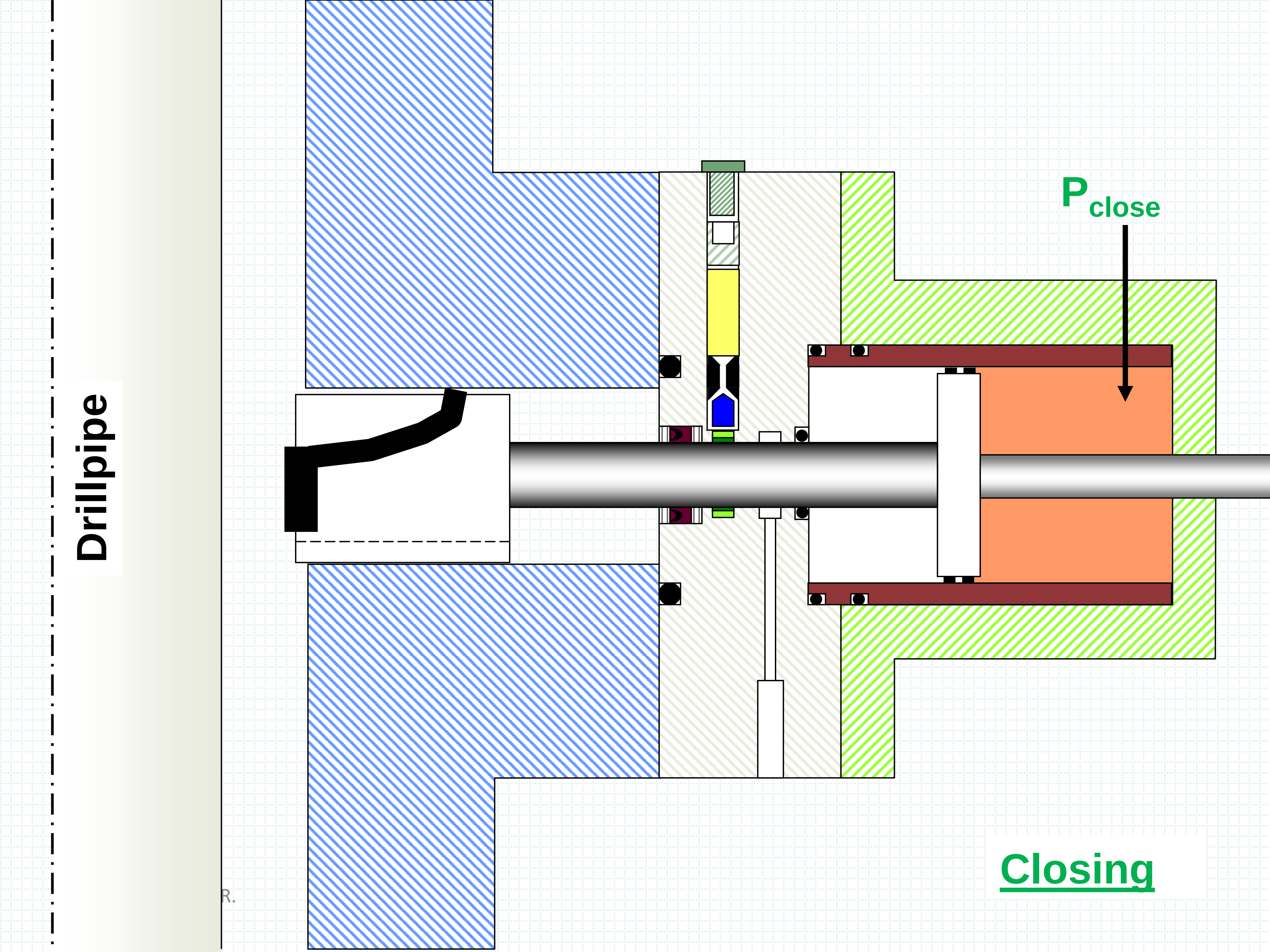


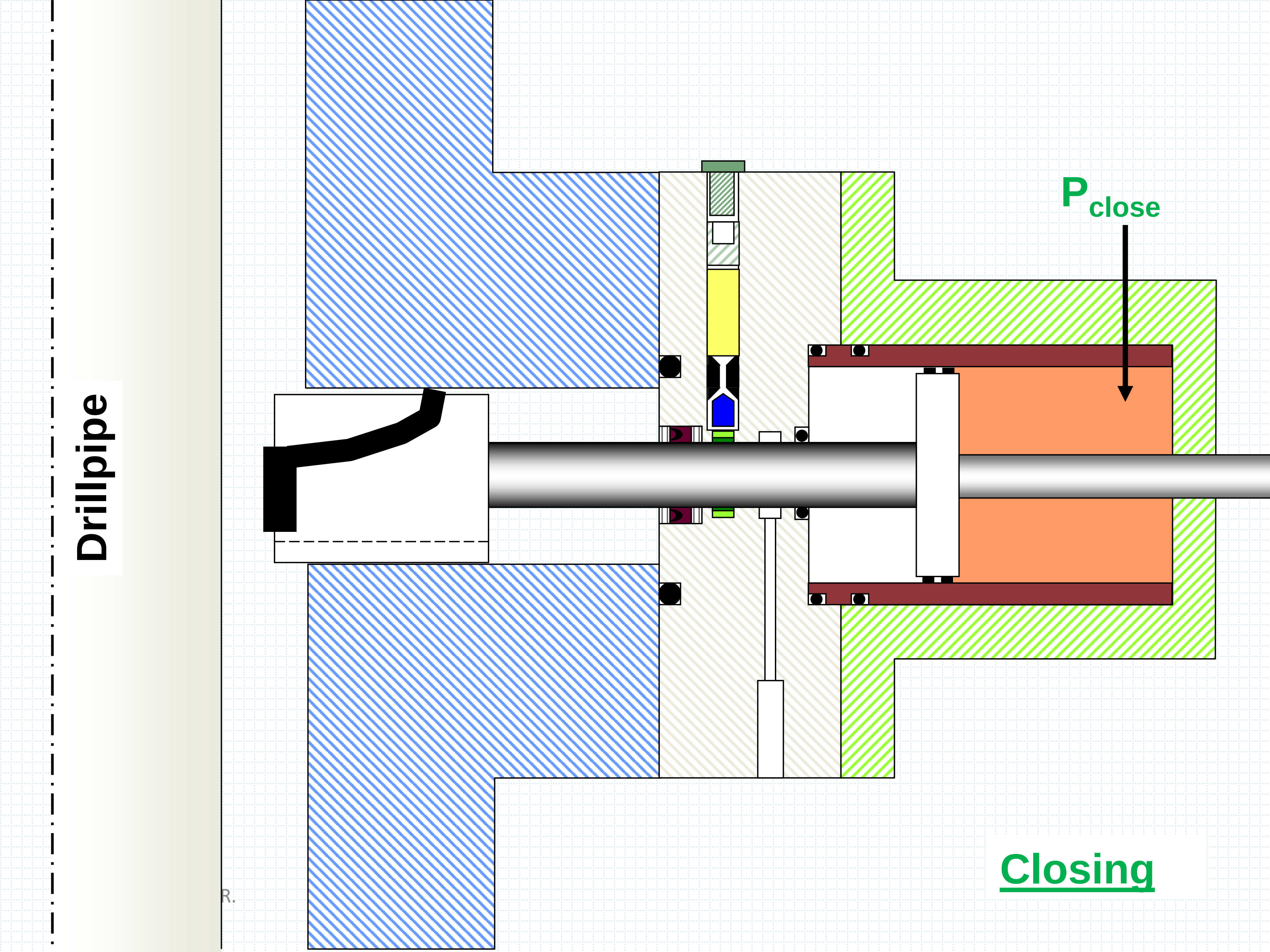
Drillpipe
$P_{close}$
Closing
R.


Drillpipe
$P_{close}$
Closing

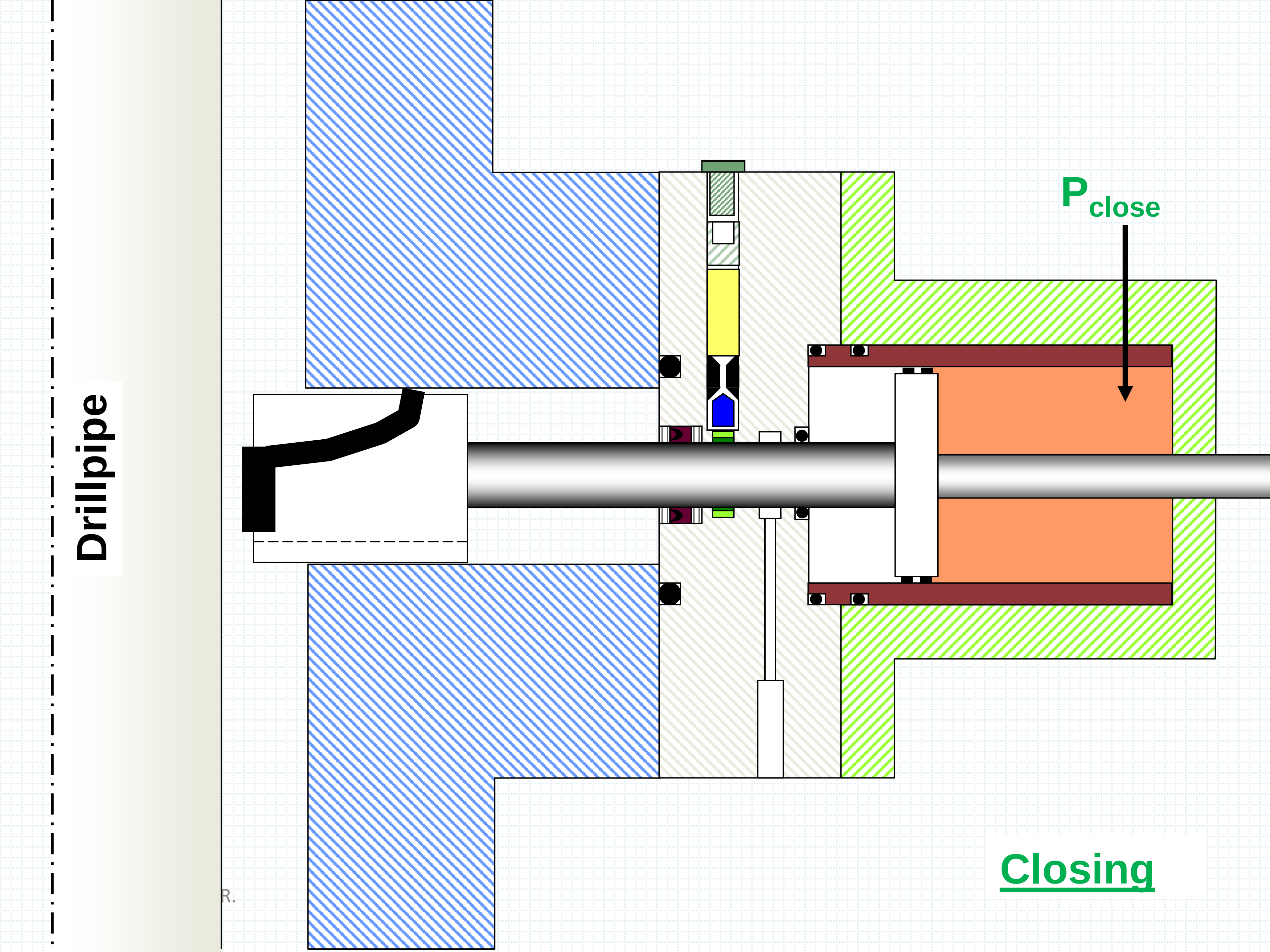


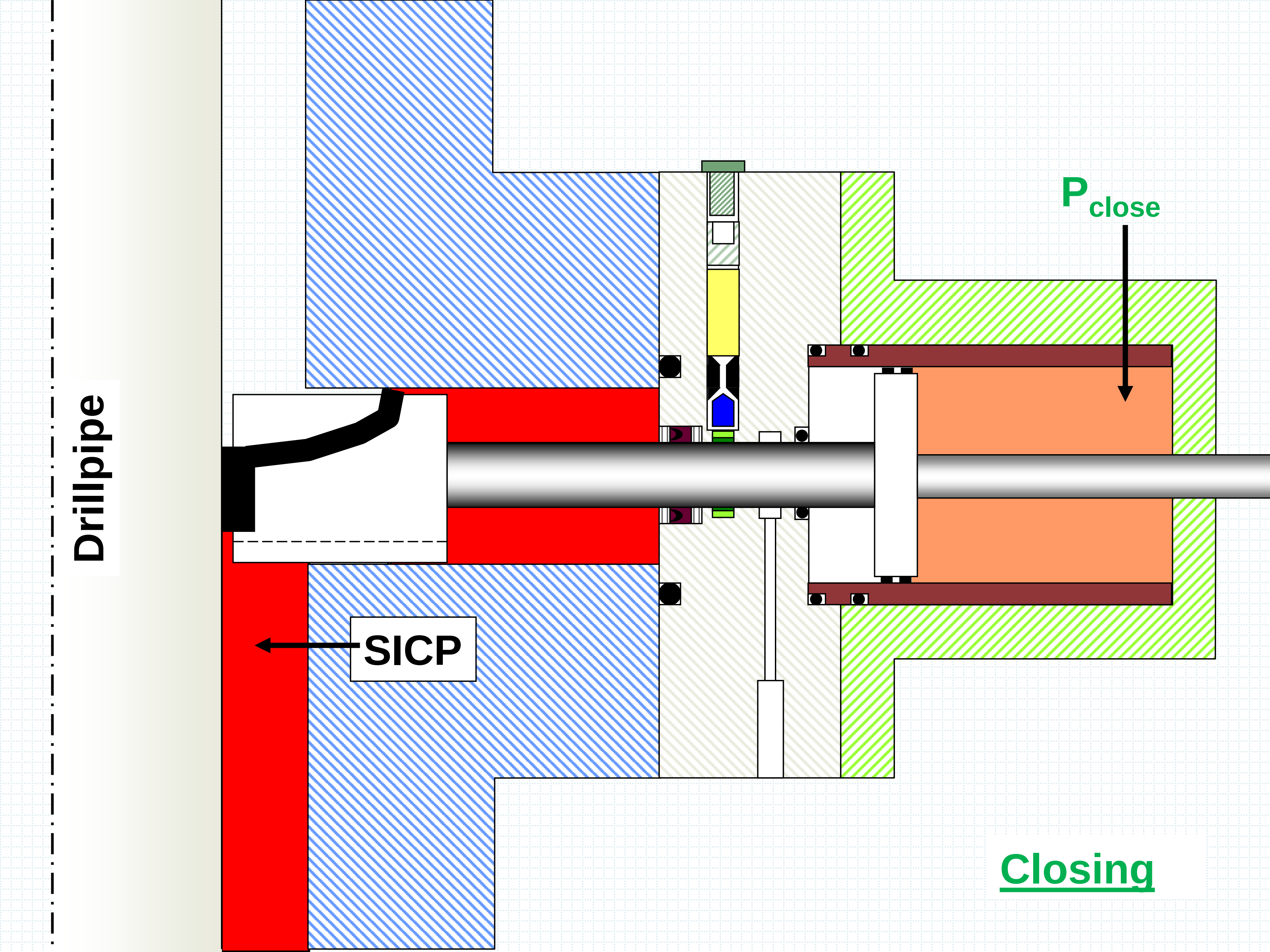
Drillpipe
$P_{close}$
Closing
R.

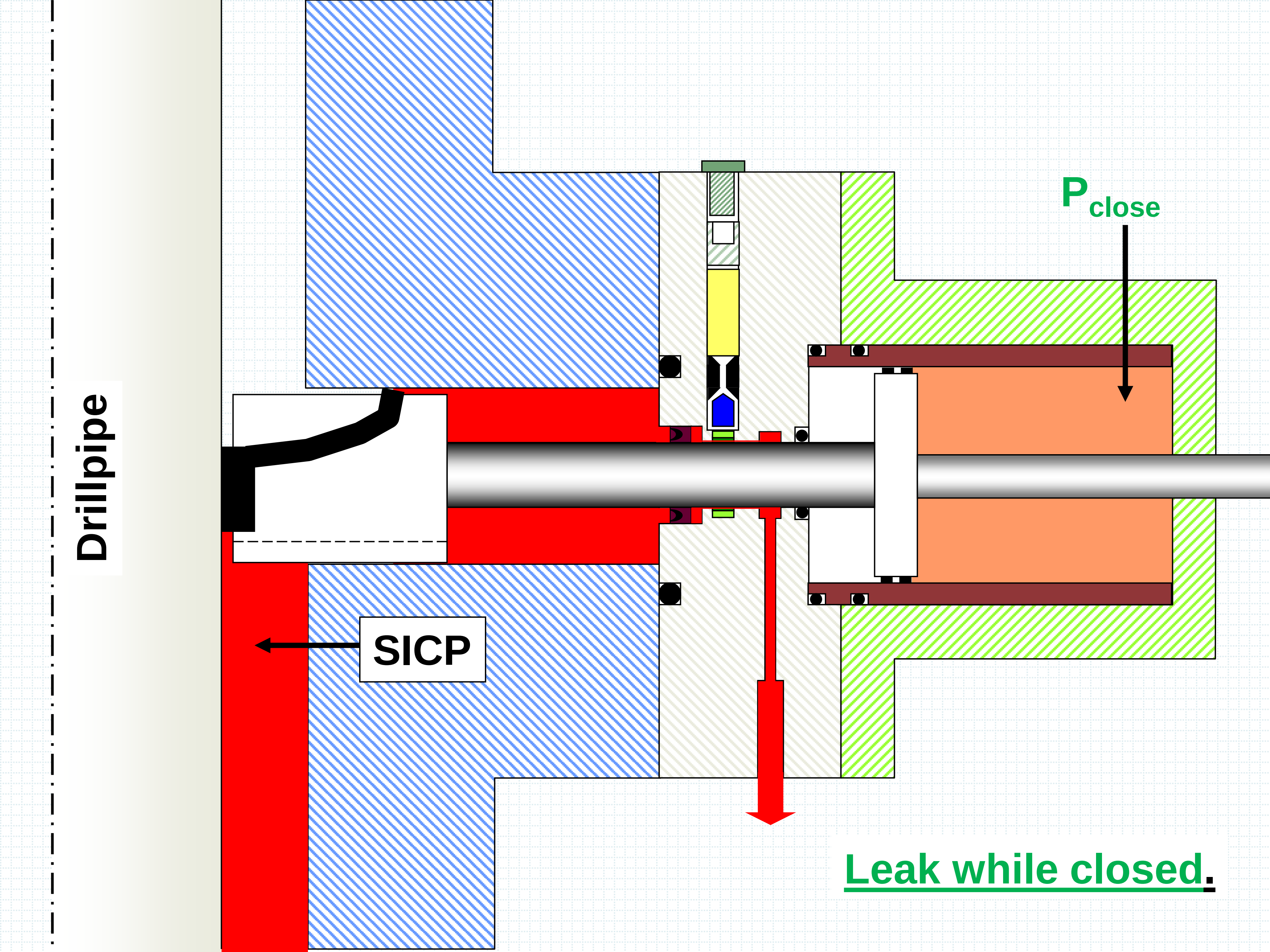

Drillpipe
Pclose
Closing

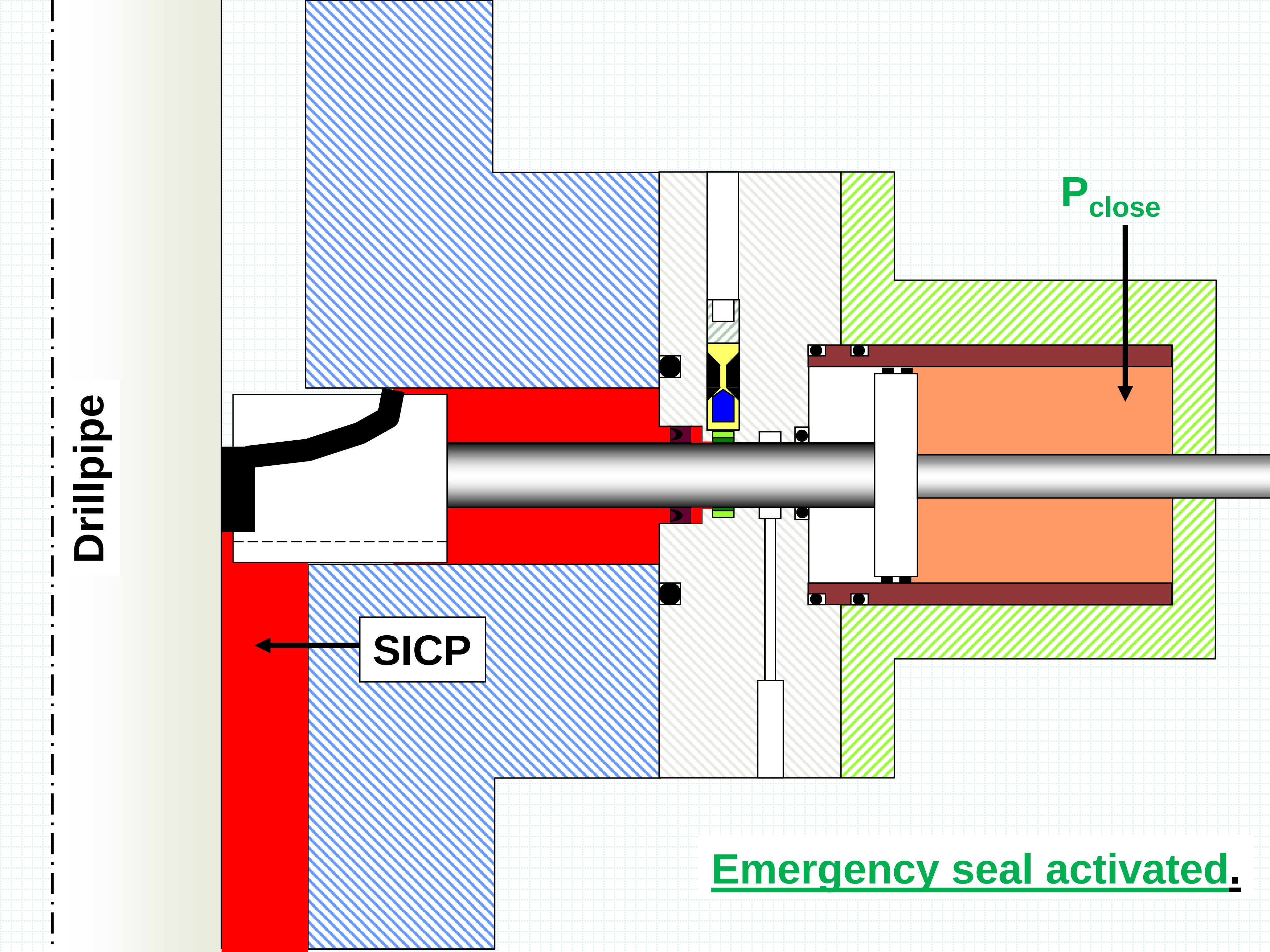



# Closing


Drillpipe
$P_{close}$
Closing
R.


Drillpipe
$P_{close}$
SICP
Closing


Drillpipe
SICP
P close
Leak while closed.



Pclose
Drillpipe
SICP
Emergency seal activated.

# WIRELINE

**Objective:**

***Provide general knowledge of WL types & its operations, WL equipments & its functional requirements, operational parameters & well control requirements.***

# WIRELINE

Well Intervention technique of conveying tools & instrument downhole.

The most efficient & practical method to diagnose well problems.

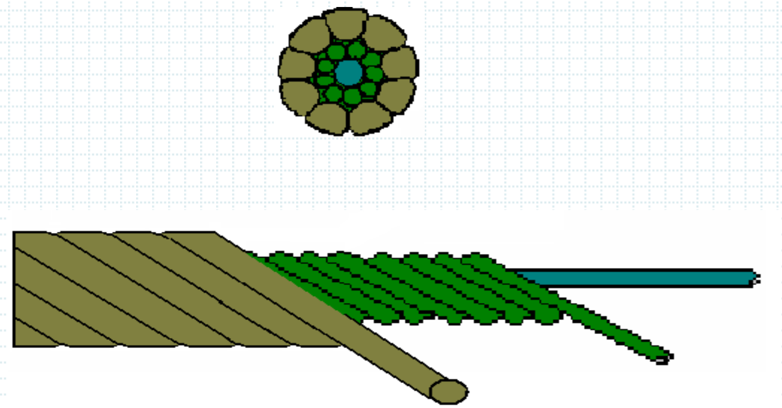
Easily junk wells if not properly manage.

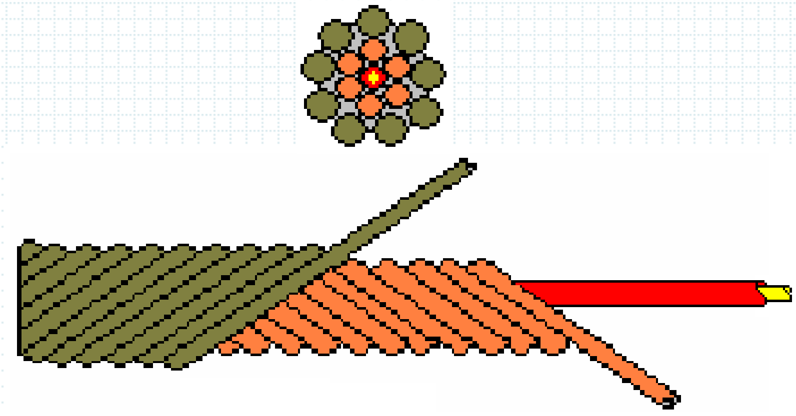
Wireline Category:

1. Slickline or Solidline or Pianoline
2. Braidedline
3. Electricline

4- Digital Slickline

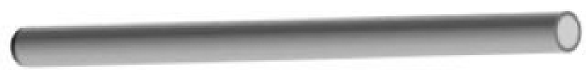
5- Fibre optic cable





# WIRELINE TYPES

1. Slick Line

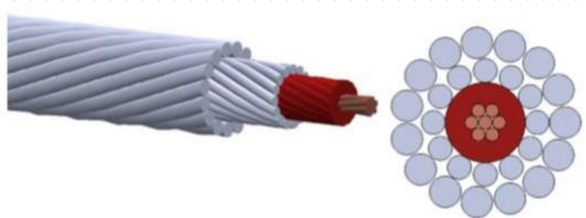


2. Braided Line


Conventional
16 strands


Dyform 19 strands
Low twist tendency
high breaking strength
smooth external

3. E-Line



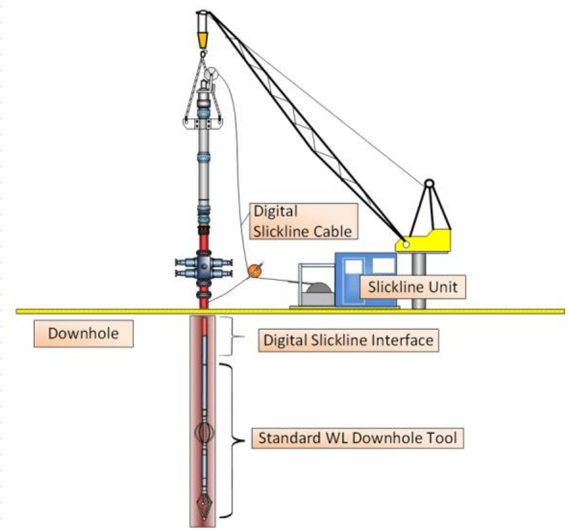
Mono-conductor
Cable3/16” or 7/32”

Multi-conductor
Cable - ¼” , $^{5}/_{16}$”, $^{7}/_{16}$” and ¾”

# Wireline types

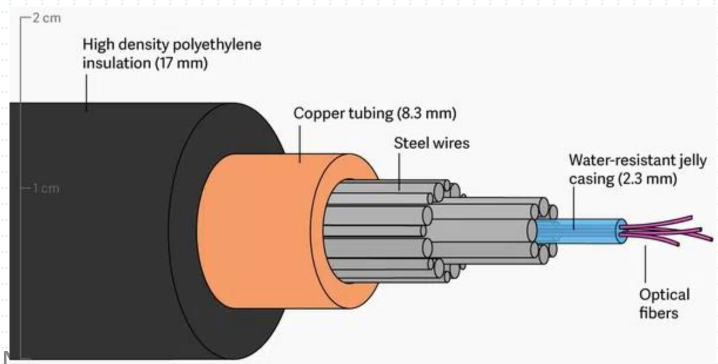
## 4- Digital Slickline

A digital slickline has an integral coating for digital two- way communication and is deployed using a standard slickline unit and pressure control equipment. Sensors in the downhole digital cartridge measure cable tension, detect shock and well deviation, and monitor internal temperature.




## 5- Fibre optic cable

Fiber optic cables have transformed telecommunications and connectivity. Fiber technology is a game changer. Via fiber cables, signals can be sent across the world at the speed of light.



# SLICKLINE

Single strand wire with wire OD range from 0.105”, **0.108”, 0.125”** , 0.140, 0.160”& 0.190”,0.260”

- Wire selection depends on the severity of the operation load.
- It’s robust & versatile in service

**Mechanical application;**

- Removal of well obstruction
- Repair mechanical failure
- Bringing other zones in-line
- Setting downhole FCD

**Advance application by running;**

- Recorded Real Time Data Acquisition Instrument
- Recorded Tubing Caliper
- Perforators
- Tubing Cutter
- Chemical Deployment Tools
- Explosive Jar etc

# BRAIDEDLINE

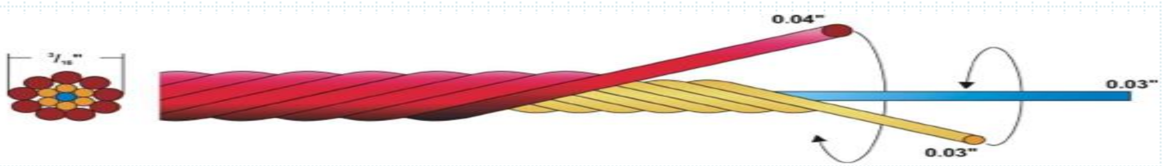
Multi-strands wire cable OD range from $^{3}/_{16}$", $^{7}/_{32}$", ¼" & $^{5}/_{16}$".

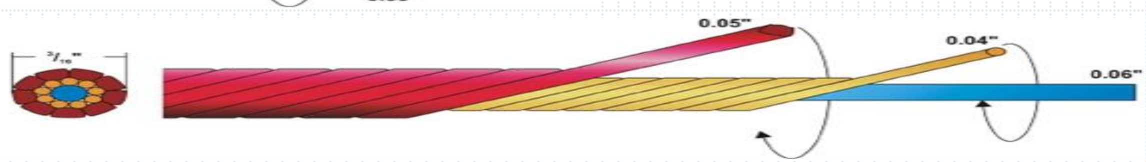
- Wire selection depends on the severity of the operation load required.

**Application:**

- Retrieval of downhole tools with slickline failure.
- Fishing job
- Swabbing

conventional type




Dyform type

20% increase in breaking load due to more steel in the same diameter , smooth external surface

# ELECTRICLINE

Multi-strands wire cable range from $^{3}/_{16}$”, $^{7}/_{32}$”, ¼” & $^{5}/_{16}$”, $^{7}/_{16}$” & ¾” OD with single or multiple conductors.

**Application by running;**

- SRO Pressure & Temperature Recorder
- SRO Electronic Logging Instrument for open hole & cased hole.
- Coring Services
- Packer setting
- Perforations etc.

# WIRELINE EQUIPMENT

## 1 Surface Equipment

- 1.1 WL Unit
- 1.2 Power Pack
- 1.3 Hoisting Unit

## 2 PCE

- 2.1 Quick Union
- 2.2 Stuffing Box / GIH
- 2.3 Lubricator
- 2.4 BOP
- 2.5 Tree Adapter

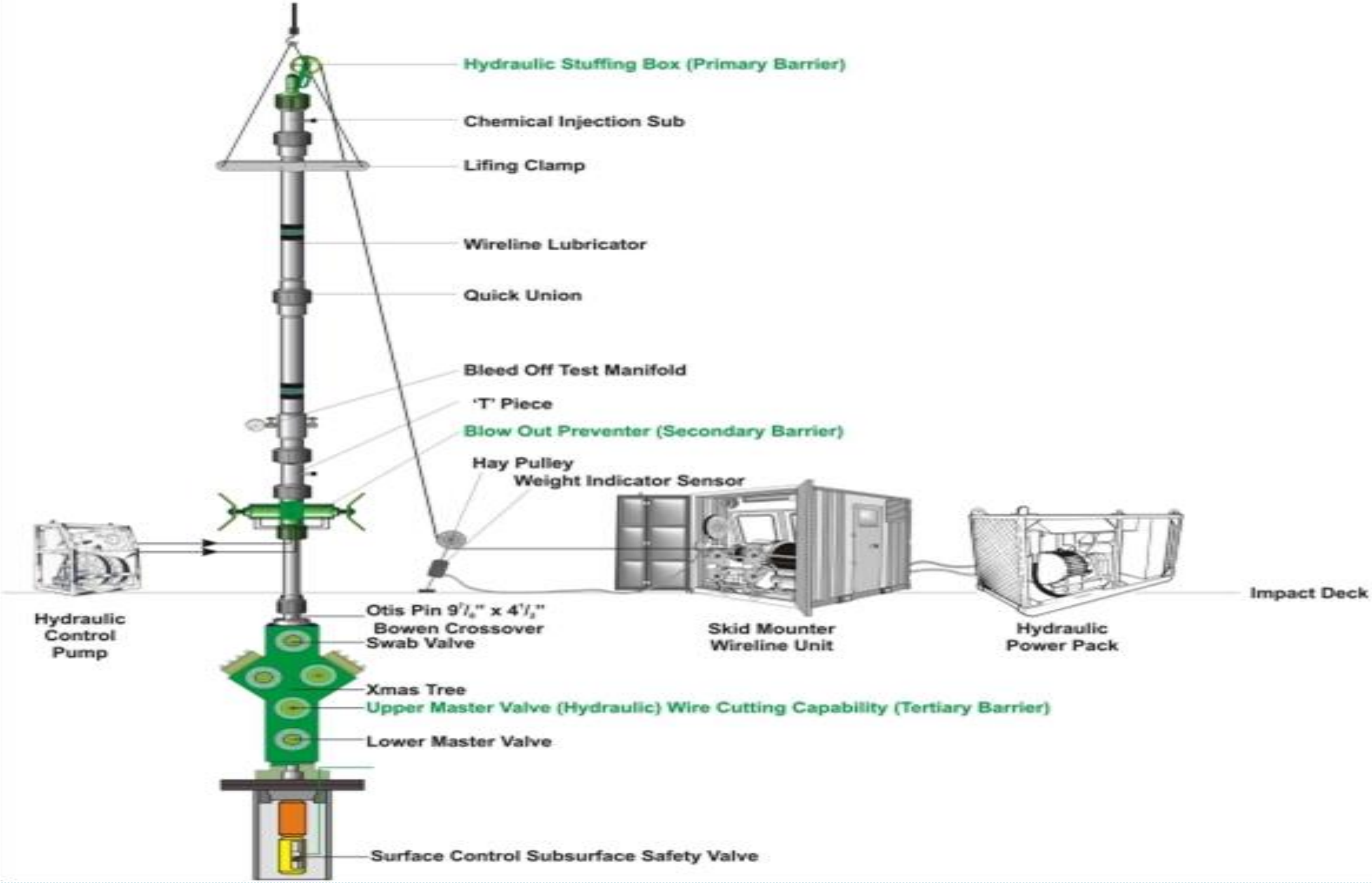
## Optional PCE

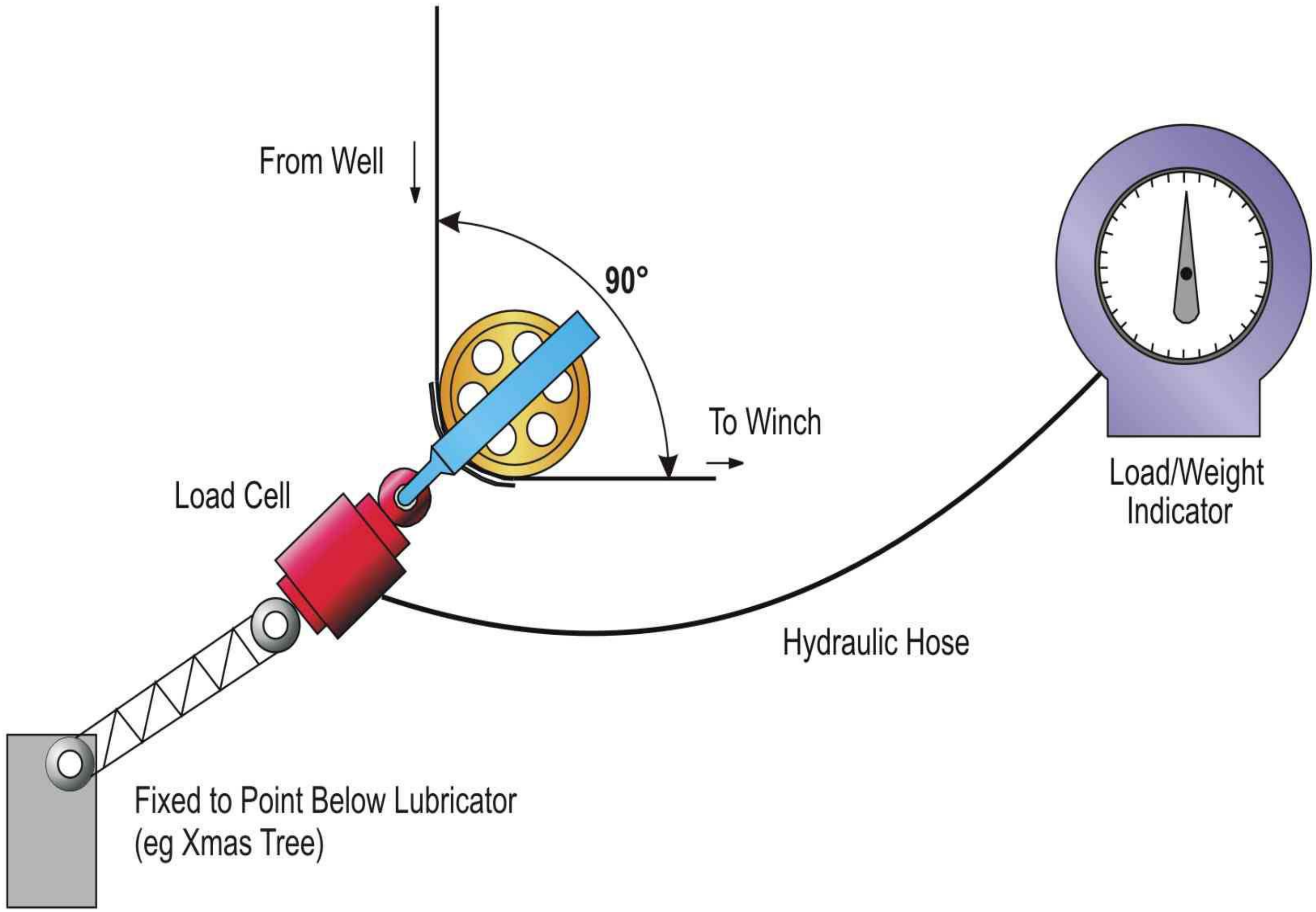
- 2.6 Tool Trap
- 2.7 Tool Catcher
- 2.8 Injection Sub
- 2.8 SCU
- 2.9 Pumping Tee

## 3 Subsurface Equipment (SSE)

- 3.1 Toolstring
- 3.2 Service Tools
- 3.3 Tubing Conditioning Tools
- 3.4 Running Tools
- 3.5 Pulling Tools
- 3.6 Shifting Tool
- 3.7 Kick Over Tool
- 3.8 Overshot
- 3.9 Spear
- 3.10 Flow Control Device

# Surface rig up




From Well
90°
To Winch
Load Cell
Load/Weight
Indicator
Hydraulic Hose
Fixed to Point Below Lubricator
(eg Xmas Tree)

# SURFACE EQUIPMENT

## 1.1 Wireline Unit

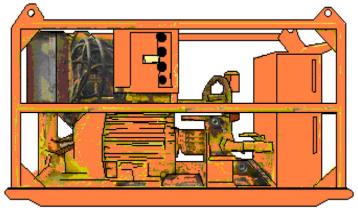
- Single Drum or Dual Drum
- Skid Mounted / Containerized / Helicopter Unit

**Component**

- Operating Panel – Direction Lever, Brake, Hydraulic Valve, Gauges etc
- Measuring Device
- Weighing Device
- Level Wind

## 1.2 Power Pack

- Diesel or Electrical Powered
- Zoning Classification

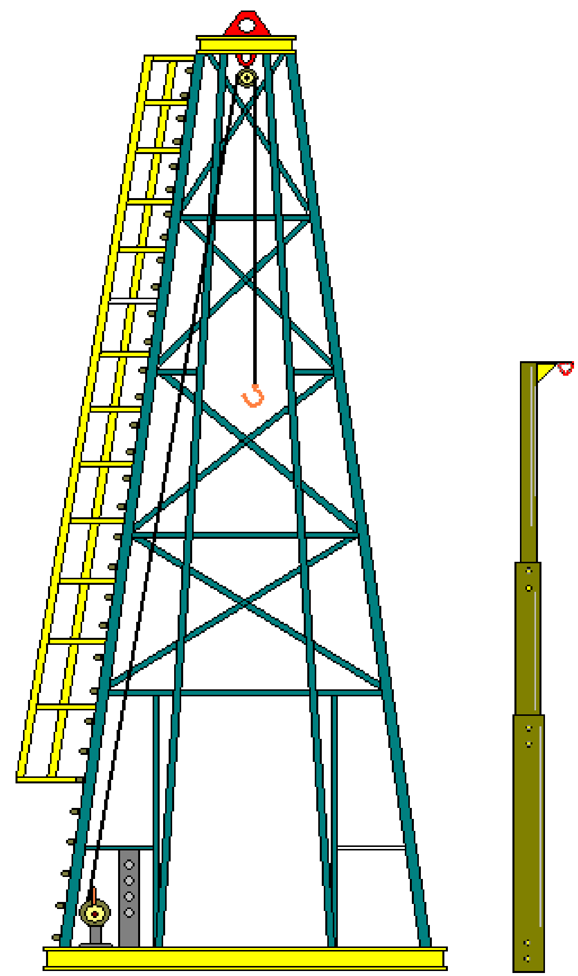


Zone 0 – Continuous present of flammable agent.
Zone 1 – Flammable agent likely to occur in normal operation.
Zone 2 – Flammable agent not likely to occur in normal operation.

# SURFACE EQUIPMENT

## 1.3 Hoisting Unit

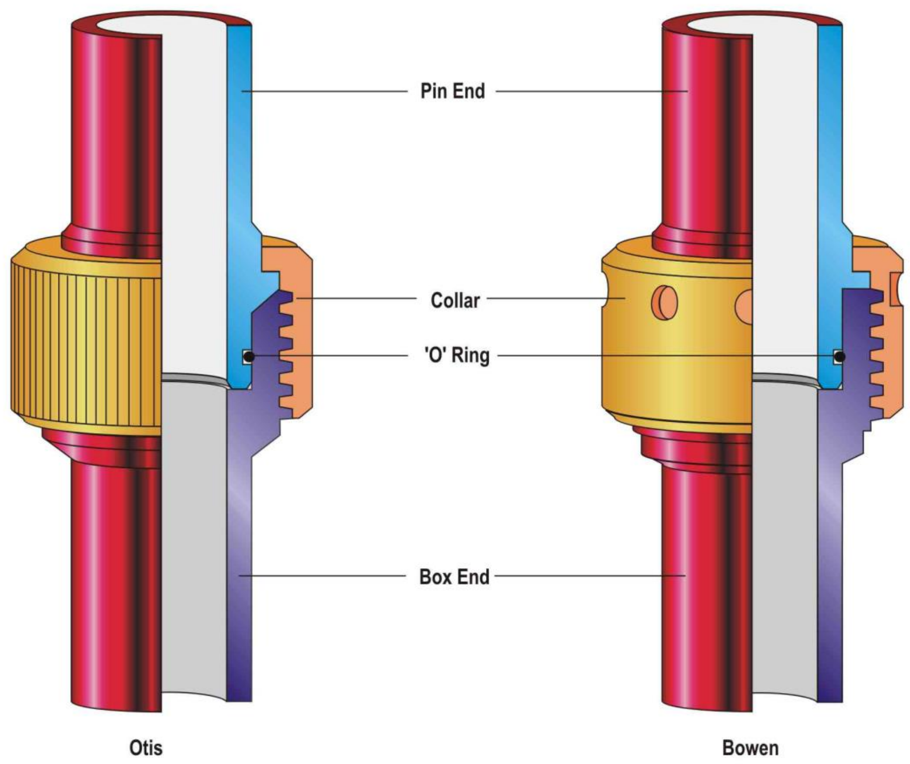
- Pedestal Crane
- Mast
- 'A' Frame
- Derrick
- Gin Pole
- Rig



## Operational Considerations:

- Length of Lubricators, Risers & Stack Up.
- Height of Hoisting Unit
- Fishing Job
- Toolstring Length

# PCE – QUICK UNION




- Thread Connection:

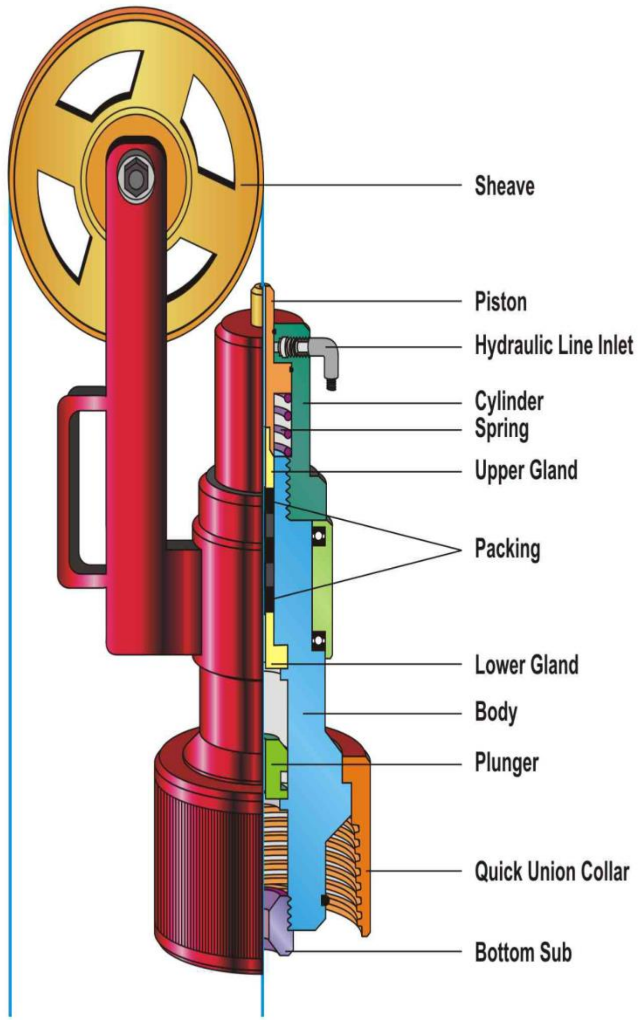
  5¾ - 4 ACME Thread

- Hand Tight NO wrench
- O-Ring seal
- To be kept clean at all time.
- Check thread & o-ring before make-up.

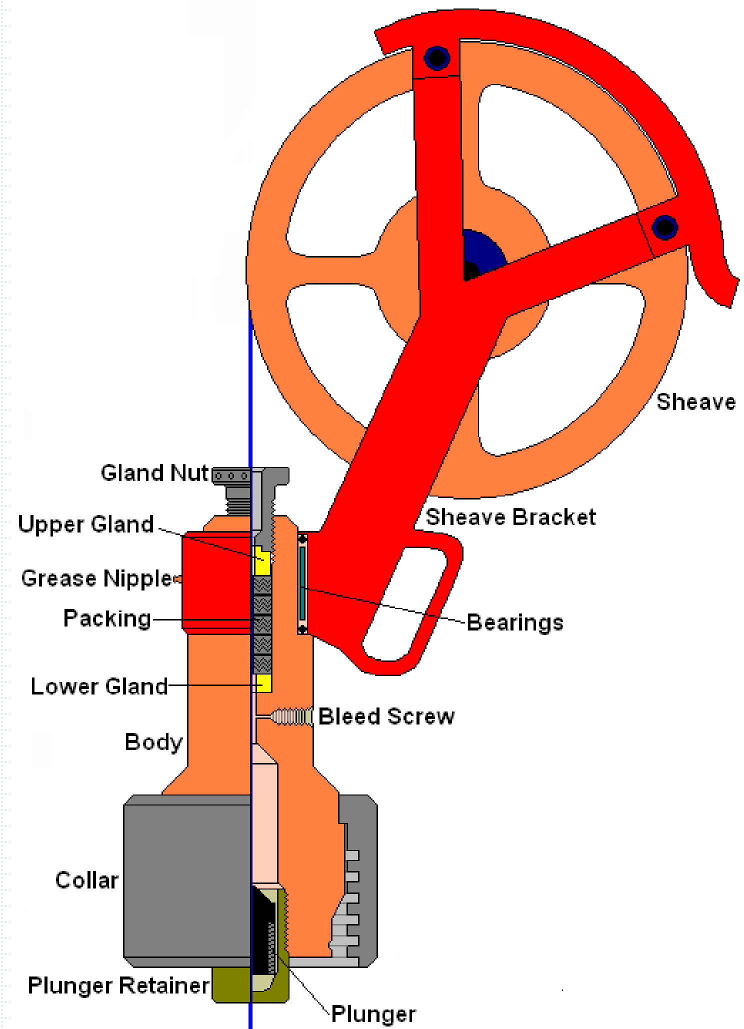
# EXPLOSIVE DECOMPRESSION OF ELASTOMERS/ SYNTHETIC RUBBER SEALS

- Rapid gas decompression, also known as *Explosive Decompression* often occurs when high-pressure gas molecules carbon dioxide in natural gas, $CO^2$ , in particular] migrates into an elastomer at a compressed state
- When the pressure surrounding the elastomer is suddenly released, the compressed gas inside the elastomer try to expand and exit the elastomer.
- Explosive decompression is major risk for any elastomeric seal operating in a high-pressure gas environment, over 1,000 psi.
- The gas gets trapped inside the seal's micro pores.
- If the seal faces a pressure change as with rapid decompression , this trapped gas rapidly expands in an effort to match external pressure
- This gas expansion will cause seal damage such as swelling tears, holes, blisters and cracks
- ***Note: Seals*** damaged by explosive decompression should ***always be discarded and not reused.***

# PCE – STUFFING BOX



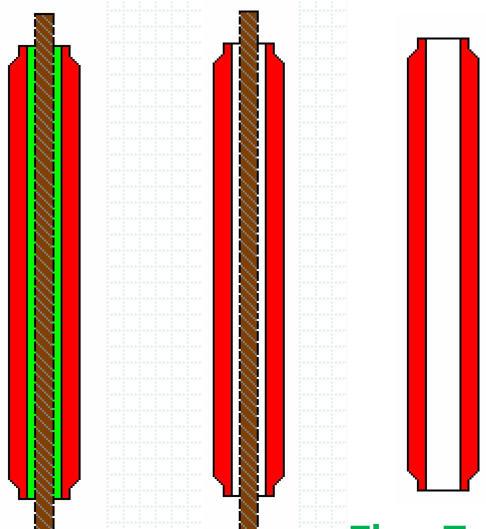

Hydraulic stuffing Box




Manual Stuffing Box

# PCE – GREASE INJECTION HEAD

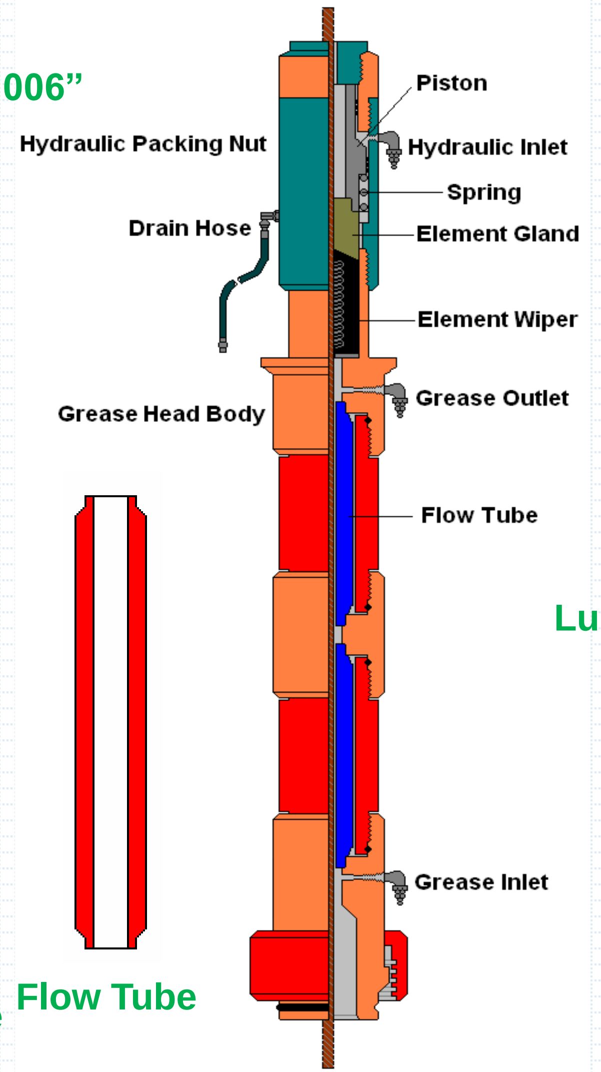
Clearance: 0.004" – 0.006"



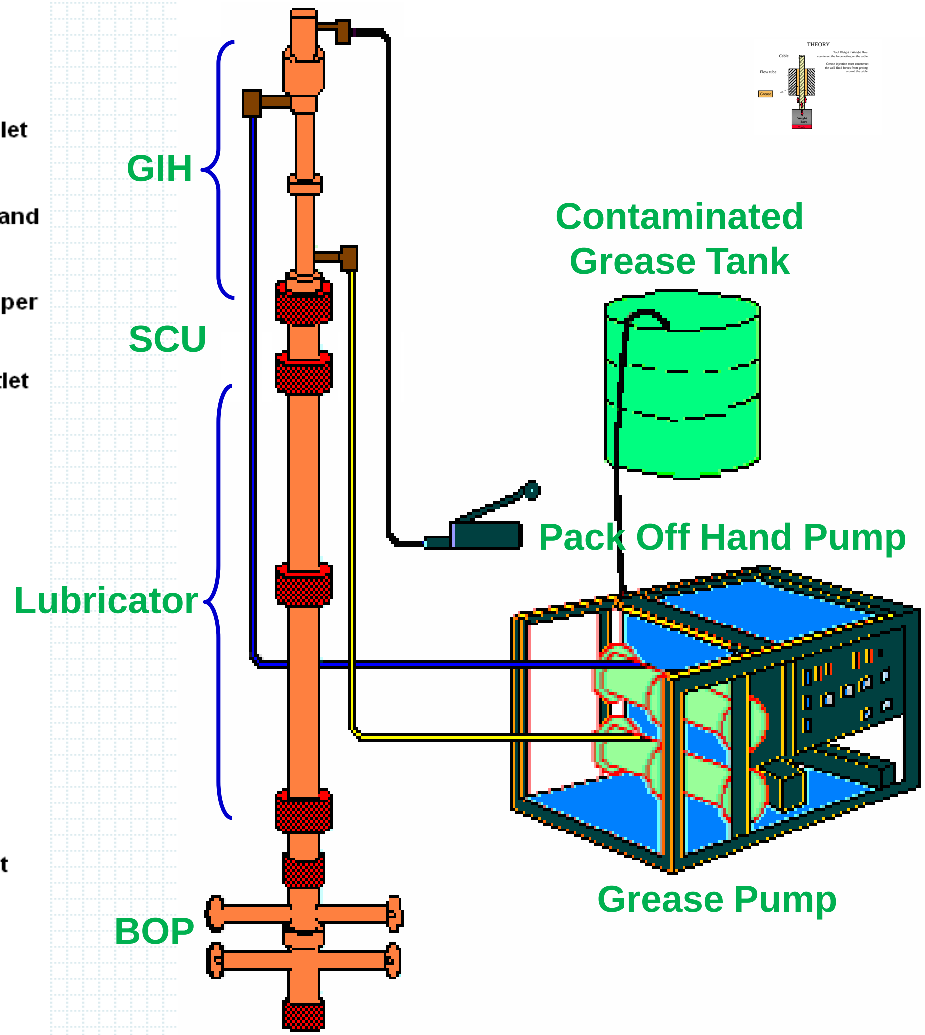
Flow Tube, Cable & Grease

Flow Tube & Cable

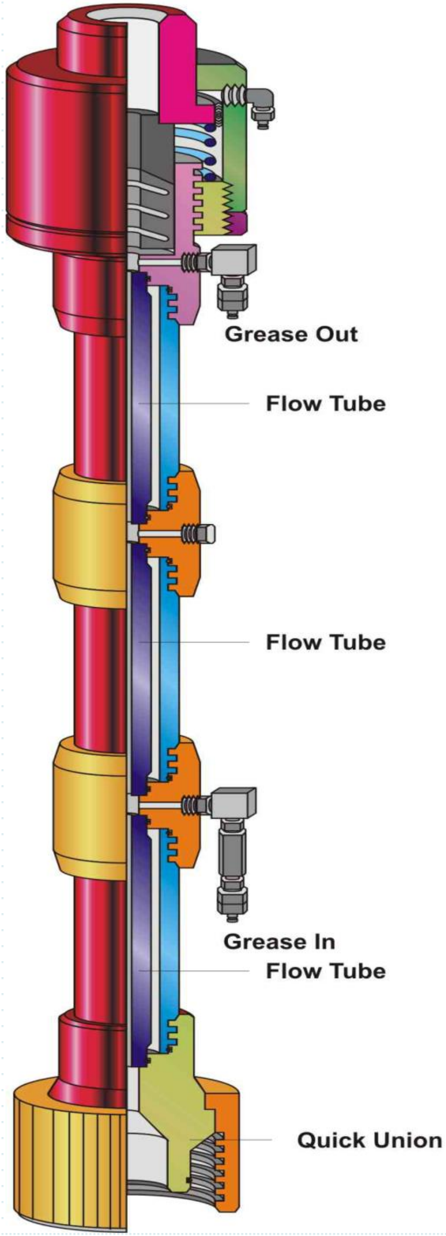
Flow Tube

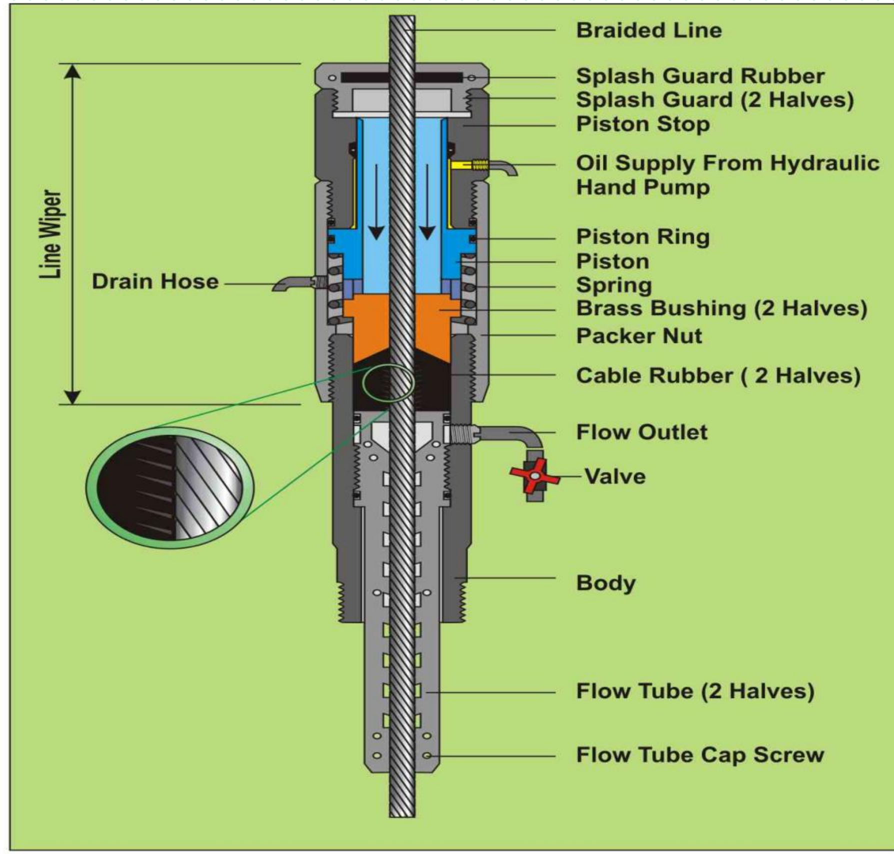



GIH




GREASE SYSTEM


Grease Out
Flow Tube
Flow Tube
Grease In
Flow Tube
Quick Union

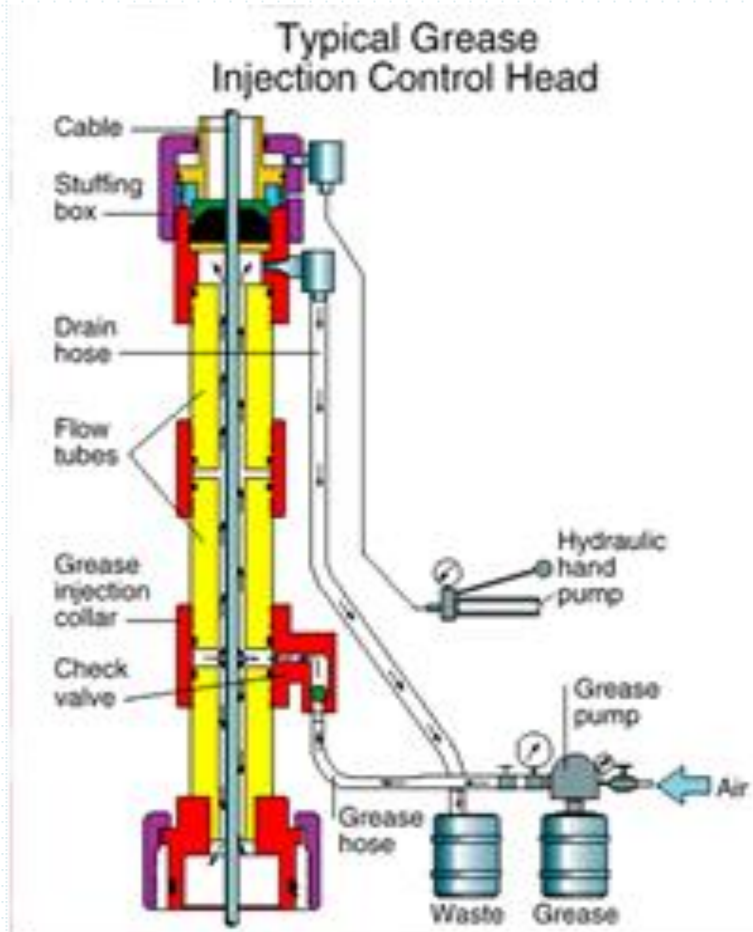

Braided Line
Splash Guard Rubber
Splash Guard (2 Halves)
Piston Stop
Oil Supply From Hydraulic Hand Pump
Line Wiper
Drain Hose
Piston Ring
Piston
Spring
Brass Bushing (2 Halves)
Packer Nut
Cable Rubber ( 2 Halves)
Flow Outlet
Valve
Body
Flow Tube (2 Halves)
Flow Tube Cap Screw

# FLOW TUBE

| # of Solid Flow Tubes<br>@ 2500psi gradient in Oil<br>@ 2000psi gradient in Gas | Well content Type | MPWHP (psi) |
|---|---|---|
| 3 | Liquid | 0 – 5000 |
| 4 | Gas | 0 – 5000 |
| 4 | Liquid | 5000–7500 |
| 5 | Gas | 5000–7500 |
| 5 | Liquid | 7500–10000 |
| 6 | Gas | 7500 – 10000 |

# GREASE INJECTION HEAD

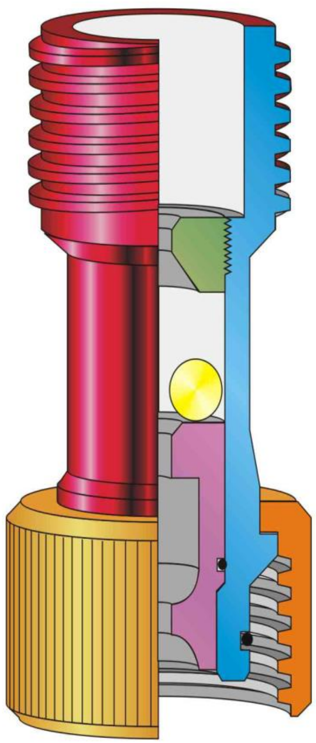
- Grease seal is a system which consist of;
- 1. Flow tubes matching to cable
- 2. Grease injection pump( able to pump at least MPWHP +30%)
- 3. Grease injection hose 10K WP, with check valve at injection point.
- 4. Grease return line 10K WP, with 'T' at the end
- Grease should be injected min 1.2 X WHP

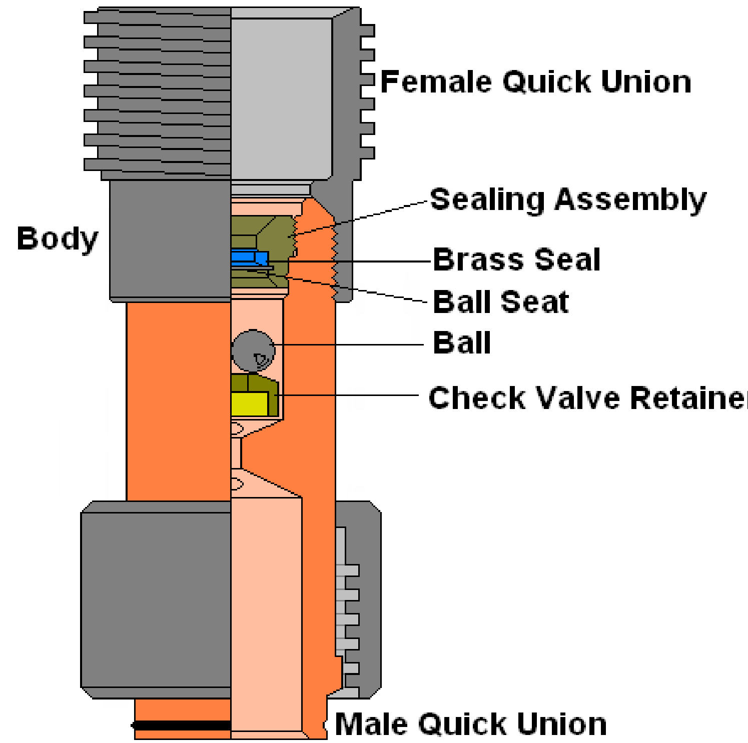


# GREASE SEAL LEAKS DUE TO

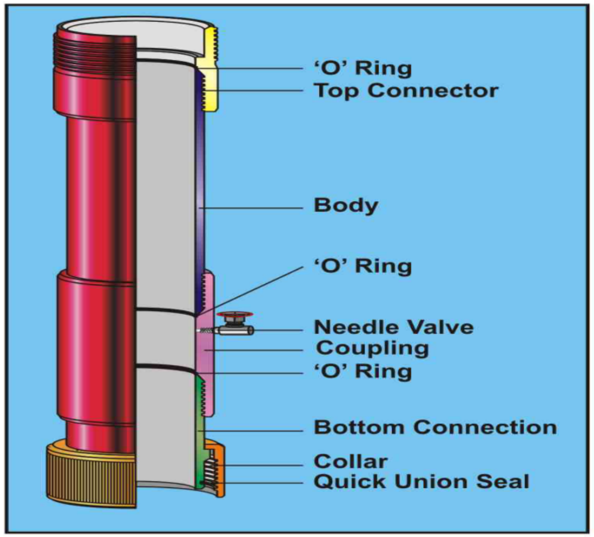
- Pulling out or running in too fast
- Setting the grease injection pressure too low
- Increase in well head pressure , e.g after perforation
- Not enough grease in the tank supply
- Restriction in the grease supply system
- low grease pump air supply pressure
- Contaminated grease becomes thinner
- Incorrect flow tube ID or worn flow tubes
- Insufficient flow tubes length
- Grease type incompatible with conditions e.g. ambient temperature or freezing due to gas escape

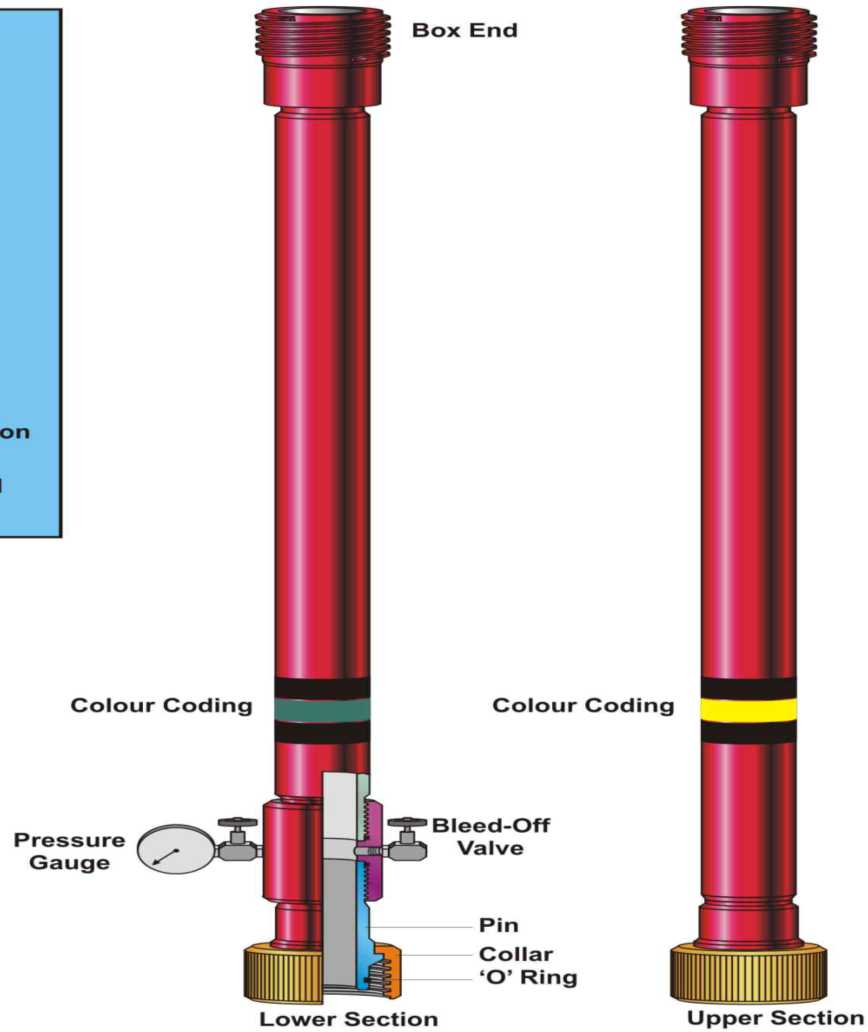
# PCE – SAFETY CHECK UNION





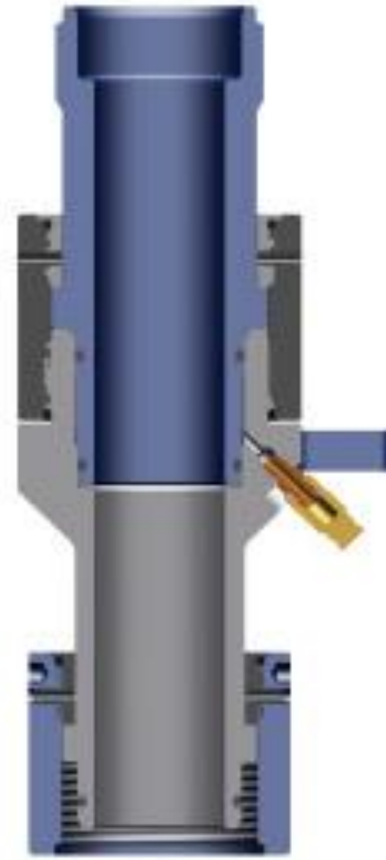
# PCE – LUBRICATOR ( OLD TYPE )



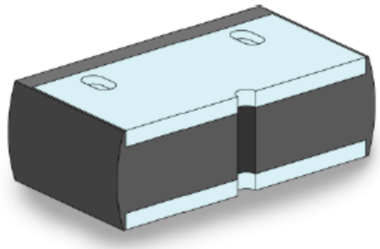
# PRESSURE TEST SUB

The In-Situ Pressure Test Sub (sometimes called Quick Test Sub) was originally designed to reduce the time spent pressure testing a lubricator rig up, prior to exposing it to live well pressure. In situations where the lubricator has to be tested each time the bottom connection is broken, much time can be lost.

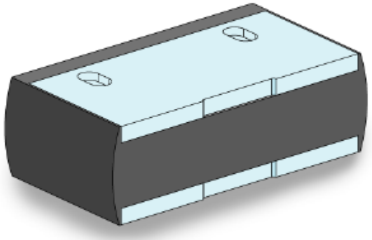


# BOP (Wireline Valve):`

- **Inner Seal Types:**
  - Normal
  - Multiline
    - While conventional seals are sized for one line size only, the Multi Line Ram Seal has been designed to seal a range of line sizes from blind through to 0.320” braided line using a specially developed elastomer.. Under development is a ram seal, which will seal from 5/16” through to 15/32” braided wireline. The ram seals are directly interchangeable with conventional seals in Elmar’s Wireline Valves with up to 10,000 psi working pressure rating and -29°C to 121°C, H2S service.
  - Constrictor Multiline:
    - In the event of unexpected cut and drop, it can seal blind, contain wellbore fluids for at least 12 hours and seal successfully on any wireline up to 5/16”.
    - The same seal is used for testing of both low and high end temperature range.
    - The Constrictor can close on the wireline cable without shearing, allowing crews to perform remedial work on the cable.
    - Fits existing NOV Elmar Q Guide™ rams and can be retrofitted to an existing wireline valve while in operation and used repeatedly without degradation, making it suitable for fishing operations.

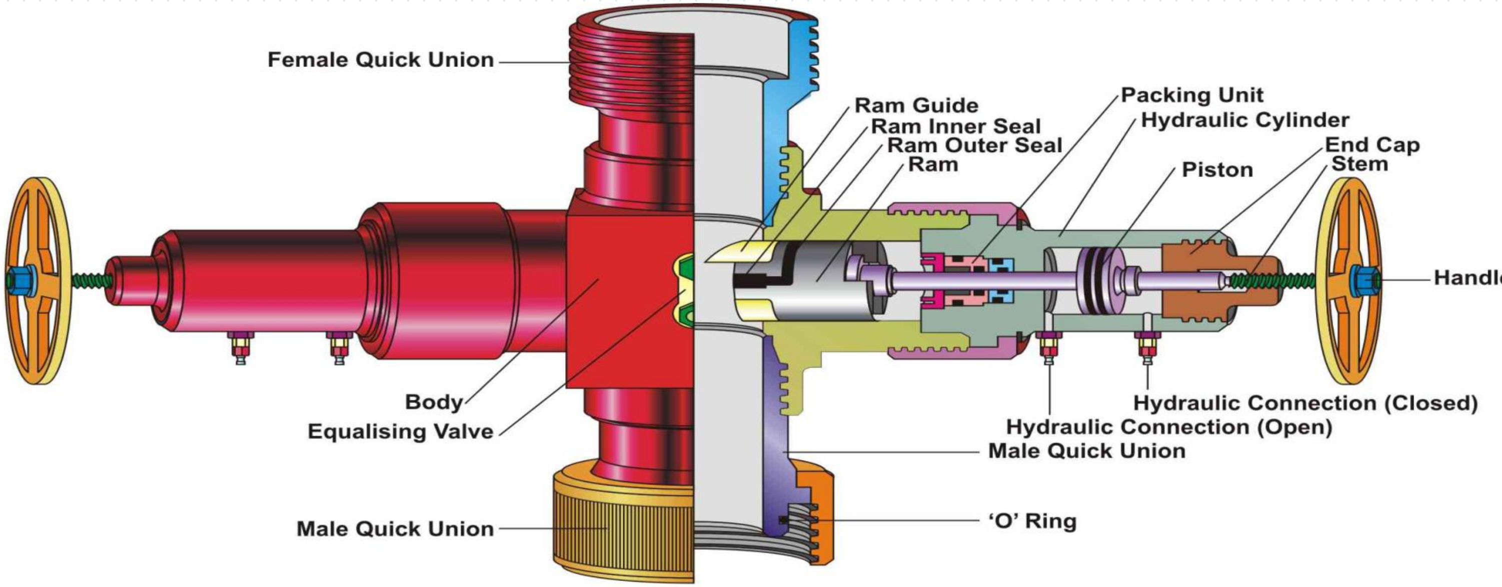


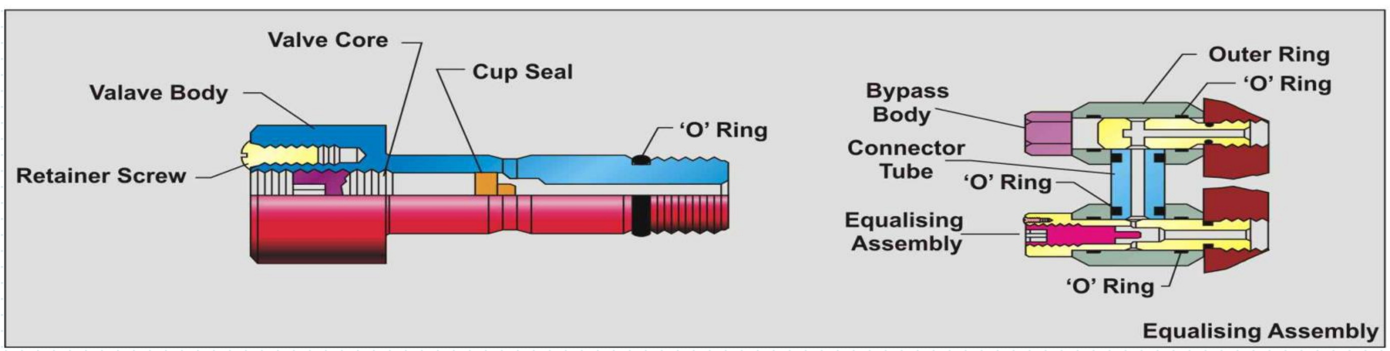
Electric line Inner Seal



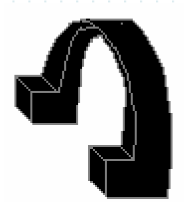
Slick line Inner Seal

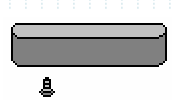
# PCE – BOP



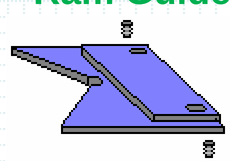

Ram Outer Seal

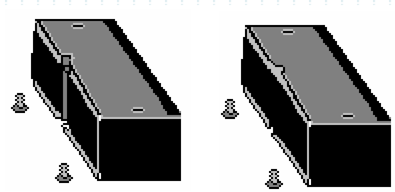




Ram Key

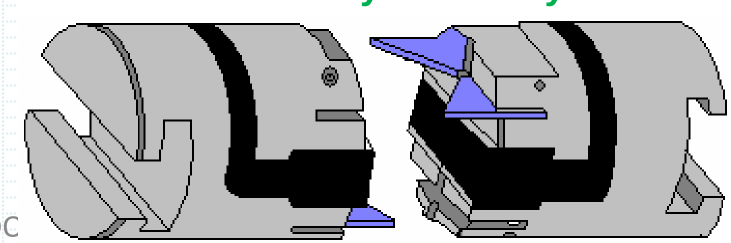
Ram Guide



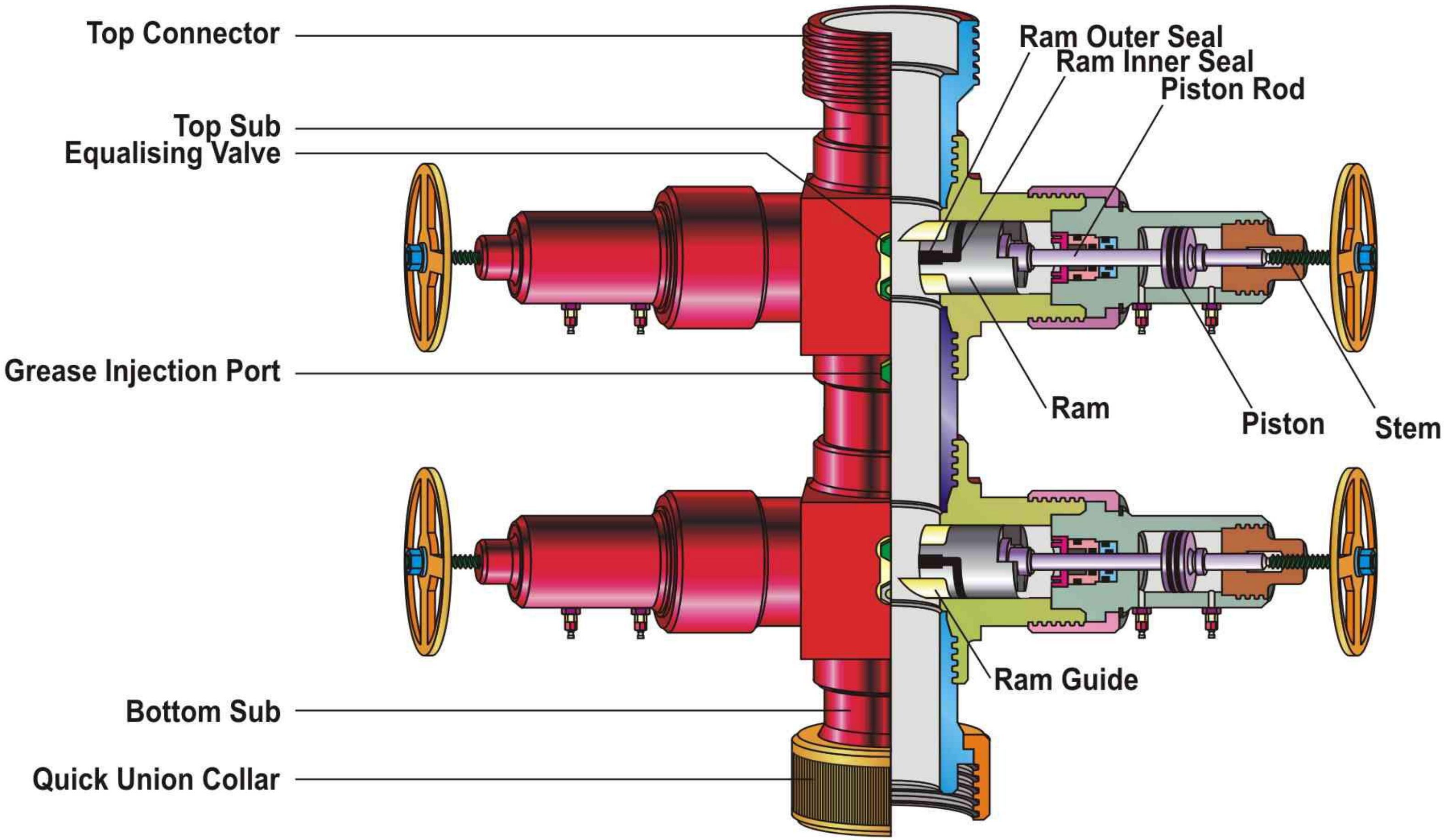


Ram Inner Seal

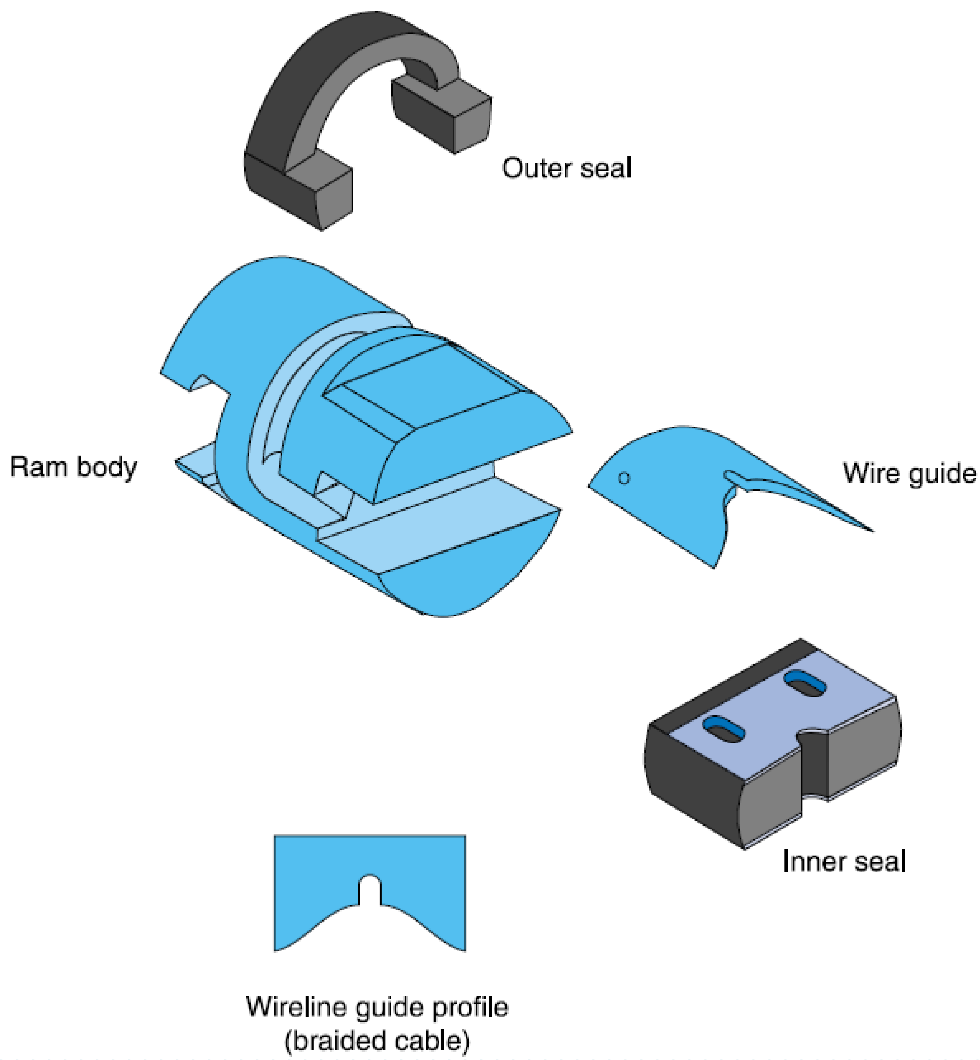
Ram Body Assembly

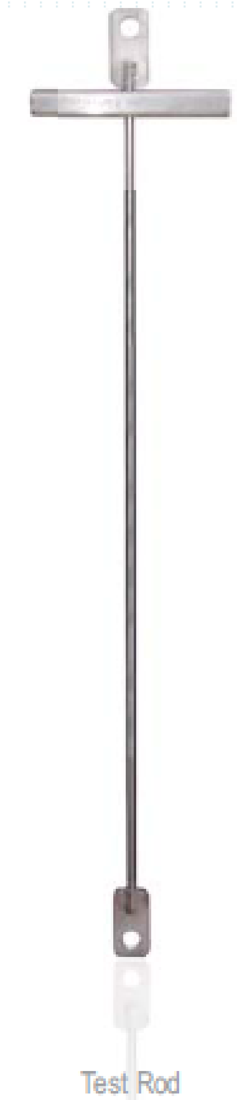


# BRAIDED BOP


Top Connector
Top Sub
Equalising Valve
Grease Injection Port
Bottom Sub
Quick Union Collar
Ram Outer Seal
Ram Inner Seal
Piston Rod
Ram
Piston
Stem
Ram Guide

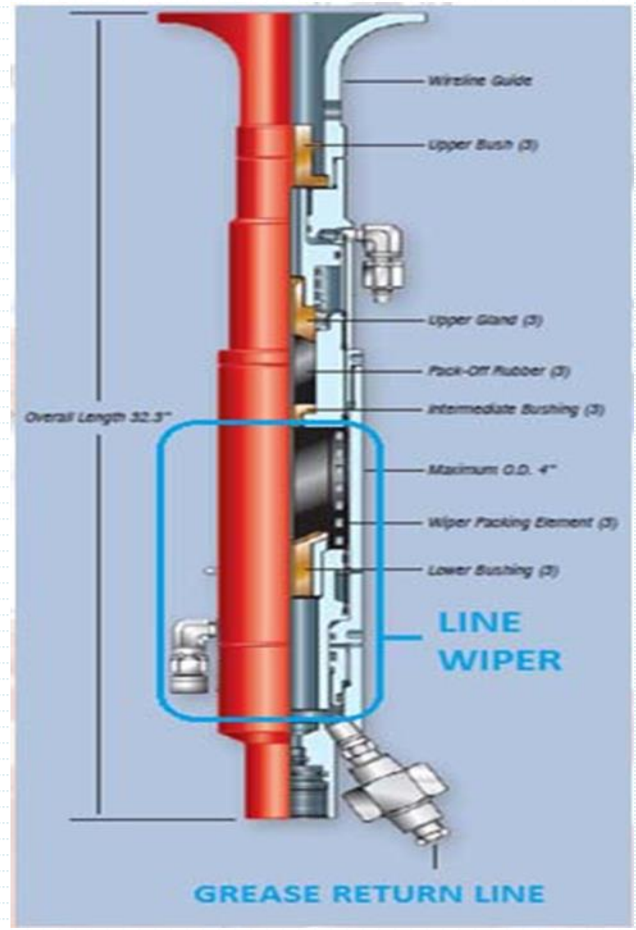
# Braided Line  Ram Assembly and Test Rod



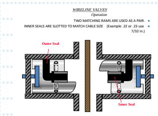



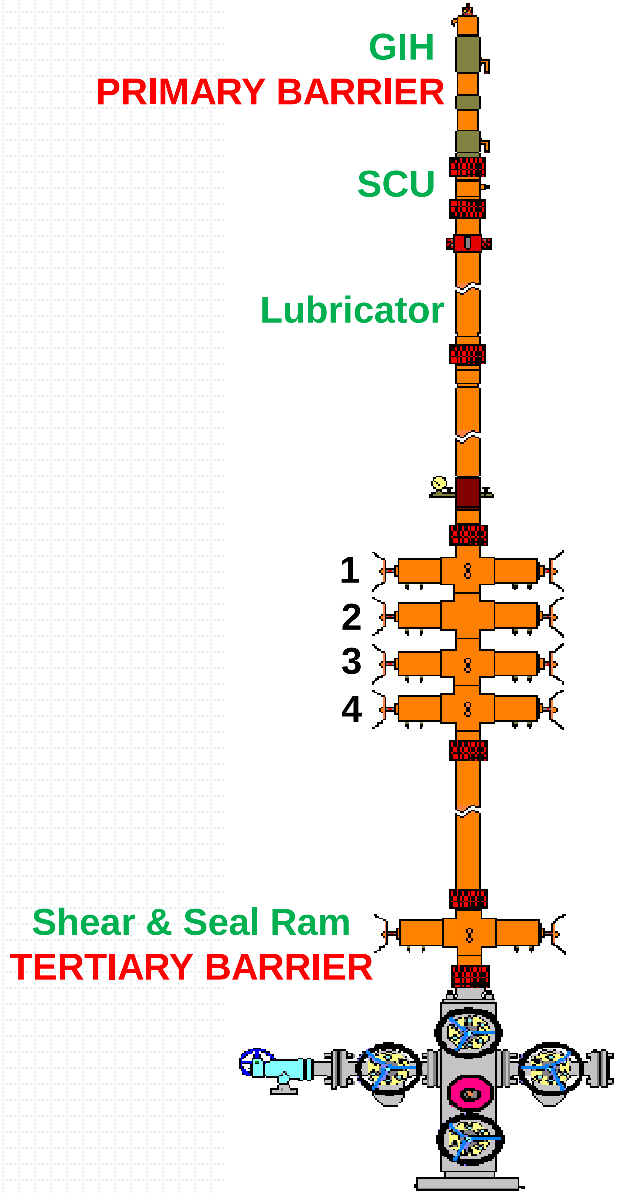
# LINE WIPER

- Used only to wipe excess grease off line
- Not a pressure control device
- **Can not be used to SEAL WELL PRESSURE**



# BRAIDEDLINE PCE






**SECONDARY BARRIER**

1 Braidedline Ram

2 Braidedline Inverted Ram

3 Blind Ram

4 Shear Ram

# Shear/Seal BOPs – Slickline/Braided Line

It is always a ***Tertiary Barrier BUT……………………………***

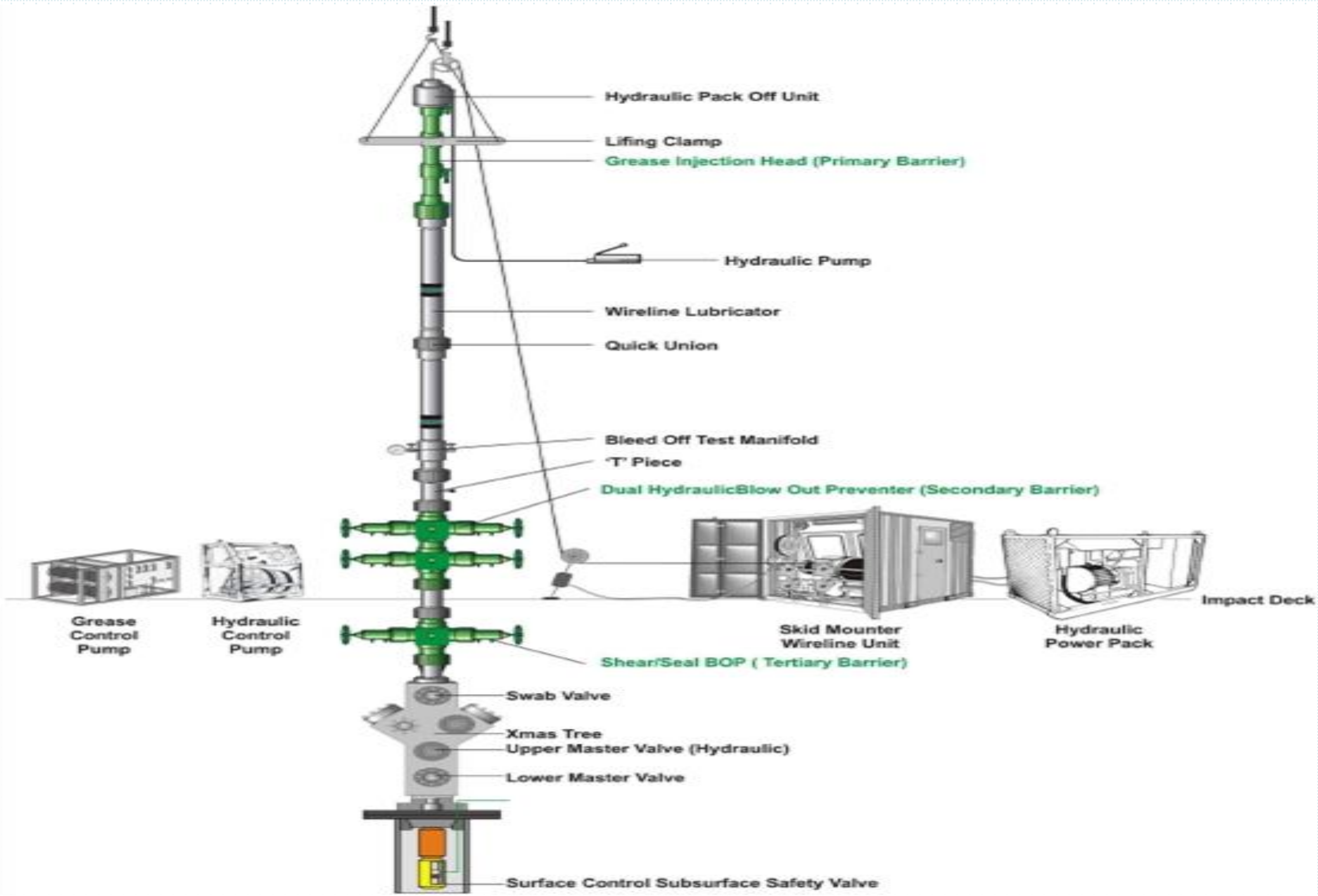
- The Shear/Seal BOP is flanged between Xmas tree and wireline BOP and often *immediately above the Xmas tree* or as close to the tree as possible.
- The Shear/Seal BOP is suitable for cutting *both* Slick Line and Braided Line/E-Line. When closed, it holds pressure from below the shear/seal BOP shall have always have an *independent control system*.
- It can cut multiple wires , usually as many as 8-10 conductive wire , but **not** tool string or running tools
- After the wire has been cut by a shear/seal BOP, the wire will usually fall **below** the Xmas tree valves, permitting the UMV to be closed.

**Note:** If the Xmas Tree UMV has wire cutting ability, we would use it instead.

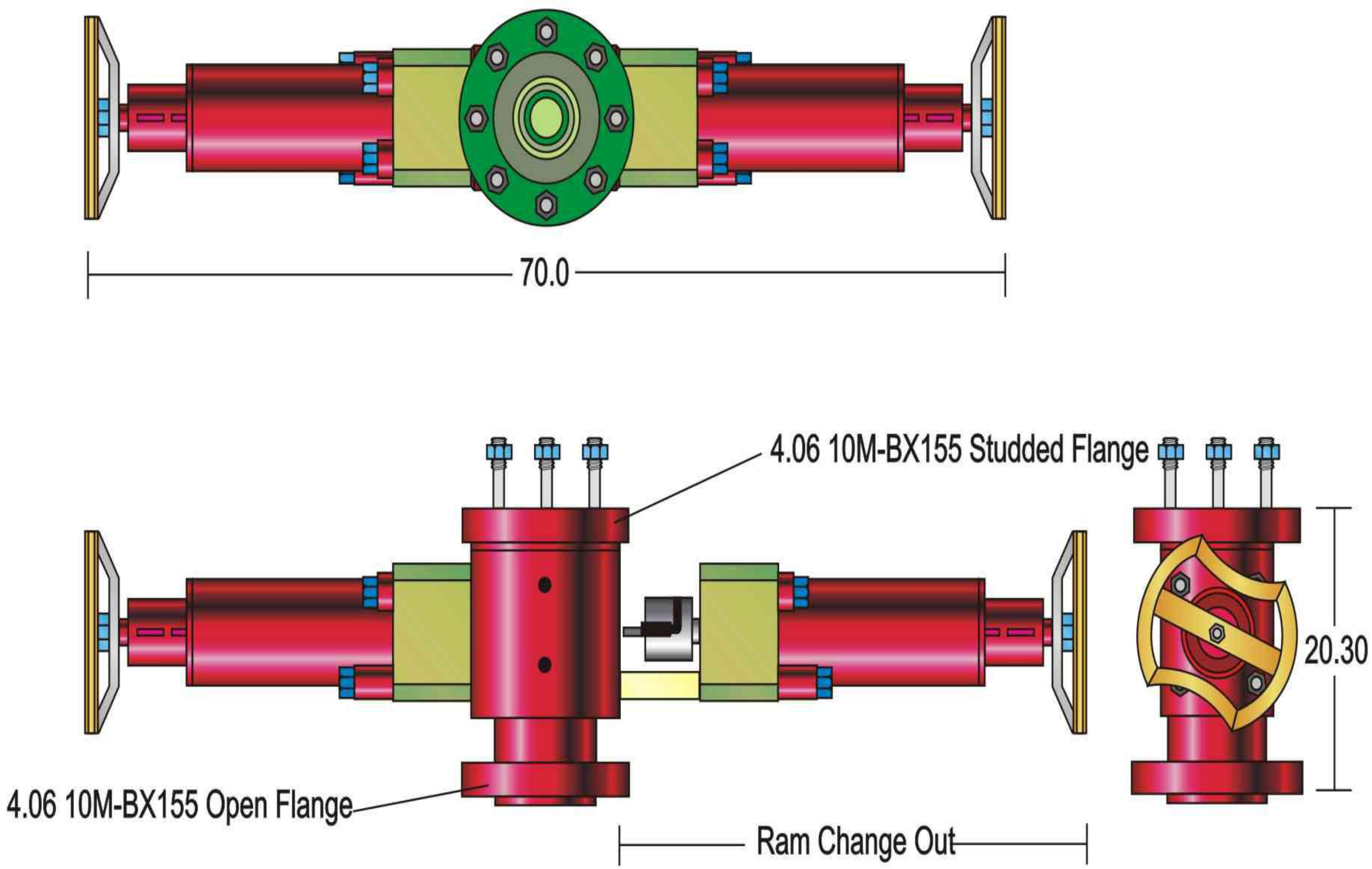
In an emergency a standard gate valve will cut up to one strand of 7/32" wire, but may be damaged in the process.

In this case the swab valve should be used to cut as it is easily replaced.

# Braided line rig up



# SHEAR /SEAL BOP



# WIRELINE PARAMETERS

Five considerations required serious attention prior wireline operations.

1. **Job Planning**
2. **Rig-up & Rig-down**
3. **Pressure Testing**
4. **RIH Considerations**
5. **Emergency Response**

# Wireline Parameters

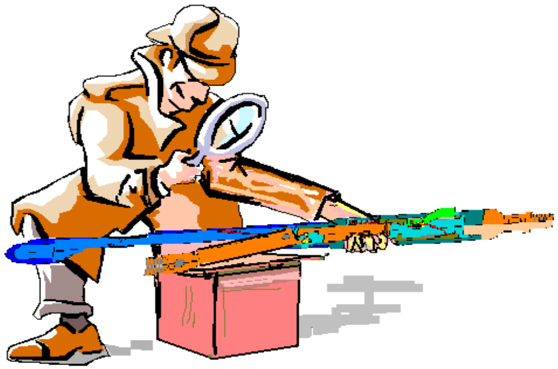
**1. Job Planning:**

- Obtain Well Information
- Study Well History
- Tools & Equipment Inventory
- Pre-Run Checks on Tools & Equipment
- Personnel Knowledge & Skills
- Safety Devices & Equipment

# WIRELINE PARAMETERS

**2. Rig Up & Rig Down:**

- Effective Communication
- Working Location
- Weather Condition
- Hazardous Environment for Material Selection
- Pre-Run Checks on Tools & Equipment
- Back Up & Spares for Tools & Consumable



# WIRELINE PARAMETERS

## 3 Pressure Testing

- Minimum is Max SITHP . 1.2 is recommended & apply slowly and in stages. Must apply low pressure test and high pressure test
- PT medium; 60:40 ratio by volume of fresh water & glycol respectively.

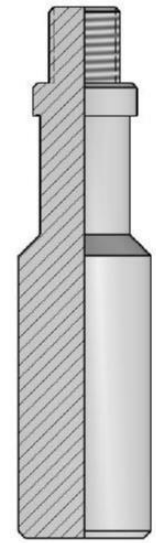
***(This would prevents "Emulsion" formation in oil wells or "Hydrates" if it would be discharged into wells)***

## De-Pressurize

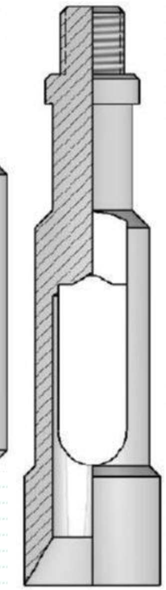
- Bleed off through production process venting system.
- If not possible, bleed off with extreme care, no personnel & source of ignition down wind.
- De-pressurizing must be a **Controlled Operation**.
- **PATIENCE** is the virtue of this case & **NO ACCELERATION** is permitted. Ensure that equipment is earthed to prevent ignition by static electricity.

# The importance of Slickline Drift Runs

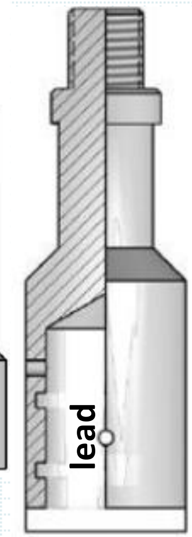
- Provides invaluable information with respect to down hole conditions in the well, and *must* be conducted as the first run in any well .
- Evaluate Hold-Up Depth-HUD and tubing end with ***tubing end locator and sample bailer***
- Establish that the tubing is clear and accessible to required depth with ***gauge cutter***
- Get a “picture” of wireline fish run a ***Lead Impression Block***
- Gauge and scrape clean paraffin, wax and other debris from inside tubing wall; run as **Gauge Cutter**
- Establish liquid level in tubing, with **Blind Box**
- Verify wireline depths of landing nipples and associated wireline Pulling and Running weights
- Minimize risk damaging expensive down hole tools or causing fishing job

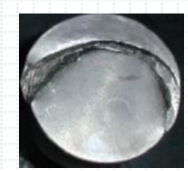


**Blind box**



**Gauge cutter**



**Lead impression block –LIB**

# Cutter Bar

Wireline Slickline cutter bar makes it possible to cut a conventional wireline from a stuck tool string. The Cutter bar is constructed of a solid length of bar with a slot and guide plate through which the wireline is passed the tool is then dropped downhole in the conventional manner.If the rope socket is clean a cutter bar type Cutter bar will cut directly on the rope socket.

# THE END
# &
# THANK YOU